

Lame Duck by Primary: Effects of Electoral Incentives on U.S. House Representatives

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Abstract

How do electoral incentives motivate representatives to exert effort and shape policy positions? While reelection is widely recognized as representatives' primary motivation in democratic politics, accurately estimating its effects is challenging because representatives self-select into rerunning or retiring decisions. Leveraging legislators' lame duck status after primary elections, this article estimates how electoral incentives affect U.S. House Representatives. Compiling data from 1955 to 2022, I compare legislators who win primaries and remain motivated for reelection to those who lose and exit the race. Through regression discontinuity and difference-in-differences designs, I find that electoral incentives shape legislative effort but not policy positions. Primary losers sharply reduce roll-call participation and cut bill sponsorship, cosponsorship, and constituent communication by roughly a third to two-thirds of their own baseline rates. Yet neither winners nor losers adjust their policy positions. Ideological and party-line adherence remain stable, and the estimates rule out a party-line shift of even two votes in a hundred. Electoral incentives govern how hard representatives work, not where they stand.

Keywords: Electoral accountability, electoral incentives, legislative politics, U.S. House Representative

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1 Introduction

Does the pursuit of reelection prompt representatives to work harder or shift their policy stances? Numerous scholars have emphasized the critical role of electoral incentives in democratic systems, arguing that the desire to secure reelection motivates officeholders to remain closely aligned with their constituents' interests and expectations. Elected officials may be motivated by goals other than reelection such as advancing their preferred policy agendas or heightening their standing within political parties, but the prevailing view holds that “[the electoral goal] must be the *proximate* goal of everyone, the goal that must be achieved repeatedly if other ends are to be entertained” (Mayhew, 2004). In this way, a drive for reelection serves as a fundamental mechanism of democratic accountability, ensuring political leaders' responsiveness to voter preferences and setting democratic governance apart from autocratic rule.

Despite the importance of electoral incentives in democratic politics, however, existing research offers conflicting theoretical perspectives and mixed empirical findings on what specific behaviors these incentives elicit from representatives. In theory, electoral incentives could pull a representative's behavior in opposing directions, or fail to move it at all when other forces dominate. Representatives may strive to meet voters' demands to secure reelection (Mayhew, 2004; Fenno, 1978). At the same time, they may (or may not) feel pressure from party leaders to act otherwise as they control crucial campaign and career resources (Rohde, 2010; Aldrich, 2008; Cox and McCubbins, 2005; Krehbiel, 1993). Alternatively, if representatives hold strong personal ideological commitments, electoral incentives may induce little change in their behavior at all (Alesina, 1988; Osborne and Slivinski, 1996; Besley and Coate, 1997; Poole and Rosenthal, 1985).

A core challenge in empirically evaluating theoretical claims about the impact of electoral incentives lies in disentangling these incentives from legislators' strategic decisions about whether to seek reelection or retire. The effect of electoral incentives is a contrast within the same legislator: behavior with the prospect of facing voters again, and behavior without it.

Behavior without that prospect is observable only among incumbents who are leaving office, so empirical analyses compare the behavior of incumbents who run for another term with the behavior of those who step down. However, rerunning or retiring decisions are shaped by a range of personal and political factors, making it difficult to attribute observed behavioral differences solely to the presence or absence of reelection motives. As a result, much of the literature has relied on theoretical assumptions and correlational evidence rather than on research designs that separate the effect of electoral incentives from selection into leaving office. Previous scholars have addressed this selection bias through institutional levers—term limits, electoral reforms, party control of renomination. Yet each removes electoral incentives in a way that is either anticipated far in advance or itself a strategic party decision, and many elective offices, including the U.S. House, offer no such institutional lever at all, so robust evidence on the behavioral effects of electoral incentives has remained correspondingly limited.

This article addresses this empirical challenge in the context of the U.S. House of Representatives by exploiting the unique opportunity presented by incumbent legislators’ lame duck status after primary elections. By collecting and combining data from over 3,800 incumbent legislators and their primary election contests between 1956 and 2022, I compare the post-primary behavior of primary election winners, who remain motivated for the upcoming general election, with that of primary election losers, who are disqualified from running for reelection. To ensure comparability between winners and losers, I employ two complementary research designs. A regression discontinuity design compares incumbents who narrowly win or lose primary elections, and a difference-in-differences design contrasts legislators’ behavior across terms in which they lost or won renomination.

The findings from both designs provide consistent evidence that electoral incentives shape how much effort legislators invest. After losing primary elections, legislators sharply reduce their roll-call participation, and the timing of the decline points to the primary itself: winners and losers are nearly indistinguishable until the closing weeks of the campaign, but the gap

widens discretely once the primary determines who remains a candidate and who does not, and persists through the general election. The withdrawal is not confined to the roll-call votes that others schedule: bill sponsorship, cosponsorship, and e-newsletter communication with constituents all decline in step, each activity falling by roughly a third or more of a legislator's own rate. The e-newsletter decline follows the same post-primary pattern as roll-call participation.

The withdrawal tracks neither the competitiveness of the upcoming general election nor the length of the career at stake: even where that election is a foregone conclusion, primary winners keep up their effort and losers fall behind them as surely as in the closest races, and younger losers, with the longest careers still ahead of them, withdraw as sharply as those near retirement. What losing a primary removes is candidacy itself—having a next election to face—not the marginal return of campaigning or the value of a career still to protect.

On the other hand, the findings from both designs show that electoral incentives do not meaningfully affect legislators' policy positions. Both primary election winners and losers remain largely consistent in terms of ideological extremity and party-line adherence following primary elections, contrary to a long line of theory in which the positions legislators take respond to electoral pressure (e.g., Downs, 1965; Mayhew, 2004). This stability is not an artifact of imprecision: the estimates are sharp enough to detect a party-line shift as small as two votes in a hundred, yet none appears. To investigate the extent of this stability, I further restrict the analysis to bills that receive greater attention from party leaders, bills that scholars have marked as significant, and contested bills that are close to passing or failing. The results show that neither primary election winners nor losers significantly change their policy positions even for these bills, underscoring the substantive strength of the null findings.

These findings carry several implications. By separating the margin of legislative behavior that responds to electoral incentives (how much effort to supply) from the one that does not (what positions to take), they sharpen the foundational assumptions of electoral accountability: the effort results reinforce Mayhew's premise that reelection is the proximate

motivation (Mayhew, 2004), while the policy-position null cuts against the corollary that legislators’ roll-call positions are themselves electorally driven—and, more broadly, against the Downsian account of electoral competition (Downs, 1965). The null instead fits models in which legislators’ ideological stances are largely fixed (Alesina, 1988; Osborne and Slivinski, 1996). The stability of incumbent positions also bears on debates over party influence (Cox and McCubbins, 2007; Krehbiel, 1993), the sources of polarization, and the reliability of roll-call-based ideology measures: it suggests that polarization operates more through electoral replacement of legislators than through within-term repositioning (Barber and McCarty, 2015; McCarty, 2019; Moskowitz, Rogowski and Snyder, 2024), and that scores such as DW-NOMINATE (Poole and Rosenthal, 1985; McCarty, 2016) capture stable preferences rather than artifacts of immediate electoral pressure.

1.1 Theoretical Considerations on Legislators’ Behavior

To understand how electoral incentives affect legislators’ behavior, this article focuses on two key aspects: effort-related behavior and policy positions based on roll-call votes. Standard models of electoral accountability suggest that electoral incentives drive elected officials to increase their effort levels in hopes of securing reelection (Barro, 1973; Ferejohn, 1986). However, when representatives must allocate limited resources across multiple tasks, it is less clear which specific activities will be prioritized (Fenno, 1973; Holmstrom and Milgrom, 1991). For example, electoral incentives may push legislators to attend roll-call votes to avoid perceptions of shirking, while also pulling them away from such floor activity toward campaigning and district demands.

When it comes to legislators’ policy positions, theoretical predictions diverge. Some models predict electoral incentives pull candidates toward the median voter (Downs, 1965; Calvert, 1985), while others suggest legislators shift toward positions that mobilize core supporters (Adams and Merrill, 2003; Fiorina, 1974).

A different theoretical tradition, or citizen-candidate and commitment-based models,

instead assumes that legislators’ policy positions remain largely fixed: candidates are treated as ideologues who steadfastly adhere to their preferred ideological stances (Alesina, 1988; Osborne and Slivinski, 1996; Besley and Coate, 1997). According to this strand of models, electoral incentives would have no effect on legislators’ policy positions.

It also remains unclear how much parties influence their legislators through electoral incentives, given the estimation challenge of legislators’ selection into parties (Smith, 2007; Krehbiel, 1993; Cox and McCubbins, 2007). If party influence exists, legislators may adopt party-line stances when seeking reelection, since the party supplies campaign resources. Donors and interest groups may exert similar pressure, each steering legislators toward the positions it favors, though evidence of their influence is mixed (Stratmann, 2002; Ansolabehere, de Figueiredo and Snyder, 2003; Fowler, Garro and Spenkuch, 2020). Because these external pressures may pull in different directions, the net effect of electoral incentives on policy positions is unclear.

Comparing rerunning and leaving legislators is the most natural empirical strategy, as the former faces the prospect of an upcoming election while the latter does not. However, even setting aside selection differences between the two groups, its interpretation hinges on the motivations and incentives that drive not only the former but also the latter. Leaving legislators are often expected to reduce their legislative activity or coast through their remaining term, but alternative motivations can keep them active in office. For instance, they may anticipate post-Congress appointments to federal or local office, or prepare for careers in the private sector or academia. Personal policy preferences or ideological commitments may also keep them active; freed from electoral pressures, they may pursue these commitments more openly than before, so the comparison recovers the effect of electoral incentives only net of these substituted motivations. In particular, if leaving legislators themselves remain motivated by future electoral concerns—say, sustaining a record for a later bid—the comparison reflects only a difference in horizon, not the presence or absence of electoral incentives, and cannot identify their effect.

These alternative motivations complicate the interpretation of this comparison as a clean test of electoral incentives. The major complication depends on what causes leaving. Voluntary retirements and anticipated departures by term limits let alternative motivations work through anticipatory behavior, making it difficult to isolate the effects of electoral incentives. A less anticipated event like a primary defeat closes this anticipatory channel, but the comparison’s cleanness still depends on what motivates the unanticipated leavers themselves. Reading the rerunning-versus-leaving comparison as a clean test of electoral incentives therefore requires a closer look at leaving legislators’ motivations and trajectories, which the next subsection takes up as part of its review of previous research.

1.2 Previous Research on Electoral Incentives

Estimating the effects of electoral incentives is challenging because it requires a credible counterfactual for what a legislator would do without them. Previous research has approached this challenge through several distinct designs, each constructing the counterfactual in a different way.

Earlier literature on shirking examined whether legislators behave differently in their last term, when they are not seeking reelection (Lott, 1987; Lott and Davis, 1992; Zupan, 1990). Because leaving legislators do not face electoral incentives, they can serve as a counterfactual, but only if they are otherwise comparable to those seeking reelection. If retirement decisions are strategic, or if retirees differ in quality or ideology from those seeking reelection, simple comparisons would introduce selection bias. Because these studies relied on such comparisons, they could not separate the effect of electoral incentives from systematic differences in who leaves office.

Building on earlier research, scholars have attempted to address this selection bias through various research designs. These approaches include leveraging term limits (Alt, Bueno de Mesquita and Rose, 2011; Fourniaies and Hall, 2022; Besley and Case, 1995), electoral reforms (Motolinia, 2021), and parties’ candidate nomination processes (Fiva and Nedregård, 2024);

each relaxes the selection problem, but none applies to the individual reelection incentive of U.S. House incumbents.

Similar to this article, Fiva and Nedregård (2024) employ party renomination decisions in Norway to compare legislators’ within-term behavior following the renomination outcome. Both studies find that legislators who fail to secure renomination tend to reduce their effort-related activities while largely maintaining their policy positions. Aside from the comparability concerns inherent to party-led renomination decisions in Fiva’s setting, the two works are situated in markedly different political contexts. In Norway, party leaders decide which incumbent legislators are renominated, a process characterized by strong party discipline and limited personal campaigning. In contrast, in the United States, incumbents must compete in primary elections that emphasize individual candidacies. Consequently, while Fiva and Nedregård (2024) highlight the influence of party leadership in shaping legislators’ behavior, this article centers on how individual legislators pursue reelection under a more candidate-driven electoral system.

Using a term-limit design with state legislators, Fourinaies and Hall (2022) also report similar findings: termed-out legislators reduce productivity yet maintain policy positions. The key difference is electoral salience. U.S. House races attract far more public attention than state-level contests, and past work attributes the policy-position null among state legislators specifically to that low salience (Rogers, 2017, 2023). The finding that the same null obtains in the high-salience U.S. House context is therefore striking, contrary to claims from earlier studies (e.g., Ansolabehere, Snyder and Stewart, 2001; Griffin, 2006; Fiorina, 1973).

Compared to a term-limit framework, this article’s use of primary elections offers a more precise way to differentiate between legislators motivated by electoral considerations and those who are not. As most House Representatives exit electoral politics after a primary defeat (Table 1), comparing primary winners and losers provides a clearer gauge of the impact of electoral incentives on legislative behavior. By contrast, termed-out state legislators may

Table 1: Career Trajectory After Losing in Primary Election

Run for Elections Later	Non-elected Public Position	Non-public Career	Retire or No Records
28 (17.50%)	27 (16.88%)	63 (39.38%)	42 (26.25%)

Note: The table shows the distribution of House representatives’ post-Congress career trajectories after losing in primary elections. Only 17.5% of them ended up running for an electoral position again, while the majority pursued non-electoral or non-public roles, retired, or left no trace of their activities after leaving Congress.

still respond to future political prospects and party pressure: Steen (2006) shows that those hitting state-legislative term limits are more likely to run for Congress than those still eligible to serve, Fourniaies and Hall (2022) find stronger productivity effects in states where term limits prohibit running for the other chamber, and Olson and Rogowski (2020) document continuing party pressure on termed-out legislators. These ongoing incentives confound any attempt to isolate the effect of reelection pressure.

Other studies compare general-election winners and losers in brief November–December postelection lame-duck sessions (Jenkins and Nokken, 2008*a,b*; Schönenberger, 2024), and they face their own limitations. Such sessions see little new legislative activity since bills introduced then expire with the outgoing Congress, and what business does occur is largely confined to must-pass items—appropriations, the National Defense Authorization Act, and continuing resolutions—shaped by strategic party agenda-setting. Primary losers, by contrast, are observed over the months between the primary and the general election, when the House conducts its ordinary business and member-initiated activity continues, not in a truncated postelection window. Furthermore, general-election winners with a two-year horizon are less immediately motivated by reelection than primary winners, who face a general election within months and provide a sharper baseline.

Finally, Cowburn and Sältzer (2025) employ a research design most similar to this article’s by comparing winners and losers of primary elections among U.S. House representatives. They analyze legislators’ language usage on X (Twitter) in 2020 and find that Democratic

primary losers took significantly more moderate positions after their defeat, whereas no such effects were observed among Republicans. By contrast, this article examines a wider array of legislative behavior, focusing on various effort-related activities and policy positions based on roll-call votes, over a more extensive period from 1955 to 2022. The outcome constructs also differ: their measure is positioning expressed in campaign rhetoric, which legislators can recast freely, while this article’s is the roll-call record itself—a difference that can reconcile their post-defeat moderation with the position stability documented below.

2 Data on Primary Elections and Legislator Behavior

In the United States, primary elections typically occur several months before the November general election—as early as March and as late as September—providing parties an opportunity to select their preferred candidate to compete for the office. For incumbent legislators, these contests are especially significant: a primary win maintains their eligibility and drive to compete in the general election, while a defeat effectively disqualifies them from running for reelection.

This article exploits these features of primary elections by comparing the post-primary behavior of incumbent House representatives who win their primary contests with that of those who lose.

Pettigrew, Owen and Wanless (2014) provide primary election results for U.S. House Representatives from 1956 to 2010, and Miller and Camberg (2020) provide them from 2012 to 2018. These data do not include runoff elections, so I manually collected them from state archives and past newspapers; I also collected results for 2020 and 2022 from state archives. The combined data cover U.S. House Representatives from 84th to 117th Congress, or 1955 to 2022, a total of 14,582 Congress-legislator units, and 2,673 distinct legislators. Of them, I focus on 3,860 Congress-legislators (1,546 distinct legislators) who competed in primary elections with at least one other challenger. There were 3,700 victories and 160 defeats.

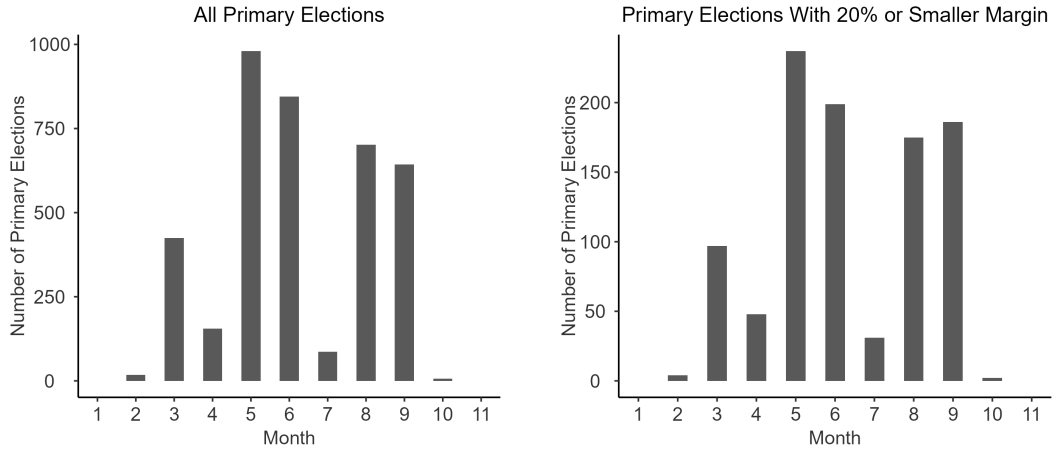
To measure policy positions, I use roll-call votes, including ideological positions on a liberal-conservative scale and party-line stances. For effort-related behavior, I examine legislative activities that proxy for legislator effort, drawing on previous literature (Dal Bo and Rossi, 2011; Titunik, 2016; Fourinaies and Hall, 2022). These activities include roll-call participation, bill sponsorship and cosponsorship, and the frequency of e-newsletters sent to constituents.

The House representatives' information and roll-call voting records are collected from voteview.com (Lewis et al., 2024). Sponsorship data combine the Congressional Bills Project's records for 1973 to 2014 (Adler and Wilkerson, 2020) with sponsor records I parse from official Congress.gov bulk data for 2015 to 2022; the Congress.gov records themselves reach back to 2003, and across the six overlapping Congresses the paper draws from the Bills Project (the 108th to 113th), per-legislator counts from the two sources correlate at 0.96, with a median within-legislator ratio of 1.00; Appendix E details the roster restriction and identifier repairs applied to these records. Cosponsorship records combine James Fowler's cosponsorship data for 1979 to 2004 (Fowler, 2006) with records I collect from official Congress.gov bulk data for 2005 to 2022, covering all four types of House bills and resolutions. E-newsletter data come from DCinbox (Cormack, 2024) and cover 2010 to 2022. The designs compare legislators who still face an upcoming general election with legislators who do not, and that contrast is defined only until the general election is held: once it passes, the winner group itself mixes in general-election losers, whose reelection incentives are also extinguished. All outcomes are therefore measured from the start of the Congress through each legislator's own general election date, and the post-general lame-duck weeks are reported separately, as an explicitly labeled extension (Appendix G).¹

Examining post-primary behavior requires knowing the exact dates of primary elections. The Federal Election Commission provides primary election dates for all states back to 1996,

¹Extending the roll-call outcomes over the full Congress, including the post-general lame-duck session, moves the vote-level difference-in-differences participation estimate from 0.105 to 0.115 in absolute value—10.5 to 11.5 percentage points of roll-call participation—both precisely estimated. The appendix opens with a note on the window convention each exhibit uses.

Figure 1: Monthly Distribution of Primary Elections



Note: The left figure shows the month distribution of all primary elections, while the right figure shows those with less than 20% point margins. In both cases, June was the median month.

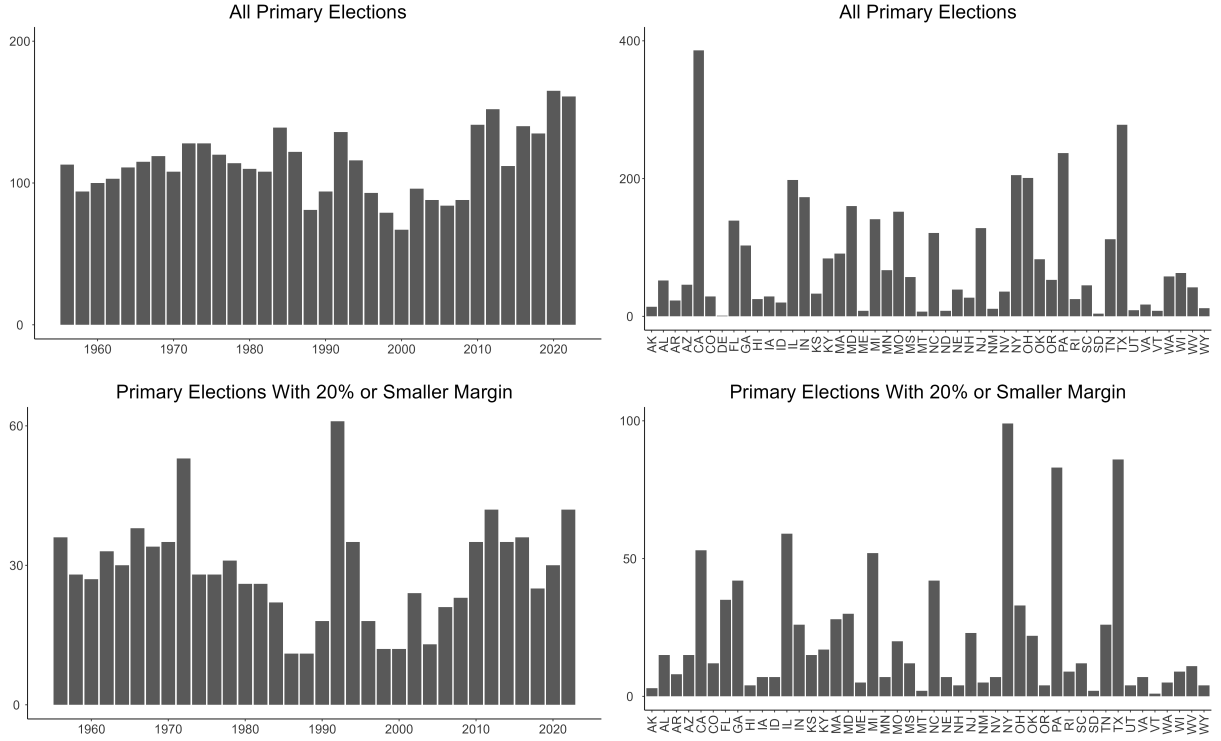
and a few state archives provide earlier dates. For the other years and states back to 1956, I manually collected them from historical newspaper articles.

Figure 1 and Figure 2 show the distribution of primary election dates by month, state, and time. The median month of primary elections was June. A total of about three and a half million roll-call votes were cast by the legislators in the analysis, and 24 percent of those were cast after each legislator’s primary election date.

3 Empirical Strategy

To examine the behavior of primary winners and losers, I employ two complementary research designs. A regression discontinuity design compares legislators who narrowly won versus narrowly lost close primary races. A difference-in-differences approach tracks how legislators’ behavior changes across terms in which they won primaries versus terms in which they lost. Both designs aim to ensure comparability between winners and losers: the regression discontinuity design exploits the as-if random outcomes of close primary elections, while the difference-in-differences approach uses within-legislator changes across terms. Each design has its own strengths and weaknesses, and together they provide more robust evidence on

Figure 2: Distribution of Primary Elections by Year and State



Note: The graphs on the left and right illustrate the number of primary elections by year and by state, respectively. The top rows display the distribution for all primary elections, and the bottom rows show the distribution for those with margins of less than 20%.

the effects of electoral incentives.

3.1 Variables and Measurement

The main independent variable distinguishes whether an incumbent House member wins or loses a primary election, capturing the presence or absence of electoral incentives for the general election. The dependent variable tracks how legislators behave after the primary, reflecting their response to the electoral incentives. Specifically, I measure within-term change in behavior from pre-primary to post-primary periods, which discounts varying base levels across legislators and absorbs any pre-primary anticipatory behavior into the baseline, thereby isolating each legislator's reaction to primary outcomes.

This article’s outcome variables fall into two structurally distinct types. Vote-based outcomes—roll-call participation, party-line voting, and ideological positions—are responses to a legislative agenda set externally by party leaders and committee chairs: legislators respond to whichever bills reach the floor. Because the composition of bills scheduled before and after the primary may differ systematically, controlling for bill-level variation is essential. In contrast, non-vote outcomes—bill sponsorship, cosponsorship, and e-newsletters—are activities where legislators can more freely set their own agenda. This structural difference shapes the dependent variable construction and estimation strategy for each type.

To analyze non-vote behaviors, I measure each activity as a weekly rate and compare the rate after the primary with the rate before it. For each legislator-Congress, the pre-primary rate divides the activity count between the start of the Congress and the primary date by the length of that period in weeks, and the post-primary rate does the same for the period between the primary and the general election. The dependent variable is the change in weekly activity,

$$\frac{B_2}{w_2} - \frac{B_1}{w_1}, \tag{1}$$

where B_1 and B_2 are the pre- and post-primary counts and w_1 and w_2 are the lengths of the two windows in weeks. Measuring rates accommodates the fact that primary dates—and hence window lengths—vary across states and years, and anchoring the pre-primary window at the start of the Congress uses each legislator’s full pre-primary record, so that short-lived fluctuations close to the campaign do not dominate the baseline. The window lengths are measured against each activity’s own opportunity set. Introducing and cosponsoring bills require the House to be in session, so for sponsorship and cosponsorship w_1 and w_2 are session-weeks—House session days, from the Congressional Record’s session calendar, divided by five—rather than calendar weeks; a calendar denominator would dilute the rate with weeks in which no bill could be introduced, and the two-month recesses and adjournments concentrate in the post-primary window. E-newsletters go out during recess as well, so their rates remain per calendar week, and roll-call participation is already measured against

the votes actually held. The session calendar, its validation against the House Historian’s legislative days, and calendar-time comparisons for every cell appear in subsection A.19. I apply this measure to legislators’ bill sponsorship, co-sponsorship, and e-newsletter counts in both research designs. The appendix reports a bounded alternative—the proportion of activity occurring after the primary, computed on windows of equal length on both sides of the primary—and discusses where the two measures agree and where they register different margins of the same response; the proportion’s exposure to pre-primary level imbalance is tested directly there (Appendix D, Table D.2).

For roll-call-based behaviors (roll-call participation, voting with the party majority, and voting with party leaders), the two research designs construct outcomes at different units of analysis but estimate the same within-legislator post-primary parameter.

For the regression discontinuity design, I construct a legislator-level dependent variable by estimating, for each Congress t , the following OLS regression:

$$\text{Vote}_{ipbt} = \gamma_{it} + \delta_{pb} + \beta_{it}\text{AfterPrimary}_{ibt} + \varepsilon_{ipbt}. \quad (2)$$

In this model, the left hand side, Vote_{ipbt} , is a binary indicator set to 1 if legislator i from party p exhibits a specific voting behavior, such as participating in roll-call voting, on bill b in Congress t , and 0 otherwise. The term γ_{it} represents legislator fixed effect and δ_{pb} accounts for bill-by-party fixed effects. The key variable, $\text{AfterPrimary}_{ibt}$, is a binary indicator that takes the value 1 if legislator i faces the roll-call for bill b after the primary election and 0 otherwise. The coefficient β_{it} captures the direction and magnitude of the change in legislator i ’s voting behavior after the primary in Congress t , and I use it as the legislator-level outcome in the regression discontinuity specification (Section 3.2.1). I extend this approach to other aspects of roll-call voting behavior by re-estimating the regression with alternative definitions of Vote_{ipbt} , such as aligning with the party majority, or voting in line with party leaders.²

²When measuring the policy positions based on roll-call votes, I follow existing measurement methods and include roll-call votes where legislators voted either “yea” or “nay,” excluding any abstained votes, so that abstention does not enter the position estimates mechanically. The estimates therefore rest on the votes

For the difference-in-differences design, I construct a bill-vote-level dependent variable by retaining the same bill-level vote data and estimating the post-primary change directly via the interaction between primary outcome and the post-primary indicator (full specification in Section 3.2.3). This avoids the intermediate step of collapsing each legislator-Congress' voting records into a single β_{it} and estimates the same within-legislator parameter; the two approaches yield quantitatively close estimates (Table A.12).

The two designs construct the vote-based dependent variables at different levels of analysis to reflect the structure of their identifying variation. The regression discontinuity design compares different legislators near the threshold, so each legislator-Congress contributes one independent observation; replicating the same running variable across a legislator's bill votes would inflate the sample without adding independent comparisons. The difference-in-differences design instead compares the same legislator across terms, so vote-level observations are natural for the within-legislator comparison and allow bill-by-party-by-Congress fixed effects to absorb bill composition differences.

These approaches for vote-based behavior share an important advantage over a simple pre/post mean comparison: the types of bills introduced before and after primary election dates may differ systematically, so a raw rate comparison could conflate genuine post-primary behavior change with shifts in the legislative agenda. Through bill-party fixed effects within Congress, these dependent variables absorb the bill-by-party level differences, ensuring that the estimated change reflects post-primary lame-duck status itself.

Nevertheless, as a robustness check, I also measure policy positions in the appendix using a simple difference of mean voting patterns between pre- and post-primary periods, with consistent results (Table A.11). The appendix also presents a descriptive comparison of bills before and after the primary. The distribution of bill topics is largely stable across the two periods, with government operations, defense, and public lands remaining the most common categories both before and after the primary (Figure A.3); similarly, weekly vote

legislators chose to cast; the bill-by-party fixed effects keep the comparison within a given bill, so a thinner post-primary vote set alters the weighting of bills, not the within-bill comparison itself.

volume, participation, and passage rates show no systematic shift around the primary date (Figure A.4). These comparisons thus reveal no notable, systematic differences in the bills considered before and after primary elections.

Roll-call extremism, an additional vote-based outcome, requires one further step beyond this shared construction: each vote is coded by the legislator’s own-party pole (conservative for Republicans, liberal for Democrats), so that higher values indicate own-pole voting. For the regression discontinuity design, I re-estimate equation (2) with Vote_{ipbt} set to a conservative-vote indicator—classifying each bill as liberal or conservative using the bill-direction algorithm of Fowler and Hall (2012)—and then sign-flip the resulting β_{it} so that higher values indicate increased conservatism among Republicans or increased liberalism among Democrats. For the difference-in-differences design, the same conservative-vote indicator is sign-flipped for Democrats—yielding the own-party-pole indicator extreme_{iv} —so that higher values likewise indicate own-pole voting. Both approaches estimate the same conceptual quantity, i.e., the within-legislator post-primary shift in own-pole voting, and I additionally examine the absolute values of these changes to assess whether legislators adjust positions in either direction even when net shifts cancel.

3.2 Research Designs

The research designs exploit the fact that an incumbent must win the party’s primary or be a sole candidate to advance to the general election. While most incumbent House Representatives secure primary election victories with relative ease, this is not always the case. Occasionally, a prominent local figure’s entry into politics poses a threat to the incumbent, and at other times, primary voters view their incumbent politician as old and incompetent, opting to replace them with a new candidate. Changes in district boundaries can also jeopardize an incumbent’s electoral prospects, as they may move loyal supporters to other districts. Consequently, some primary elections result in either the incumbent losing or facing a closely contested race. These variations in outcomes provide the identifying variation for estimating

electoral incentive effects.

3.2.1 Regression Discontinuity Design

First, I implement the regression discontinuity design. The running variable V_{it} is a legislator i 's share of the top two candidates' vote less 0.5 in the primary election during Congress t . For example, if an incumbent received 60% of the votes, and two challengers received 35% and 5%, respectively, the running variable is calculated as $\frac{60\%}{60\%+35\%} - 0.5 = 0.132$. This indicates that incumbents with positive running variables won primary elections, whereas those with negative running variables lost.

The construction of this running variable is not feasible in states with different primary election systems, so I exclude these states. Specifically, I exclude Georgia before 1962, which used a county-unit system in place of direct popular vote. I also exclude states with nonpartisan blanket primaries, such as the top-two or top-four systems used in Washington (since 2008), California (since 2012), and Alaska (since 2022). Additionally, I exclude Connecticut and Louisiana because their primary systems differ significantly; in Connecticut, party nominees are typically chosen at party conventions, making primaries rarely competitive, while Louisiana uses a nonpartisan blanket primary where winners can be elected outright.

I estimate the standard regression discontinuity specification,

$$Y_{it} = \beta_0 + \beta_1 \mathbf{1}\{V_{it} < 0\} + f(V_{it}) + \varepsilon_{it}. \quad (3)$$

Here, V_{it} is the running variable, or a legislator's top-two winning margin in the primary election, and Y_{it} captures the legislator's change in behavior following the primary election. The function $f(\cdot)$ is flexible in the running variable; given the small sample size, I primarily use a linear specification (Gelman and Imbens, 2019) with a 20-percentage-point bandwidth. In the appendix, I show that the sign and direction of the estimates remain largely stable across alternative fixed bandwidths (Figure A.2), and I report estimates at the data-driven

Table 2: Observable Characteristics of Legislators by Electoral Margin in Primaries

	(1)	(2)	(3)	(4)
	Close	Close	Close	Non-close
	Losers	Winners	Primaries	Primaries
N	159	820	979	2,881
Congress	100.08 (10.31)	100.16 (10.39)	100.14 (10.37)	101.34 (10.04)
Seniority	5.11 (3.31)	5.04 (4.01)	5.05 (3.90)	5.06 (3.90)
Age	57.99 (10.98)	56.27 (10.97)	56.55 (10.98)	55.40 (11.01)
Democratic party	0.65	0.62	0.63	0.62
Previous vote share in general	0.68 (0.15)	0.71 (0.16)	0.70 (0.15)	0.70 (0.15)
DW-NOMINATE score	−0.08 (0.40)	−0.05 (0.39)	−0.05 (0.40)	−0.06 (0.40)
DW-NOMINATE $\times (-1\{\text{Democrat}\})$	0.37 (0.18)	0.35 (0.19)	0.35 (0.19)	0.37 (0.17)
Roll-call participation before primary	0.90 (0.10)	0.91 (0.08)	0.91 (0.08)	0.94 (0.07)
Sponsored bills pre-primary	3.24 (4.04)	3.07 (4.15)	3.09 (4.13)	3.09 (3.95)
Cosponsored bills before primary	66.03 (46.83)	57.07 (41.24)	58.52 (42.29)	57.89 (45.30)
E-newsletters before primary	21.40 (38.36)	14.88 (16.74)	15.95 (21.80)	11.73 (14.67)
Voting with party majority before primary	0.87 (0.10)	0.87 (0.10)	0.87 (0.10)	0.89 (0.09)
Voting with party leaders before primary	0.88 (0.10)	0.88 (0.11)	0.88 (0.11)	0.90 (0.09)

Note: Close primaries are those with margins smaller than 20 percentage points. Standard deviations are in the parentheses. Count activities are measured over the equidistant pre-primary window and cover each activity’s record years (sponsorship 1973–2022, cosponsorship 1979–2022, e-newsletters 2010–2022, with non-users counted as zeros), so their cells rest on the correspondingly smaller subsamples.

optimal bandwidth of Calonico, Cattaneo and Titiunik (2014) (Table A.6). Because close primaries are few, those optimal windows are narrow and trade statistical power for local identification—several effort coefficients, including participation, lose significance there—so I treat them as a limitation of the close-primary design rather than as corroboration and retain the 20-percentage-point window for the main analysis. The coefficient of interest, β_1 , represents the RD estimator, capturing the effect of primary election *loss*—the removal of electoral incentives that is the substantive comparison of interest—on changes in legislators’ behavior, and it is the coefficient the tables report.

Columns (1) and (2) of Table 2 present a balance check between winners and losers in close primary elections within a 20-percentage-point vote margin. The two groups are well balanced on demographic and ideological characteristics and on the vote-based pre-primary behaviors (roll-call participation, voting with the party majority, voting with party leaders); formal tests at the threshold detect no discontinuity in any pre-determined characteristic, in freshman status or seniority, or in the length of the measurement window itself (subsection A.1, which opens the appendix with the full balance record). Pre-primary activity levels are balanced as well: in the window just before the primary and over the full pre-primary period, no activity differs significantly between close winners and close losers at the cutoff. The cosponsorship and e-newsletter point estimates in the window just before the primary are positive—close losers the more active group—but each estimate carries a standard error of its own size, making it statistically insignificant (+9.4 cosponsored bills, standard error 8.8; +9.1 e-newsletters, 6.8). One Congress earlier the differences that do appear have eventual losers the *less* active group (sponsorship, -0.17 per session-week, $p = 0.04$; e-newsletters, -0.40 per week, $p = 0.09$), while cosponsorship’s estimates, positive there as well, are smaller than their own standard errors on every unit. Both run against, not toward, the pattern that could manufacture a post-primary decline (Table A.2).

Any residual winner–loser difference, even one too small for the balance tests to detect, could in principle enter and bias the estimates through two channels: the legislators themselves may differ (eventual losers simply a different kind of representative throughout), or their activity may already be moving within the loss term as the primary approaches. The first channel is neutralized by construction and tested outside the loss term. By construction, a legislator’s baseline level does not enter the estimates by itself: the RDD is estimated on the change in each legislator’s own weekly rate, and the DiD compares each legislator’s post-primary weeks with the same legislator’s pre-primary weeks in that term; what enters is the change against that baseline. The test is whether the upcoming primary’s outcome explains behavior one Congress earlier, where the loss it assigns cannot yet operate. For

levels, the preceding balance record already answers: the only previous-Congress cells away from zero run opposite, eventual losers the less active group. For patterns, recomputing every change estimate around each legislator’s own previous-Congress primary—the same pre/post contrast, one term removed—returns null in all fourteen cells, the one marginal cell opposite in sign (Table A.21).

The second channel is examined inside the loss term, where the event studies of Section 4 confront it directly: the pre-primary leads are flat for participation and sponsorship. The remaining two outcomes, cosponsorship and e-newsletters, depart from this pattern. Cosponsorship’s leads fail flatness because eventual losers begin scaling back in the final pre-primary months while winners’ own path stays flat, a movement that breaks discretely at each legislator’s own primary date and carries losers below their own earlier baseline, the trajectory of a response taking hold rather than of a prior difference reverting (Appendix C); e-newsletters diverge modestly in the final campaign month (Appendix B). Each departure bears on the interpretation of that outcome’s own result; Section 4 therefore weighs them where those results are presented, so that estimate and caveat can be read together. Finally, both channels are tested by the same direct adjustment: each change coefficient is re-estimated with the legislator’s own pre-primary level added as a control. To the extent a coefficient reflected the first channel (a stable difference between the legislators themselves), the level measured over the entire pre-primary period would absorb it; to the extent it reflected the second (movement already underway), the level measured over the outcome’s own window ending at the primary would. None moves materially under either control (subsection A.3).

3.2.2 External Validity of Regression Discontinuity Design

External validity warrants further consideration. The regression discontinuity design results are derived exclusively from legislators who faced close primary elections, which may limit their generalizability to the broader population of legislative representatives. To address this concern, I also estimate a difference-in-differences specification (Section 3.2.3) that uses all

contested primaries, not only close ones, and produces consistent results. As a more direct check on scope, I compare the contested-primary incumbents who enter the design against the non-challenged incumbents who fall outside it—the group that most concretely stands in for that broader population (subsection A.15). The two groups differ only modestly on observable characteristics: the two district-political variables of greatest concern, general-election margin and district presidential vote, differ by just 1.1 and 2.6 percentage points. More tellingly, in a three-group difference-in-differences comparison the non-challenged incumbents behave like narrow primary winners rather than losers across twelve effort and ideology outcomes—exactly what the lame-duck mechanism predicts for legislators who never carried a re-election threat to remove, and an indication that the post-primary effort decline is concentrated specifically among legislators whose re-election incentive disappears. The following considerations further suggest that findings from close primary elections may reasonably extrapolate to legislators in general, including those facing minimal or no primary challenges.

First, representatives in close and non-close primary elections appear largely similar on descriptive characteristics. Columns (3) and (4) of Table 2 present legislators’ characteristics and behavior before the primary elections, distinguishing between those from close and non-close primary contests. Overall, the two groups appear similar across most observable variables, with e-newsletter counts showing the only statistically significant difference between the two groups.

Second, the motivations for primary challenges are similar across close and non-close contests. Using Cowburn (2024)’s classification of challenge motivations for U.S. House Representatives from 2006 to 2020 (following Boatright, 2014), I find that the underlying reasons are largely similar across the two groups. Among close primary contests within 20% point margin, the most common reason for a challenge is the perception that the incumbent is too moderate (56%), followed by concerns over age and competence (17%), the incumbent being seen as too extreme (13%), scandals (5%), and national issues (4%). This distribution

is not dramatically different from that of non-close primaries, where the most frequently cited reason remains the incumbent’s moderate ideology (45%), followed by age and competence (24%), the incumbent being perceived as too extreme (14%), national issues (4%), and scandals (under 1%).

Third, many close primary elections occur as a consequence of redistricting. Redistricting often disrupts an incumbent’s electoral advantages unexpectedly by forcing incumbents to run against one another or by splitting their loyal voter bases across districts. Because these disruptions often lie beyond the influence of legislators’ own strategic decisions or qualities, close primary contests in these instances are less likely to reflect systematic differences between legislators alone. The data confirm this pattern: on average, about 40 House representatives face a primary election decided within a 20% point margin during a redistricting year, compared to about 26 in a non-redistricting year. To the extent these challenges arise from unforeseen external shocks, the dynamics observed in close elections can reasonably generalize to legislators facing little or no primary competition; in a robustness check, I restrict the analyses to redistricting years only, with consistent results (Table A.7).

3.2.3 Difference-in-Differences Design

To demonstrate that the results are not a product of the small sample of close primary elections, I also employ a difference-in-differences design. In this research design, I compare individual legislators’ behavior across terms when they won primary elections versus terms when they lost. The advantage of this design is its ability to utilize all contested primaries rather than only close ones—the same 3,860 contested legislator-terms described in Section 2, that is, incumbents who competed against at least one challenger, now across the full range of primary margins—thereby representing a larger number of legislators and offering greater statistical power. As shown in the results, both designs yield similar conclusions, reinforcing the findings and providing robust evidence. The wide-margin pool does not drive the agreement between the designs: excluding near-uncontested landslide winners, those with winning

margins above 45 percentage points, leaves the vote-level participation estimate unchanged at 0.105 in absolute value—about ten percentage points of roll-call participation—and restricting the panel to close races decided within twenty points yields 0.103.

The difference-in-differences specification differs across outcome types, reflecting the dependent-variable structure introduced in Section 3.1. For roll-call-based outcomes (roll-call participation, voting with the party majority, voting with party leaders, and roll-call extremism), I estimate legislators’ change in behavior after primary at the legislator-bill-vote level. Let Y_{ibt} denote legislator i ’s voting outcome on roll-call vote b in Congress t , PrimaryLoss_{it} an indicator for whether the legislator lost the primary in the Congress t covering vote b , and $\text{AfterPrimary}_{ibt}$ an indicator for whether the vote occurred after that primary’s election date. I estimate

$$Y_{ibt} = \beta_0 + \beta_1 \text{PrimaryLoss}_{it} + \beta_2 \text{AfterPrimary}_{ibt} + \beta_3 (\text{PrimaryLoss}_{it} \times \text{AfterPrimary}_{ibt}) + \delta_i + \gamma_{bpt} + \varepsilon_{ibt}, \quad (4)$$

where δ_i denotes legislator fixed effects and γ_{bpt} denotes bill-by-party-by-Congress fixed effects (with p the legislator’s party), which absorb bill-level and party-specific shocks. Standard errors are clustered at the legislator level. The coefficient of interest is the interaction β_3 , which identifies the difference in the post-primary change in voting behavior between terms in which a legislator lost versus won the primary, net of time-invariant differences across legislators.

For non-vote outcomes (sponsored bills, cosponsored bills, and e-newsletters)—where legislators can more freely set their own agenda, as discussed in Section 3.1—the vote-level specification in equation (4) is not appropriate: these activities are each legislator’s own output rather than responses to a common set of bills, so there is no shared bill-level structure on which to define the γ_{bpt} fixed effects. For these outcomes I instead estimate a weekly count panel. Each legislator-Congress is expanded into weekly activity counts from

the start of the Congress through the general election, on a common grid of Wednesday-to-Tuesday weeks anchored at each Congress’s first Wednesday—general elections fall on Tuesdays, so the term ends on a week boundary—and, for sponsorship and cosponsorship, restricted to the weeks with at least one House session day, per the session-time convention of Section 3.1. I estimate

$$Y_{iwt} = \beta_0 + \beta_2 \text{AfterPrimary}_{iwt} + \beta_3 (\text{PrimaryLoss}_{it} \times \text{AfterPrimary}_{iwt}) + \delta_{it} + \gamma_{pwt} + \varepsilon_{iwt}, \quad (5)$$

where Y_{iwt} is legislator i ’s activity count in week w of Congress t , δ_{it} are legislator-Congress fixed effects, and γ_{pwt} are party-by-Congress-by-week fixed effects—the count analog of the bill-by-party-by-Congress fixed effects in equation (4). Because primary dates are staggered across states, legislators of the same party are observed in the same calendar week both before and after their own primaries, so β_3 compares winners’ and losers’ within-term changes in weekly activity while absorbing party-specific agenda and calendar shocks. $\text{AfterPrimary}_{iwt}$ equals one for the weeks that begin after the primary date. House primaries are nearly all held on Tuesdays—the last day of the week on this grid—so for over ninety percent of the sample the pre/post boundary falls exactly on the primary date; in the remaining states, whose primaries fall on other weekdays, the week containing the primary is coded pre-primary, and dropping those weeks leaves every estimate essentially unchanged. Standard errors are clustered by legislator; coarser time fixed effects yield the same conclusions. Because legislators differ widely in baseline activity volume and the weekly counts are sparse, equation (5) is also estimated by fixed-effects Poisson regression, which compares each legislator’s post-primary weeks against that legislator’s own baseline rate—treating the pre-primary period as data rather than as an ingredient of the outcome—and measures the change in proportional terms; these estimates are reported alongside the least-squares estimates for the count outcomes.³

³Fixed-effects Poisson estimation uses only observations that carry information about a proportional change, which is why the Poisson columns report smaller sample sizes than their least-squares twins. For sponsorship and cosponsorship the difference consists mostly of weeks in which no legislator of a party records any activity—overwhelmingly weeks the House is in recess—which enter a least-squares average as zeros but are absorbed as uninformative by the Poisson week fixed effects. The remainder, and for e-newsletters the

For the vote-based outcomes, an aggregated legislator-level specification is also reported in the printed appendix (Table A.12) as a cross-check.

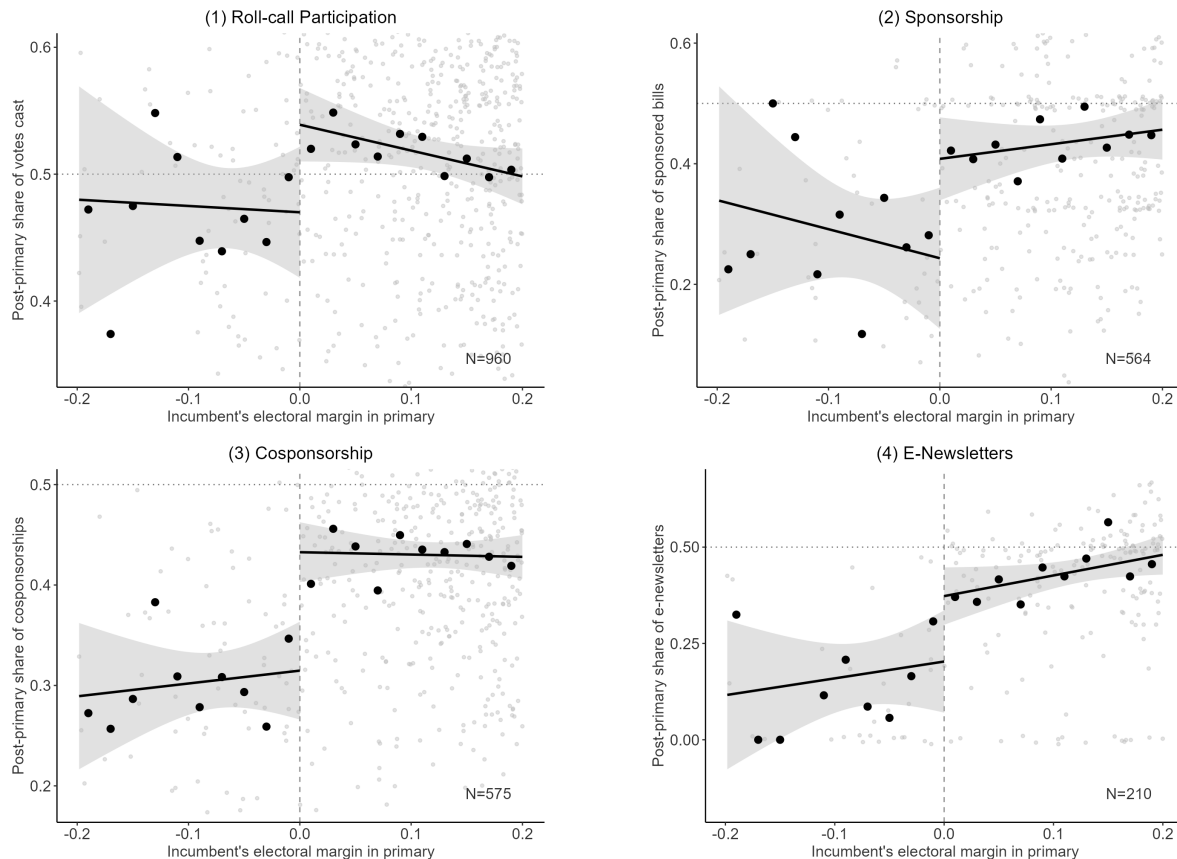
4 Primary Losers Exhibit Decreased Effort

Figure 3 presents the discontinuity for all four effort activities, each measured as the share of a legislator’s activity that falls after the primary: all four drop for legislators who fall just short of renomination, with a clear jump at the threshold between barely losing and barely winning. Measured in absolute terms rather than against a legislator’s own volume, sponsorship alone shows no discontinuity (Figure D.1). Figure 4 shows when the gaps emerge. For participation, winners and losers are indistinguishable for the first fourteen months of the term; the gap opens in the final weeks of the primary campaign, widens sharply at the primary itself, and persists through the general election; in the labeled full-term extension the gap remains—indeed grows slightly—through the post-general lame-duck weeks (Appendix G). The count outcomes break at the primary as well, each with the qualification its own paragraph below discusses.

This timing pattern is the signature of an incentive effect rather than an end-of-career drift. Because primary dates are staggered across states, the event-study panel nets out calendar-time shocks; a gradual winding down of careers would produce a slope spread over the whole term, not a flat year followed by a break aligned to each legislator’s own primary date. The modest pre-primary onset reflects the campaign itself rather than anticipatory disengagement by eventual losers: closely contested incumbents, winners and losers alike, spend the final weeks campaigning at home relative to safe winners, and the participation dip appears on both sides of the eventual result—a pattern anticipation cannot produce, since

larger part, are legislator-Congress cells with zero activity in every week (27 such cells for sponsorship, 7 for cosponsorship, 95 for e-newsletters). That drop is bookkeeping rather than sample selection: for a legislator who records no activity at all in a Congress, a proportional change is not defined—there is no rate whose movement the estimator could measure—so these cells are naturally excluded. Legislators active before the primary and silent after it, the sharpest form of withdrawal, have a well-defined proportional change and remain in the sample; only the never-active drop out.

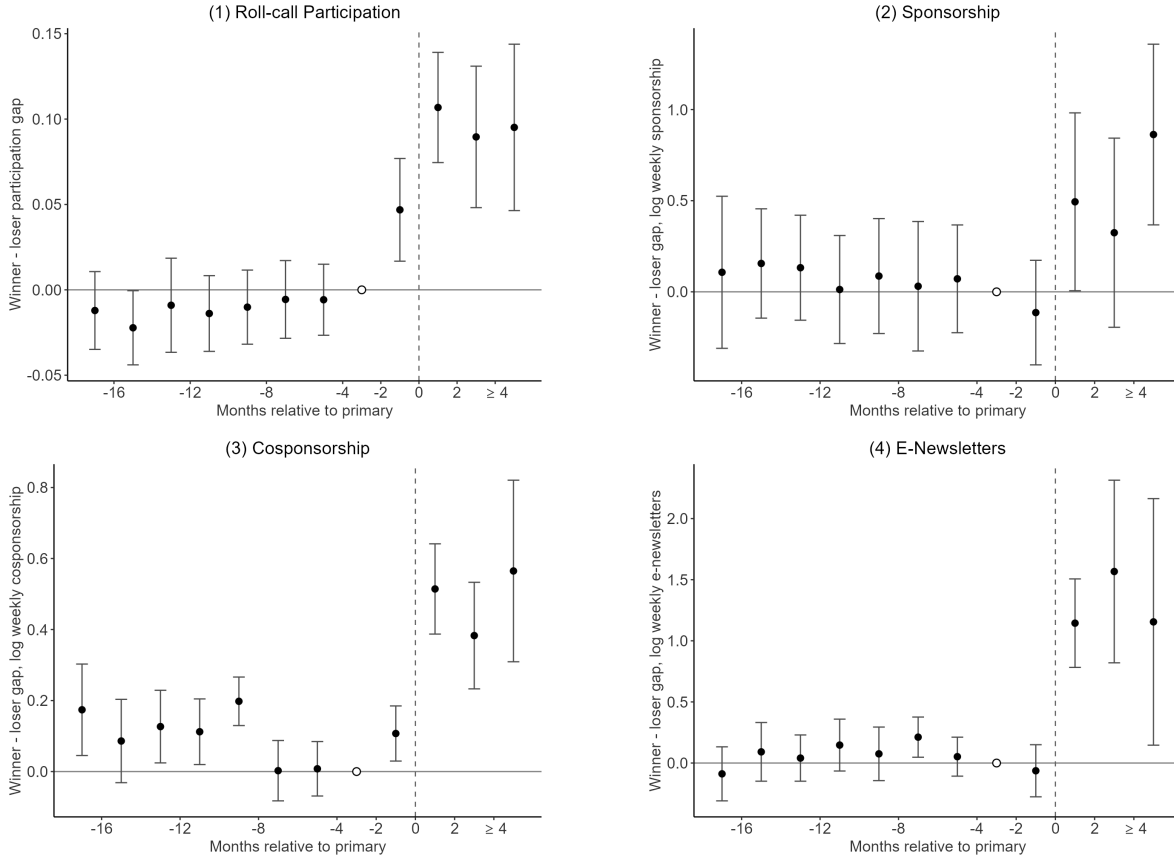
Figure 3: Regression Discontinuity Results on Legislators' Effort-Related Activities: Proportional Margin



Note: Each panel plots the share of the activity that falls after the primary, $\text{post}/(\text{pre} + \text{post})$, against the incumbent's primary winning margin; legislators to the left of the threshold lose the primary. The windows are equidistant around the primary, and the dotted line at 0.5 marks activity unchanged from before it. Large black points are averages within 0.02-point bins, small gray points the raw data jittered vertically for legibility, and the lines are OLS fits estimated separately on each side; each panel's display range is set around the binned means and fitted lines so the estimated discontinuity is legible, and raw points outside that range are not shown (fits use all observations). Sample sizes match the proportional-margin cells of Table 3; the additive-margin panels, on which sponsorship is null, appear in Figure D.1. The share is undefined for a legislator with no activity in either window, which drops one legislator-term for participation, 101 for sponsorship, 3 for cosponsorship, and 35 for e-newsletters—the least active members in each case. Sponsorship's share takes few distinct values, since the median legislator introduces about 0.16 bills per week, so its raw points fall on discrete levels.

eventual winners have no defeat to foresee. Consistent with this reading, the onset shrinks toward zero as the margin window tightens and is absent among primaries decided within ten points, races in which both sides campaign comparably hard, while the discontinuity at

Figure 4: Timing of the Effort Response: Event Studies around the Primary



Note: Each panel traces the winner–loser gap in the activity by event time relative to each legislator’s own primary date, in two-month bins with 95% confidence intervals; the open circle marks the omitted reference bin (months -4 to -2), estimates are plotted at window centers so the dashed line at the primary date separates pre from post, the edge bins pool months -16 and earlier and $+4$ and later, and the post window ends at each legislator’s general election. Panel (1) is the close-primary participation event study on the additive margin, participation’s claim margin (20-percentage-point window; full-term extension in Appendix G). Panels (2)–(4) are the proportional-margin count event studies—fixed-effects Poisson on the weekly count panels (log scale), event-time and party-by-Congress-by-week fixed effects, all contested primaries (sponsorship Congresses 93–117, cosponsorship 96–117, e-newsletters 2010–2022); the week fixed effect absorbs each week’s session exposure, and the additive-margin panels appear in Figure D.2. Standard errors are clustered by legislator. Joint tests that the pre-reference leads are flat: $p = 0.46$ (participation), 0.97 (sponsorship), < 0.001 (cosponsorship), 0.004 (e-newsletters); joint post-primary gaps: $p < 0.01$ for all four, and sponsorship’s proportional gap is deepest in the pooled final bin. Cosponsorship’s pre-primary movement comes entirely from the eventual losers’ side—winners’ own path is flat; the group trajectories are decomposed in Appendix C. The tabulated event studies behind the timing criteria appear in Table B.1 and Table C.2, with the four-week estimates reported in their notes.

the primary itself persists in every window (Appendix G). A placebo exercise reinforces the point: displacing the comparison window to arbitrary dates away from the primary produces

no effect, while the actual primary date does (subsection A.20).

Table 3 presents the effort estimates under both research designs, expressed as the effect of losing the primary. For each activity, both designs are estimated on both margins of measurement: the additive margin, the absolute change in activity, and the proportional margin, the change relative to each legislator’s own volume. The two margins have power in different places, so the table reports every combination rather than privileging one, and the paper claims what survives every informative measurement: a margin too coarse to detect the change at issue can withhold corroboration, but its null is not read as contrary evidence. For roll-call participation the two margins are two readings of the same quantity rather than independent checks—the chamber sets the denominator, the votes held, common to every legislator in a week—so its proportional estimates restate the additive ones against the base rate and are reported for completeness. Every estimate in the table carries a previous-Congress placebo, re-estimated one Congress earlier where no incentive effect exists to find; the placebos, reported in full in subsection A.21, are null throughout, and the single marginally significant cell runs opposite in sign to the estimates it checks. The two designs, resting on different comparisons and different samples, agree closely wherever both carry information. Decomposing the discontinuity confirms the pattern the design predicts: at the cutoff, winners and losers are indistinguishable in their pre-primary participation and cosponsorship levels, and the entire gap arises in the post-primary level (Table A.3).

Legislators who lose their primary election become approximately 10 to 12 percentage points less likely to participate in roll-call votes. E-newsletter communication with constituents falls by roughly two-thirds of a newsletter per week—nearly three newsletters a month—in the same post-primary pattern. The e-newsletter finding rests on the shortest record: the data cover only 2010–2022, the close-primary sample is correspondingly small, and winners’ and losers’ e-newsletter activity also diverges modestly in the final campaign month before the primary (Appendix B). A variety of placebo checks confirm the head-line finding, most decisively for participation, including a placebo-window test showing that

Table 3: Primary Loss and Legislators’ Effort-Related Activities: Two Designs, Two Margins

Regression Discontinuity				
	Roll-call Participation (share of votes cast)	Bill Sponsorship (count per session-week)	Bill Cosponsorship (count per session-week)	E-Newsletters (count per week)
<i>Additive margin: change in rate</i>				
Primary Loss	−0.119*** (0.020)	−0.024 (0.075)	−1.521*** (0.537)	−0.639*** (0.198)
Observations	926	665	578	245
<i>Proportional margin: post-primary share of activity</i>				
Primary Loss	−0.069** (0.028)	−0.165** (0.068)	−0.118*** (0.030)	−0.170** (0.072)
Observations	960	564	575	210
Bandwidth	0.2	0.2	0.2	0.2
Kernel	Uniform	Uniform	Uniform	Uniform
Specification	Linear	Linear	Linear	Linear
Difference in Differences				
	Roll-call Participation (share of votes cast)	Bill Sponsorship (count per session-week)	Bill Cosponsorship (count per session-week)	E-Newsletters (count per week)
<i>Additive margin: change in rate (least squares)</i>				
Primary Loss × Post	−0.105*** (0.014)	−0.049** (0.022)	−1.026*** (0.173)	−0.673*** (0.204)
Observations	3,520,416	236,592	206,296	96,158
<i>Proportional margin: change relative to own rate (Poisson)</i>				
Primary Loss × Post	−0.125*** (0.018)	−0.441** (0.172)	−0.387*** (0.061)	−1.223*** (0.241)
Observations	3,520,416	226,627	209,231	81,840
Unit of Analysis	Legislator-vote	Legislator-week	Legislator-week	Legislator-week
Fixed Effects	Legislator, Bill-Party- Cong.	Legislator-Cong., Party-Cong.- Week	Legislator-Cong., Party-Cong.- Week	Legislator-Cong., Party-Cong.- Week

* $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$

Note: Coefficients are the effect of losing the primary (Primary Loss = $\mathbf{1}\{V_{it} < 0\}$); standard errors in parentheses; each cell is a separate estimation. Additive-margin estimates measure absolute changes (equations (3) and (1) at the cutoff; the vote-level interaction of equation (4) and least squares on the weekly count panel of equation (5)). For sponsorship and cosponsorship the additive cells are measured over session time—cutoff rates per session-week of five House session days, and panel weeks restricted to those with at least one session day—per the measurement convention of Section 2; e-newsletters, which do not require the chamber, are per calendar week, and calendar-time estimates for all cells appear in subsection A.19.

Proportional-margin estimates measure change relative to each legislator’s own volume: the post-primary proportion post/(pre + post) at the cutoff (votes cast, for participation; windows equidistant around the primary, with 0.5 the benchmark of unchanged activity), and the same panel re-estimated by fixed-effects Poisson—the weekly count panel for the count outcomes, the vote-level panel of equation (4) for participation (log scale, with $\exp(b) - 1$ the proportional change in a legislator’s own rate; no-activity weeks and all-zero cells drop out as bookkeeping, hence the smaller count Poisson samples).

Table D.2 holds the windows fixed across margins and reports the proportion’s randomization inference. For participation the two margins are mechanically linked rather than independent—the denominator, the votes held, is common to every legislator in a week—so its proportional cells are the additive estimates translated through the base rate (the Poisson coefficient of -0.125 translates to an 11.8 percent decline in a legislator’s own rate; the least-squares estimate of -0.105 , set against the pre-primary rate it implies—the losers’ post-primary mean of 0.79 plus the decline itself—is the same 11.7 percent); they are shown for completeness of the margin map, and their agreement is arithmetic rather than independent corroboration. Winner and loser mean changes at the cutoff are 0.029 and -0.086 for participation, 0.076 and -0.611 for e-newsletters. The previous-Congress placebos are reported in full in Table A.20 (fourteen cells; the vote-level difference-in-differences placebos are not constructed), thirteen of fourteen null and the one marginal cell opposite in sign to the estimates.

Regression-discontinuity standard errors are conventional (bootstrap alternatives in Table A.22); panel standard errors are clustered by legislator. Sponsorship records cover 1973–2022, cosponsorship 1979–2022, e-newsletters 2010–2022. The weekly panel remains the primary specification for the count conclusions; sponsorship’s decline is claimed on the proportional margin, the session-time panel detecting it on the additive margin as well, and its cutoff nulls are the arithmetic consequence of its low

the effort effects are specific to the actual primary date rather than to an arbitrary short window, and a previous-Congress placebo showing that the same change is absent one term earlier (subsection A.2, subsection A.20, subsection A.21); the count results are additionally invariant to the window construction (Appendix D).

These findings closely align with conventional literature that argues legislators’ effort-related behavior is driven by their reelection motivation (Mayhew, 2004; Fenno, 1978; Cox and McCubbins, 2007). Conversely, it contrasts with predictions by scholars who contend that voters’ insufficient attention and political knowledge constrain representatives’ responsiveness and thus the effects of electoral incentives are limited (Achen and Bartels, 2017; Somin, 2016).

Nevertheless, lame duck legislators continue to participate in most roll-call votes and engage in various legislative activities, even after losing their primaries, albeit at a lower frequency than winners. After losing their primaries and until the general election, on average, these legislators still participate in 79% of roll-call votes, sponsor 1.66 bills, cosponsor 44.83 bills, and send 6.94 e-newsletters. These figures indicate that electoral incentives are hardly the sole driver of legislative activities. Their continued engagement may reflect ideological commitment, policy advocacy, reputation concerns for post-Congress careers, or a combination of these and other motivations.

Bill sponsorship, the most consequential of the count activities and the sparsest, is the outcome where measurement decides what is visible (Table 3). Measured over calendar weeks, its absolute response is barely visible: the weekly-rate discontinuity is null, and the calendar-week panel estimate reaches only the ten-percent level. Part of that faintness is arithmetic—the median legislator introduces about 0.16 bills per week, so even a decline as deep as a third of a legislator’s own rate implies an absolute change of a few hundredths of a bill—and part of it is the calendar: weeks in which the House holds no session enter a calendar-week average as mechanical zeros for winners and losers alike, diluting the gap. Measured over the weeks the House sits—the session-time convention of Section 2—the panel detects the absolute

decline (-0.049 bills per session-week, $p = 0.027$) with a clean previous-Congress placebo, while the cutoff discontinuity remains null, lacking the panel’s power. Measured against each legislator’s own baseline, the decline is detected in both designs: the fixed-effects Poisson estimate of equation (5) is -0.44 ($p = 0.010$; -0.40 among close races), and the bounded proportion falls at the cutoff (-0.165 , $p = 0.016$). Both carry null previous-Congress placebos and are sharp outliers under displaced-date randomization inference, and the cutoff estimate is unchanged by adjustment for the pre-primary level (Appendix D).⁴ The timing agrees across margins once the measurement does: the session-week event study’s post-primary gaps are jointly significant at the ten-percent level ($p = 0.056$) with flat leads, and on a base that itself shrinks toward the session’s end, the proportional gap if anything deepens late in the term (Appendix C). Nor is the proportional response a ceremonial artifact. A decomposition specified in advance, using a title classifier validated against the Center for Effective Lawmaking’s commemorative counts (Volden and Wiseman, 2014), locates the decline in substantive bills: the substantive-only Poisson estimate is -0.44 ($p = 0.016$) and the substantive-only proportion at the cutoff agrees in direction, while commemorative bills, seven percent of the records, are too sparse to be informative on any margin (Table C.3). Sponsorship, then, declines as the other three activities do, by between roughly a third and a half of a legislator’s own rate: the Poisson estimate of -0.44 translates to a decline of 36 percent ($\exp(-0.44) - 1$), and the cutoff estimate places the post-primary share of activity at 0.335 against the unchanged benchmark of 0.5, a post-to-pre ratio of about one half. The claim is stated on the proportional margin, where every powered design detects it; the session-time panel corroborates it in absolute terms, and the remaining nulls—the cutoff cells—are the arithmetic consequence of the base rate rather than evidence against the response. The proportional characterization emerged from mapping the outcome’s sensitivity

⁴A cross-sectional Poisson discontinuity—the proportional analog of the level RDD—cannot arbitrate between the margins: comparing across legislators, its minimum detectable effect is roughly twice the size of the panel estimate, and even cosponsorship, whose panel estimate is the strongest of the three count activities, returns a null under it. Its silence for sponsorship is non-corroboration by an uninformative design, not contradiction.

to measurement rather than from criteria specified in advance; it is claimed on the validation reported above.

Cosponsorship, with records covering 1979–2022 and measured as session-week rates, also declines when the reelection incentive is removed, on both margins and in both designs (Table 3). Losing a primary is associated with significantly less cosponsoring at the discontinuity, no such gap appears one Congress earlier, and the panel estimate holds among close races. At the narrow data-driven bandwidth the discontinuity itself is too imprecise to be informative, as for all count outcomes (Table A.6); the close-race panel estimate and the placebo-window test do not depend on the window choice (subsection A.20). The timing evidence carries one qualification that participation’s does not: the event-study leads are not jointly flat, so cosponsorship fails the timing criterion I specified in advance, and the timing decomposition that follows is accordingly post-hoc. That decomposition attributes the lead violation entirely to the losers’ own path. Winners’ cosponsoring is flat throughout the term. Eventual losers rise above their final pre-primary pace over the contest season, peak three to four months out, and converge back to that pace as the primary nears; at the primary they drop discretely, to nearly one cosponsorship per week below it, and stay roughly a third below their own rate through the general election ($p = 0.004$; Appendix C). Read as mean reversion, that path has the wrong destination: reversion returns unusually active legislators to their own normal pace, and losers end the term well below theirs. A prior-year comparison fixes that normal from outside the contest season: measured against the same calendar window one year earlier, close losers fall to well under half of their own prior-year cosponsoring volume while winners hold near two-thirds of theirs—a gap that is precisely estimated ($p = 0.004$) and absent one Congress earlier (Table D.3)—and the retained proportion estimate is essentially unchanged when the pre-primary level is controlled (Appendix D). The trajectory, then, reads as a withdrawal that begins as the loss becomes foreseeable and breaks discretely at the primary itself, not as a return to form. That anticipatory reading is specific to cosponsorship, and it is consistent with the campaign account of

the participation onset: the participation dip appears among closely contested winners and losers alike because campaigning removes members from the floor votes they must attend in person (Appendix G), whereas winners’ flat cosponsoring path through the same campaign weeks shows that position-taking is not mechanically interrupted by campaigning—so a pre-primary decline in cosponsorship isolates the legislators with reason to stop investing in it, those who foresee defeat.

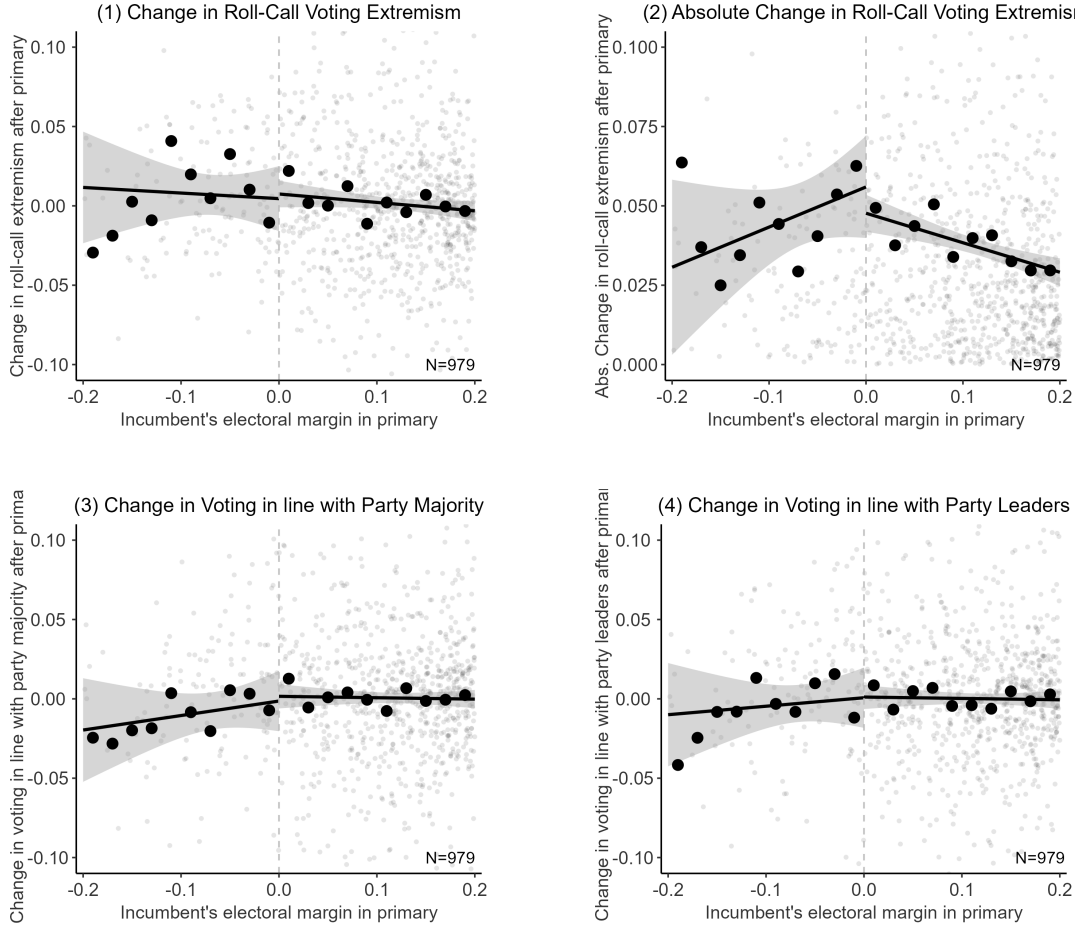
The durable contrast between the two bill-based activities is corroboration and anticipation, not raw magnitude: on the common Poisson translation the two declines are alike, roughly a third of a legislator’s own rate. Cosponsoring—cheap, discretionary position-taking—falls on both margins at the cutoff and in the panel, begins its decline as the loss becomes foreseeable, and stays down through the general election. Sponsorship runs through a longer pipeline of drafting and agenda-setting and is sparse enough that only the panel has the power to see its absolute decline; where a design has power, the response appears, with flat leads and a break at the primary itself. Electoral incentives, on this evidence, take hold earlier in the activity legislators can adjust at will.

5 Both Winners and Losers Show No Significant Shift in Policy Positions

Figure 5 visualizes the regression discontinuity results for legislators’ change in roll-call voting patterns: panels (1), (3), and (4) show changes in roll-call voting extremism, voting in line with the party majority, and voting in line with party leaders, respectively. In all three, there appears to be a negligible shift in the trend at the threshold, suggesting that primary election outcomes, and thus eligibility for the general election, do not significantly shift legislators’ post-primary policy positions.

Table 4 presents the formal estimates, and the results from both designs consistently indicate negligible changes in policy positions. Across all four columns, the regression-

Figure 5: Regression Discontinuity Results on Legislators' Policy Positions



Note: Legislators who narrowly win and lose do not show a statistically significant difference in their changes in roll-call voting after the primaries. In all measures, the changes for both winners and losers are statistically indistinguishable from zero, implying that neither group significantly alters their policy positions. Each panel plots the change in the score itself. The proportional margin used for the effort activities has no counterpart here: a position has no volume to normalize, and an extremity score is measured from a conventional origin rather than a true zero, so a share or a percentage change of it carries no meaning.

discontinuity coefficients are at most 0.01 in magnitude, and the difference-in-differences estimates are likewise small, suggesting minimal differences in how winners and losers adjust their positions. The table also reports the mean changes in policy positions for winners and losers separately, and in every case, these mean changes are close to zero. This pattern indicates not only that winners and losers change their behavior similarly after primaries, but that neither group meaningfully alters its policy positions between pre- and post-primary

periods.

These results suggest limited effects of electoral concerns on legislators’ ideological positions, contrasting with the conventional view of position-taking as central to the electoral connection and competition (Mayhew, 2004; Downs, 1965). The null instead aligns with empirical evidence of rigidity in candidate positions (Hall, 2019; Rogers, 2017; Poole and Rosenthal, 2014; Lee, Moretti and Butler, 2004) and formal models that assume fixed policy positions (Alesina, 1988; Osborne and Slivinski, 1996).

One might worry that a null on the average shift masks offsetting individual-level movement: legislators within the same party could move in opposing ideological directions after primaries, leaving the aggregate unchanged even as individuals move substantially. The natural test is to examine absolute change in roll-call extremism, which captures movement in either direction without cancellation. Graph (2) of Figure 5 and the second column of Table 4 show a slight increase in absolute repositioning among primary losers on this measure, significant under the difference-in-differences design but not the regression discontinuity, offering weak evidence that losers change their voting patterns somewhat more than winners after primaries. These patterns are consistent with the last-period (shirking) literature, in which legislators no longer subject to a reelection constraint vote closer to their own ideological preferences (Kalt and Zupan, 1990; Zupan, 1990; Bender and Lott, 1996). The statistical significance is not robust across the two designs, however, so this interpretation remains speculative. The magnitude speaks to the original worry—that sizable individual shifts might be cancelling out in the average. Taken at face value, losers’ absolute repositioning exceeds winners’ by 0.014, on a winner base of 0.032. That excess amounts to about one additional vote in seventy switching relative to the party pole; even at its upper confidence bound, about one in forty. Repositioning large enough to hide behind a flat average would have appeared here, and it does not.

The null on policy positions itself, by contrast, is highly robust: the same result holds whether the dependent variable is the within-term regression coefficient β_{it} from equation (2)

Table 4: Numerical Results on Change in Legislators' Policy Positions

Regression Discontinuity				
	Change in Extremism	Absolute Change in Extremism	Change in Voting with Party Majority	Change in Voting with Party Leaders
Primary Loss	−0.003 (0.009)	0.008 (0.006)	−0.003 (0.008)	0.000 (0.008)
Observations	918	918	918	911
Winner Mean	0.001	0.037	0.001	0.000
Loser Mean	0.007	0.048	−0.007	−0.003
Bandwidth	0.2	0.2	0.2	0.2
Kernel	Uniform	Uniform	Uniform	Uniform
Specification	Linear	Linear	Linear	Linear

Difference in Differences				
	Change in Extremism	Absolute Change in Extremism	Change in Voting with Party Majority	Change in Voting with Party Leaders
Primary Loss	0.006 (0.004)	0.014** (0.006)	0.002 (0.004)	0.006 (0.004)
Observations	3,296,336	3,694	3,290,805	2,892,750
Winner Mean	−0.000	0.032	−0.000	0.001
Loser Mean	0.007	0.047	−0.007	−0.003
Unit of Analysis	Legislator–vote	Legislator–Congress	Legislator–vote	Legislator–vote
Fixed Effects	Legislator, Bill-Party- Cong.	Legislator, Party-Congress	Legislator, Bill-Party- Cong.	Legislator, Bill-Party- Cong.

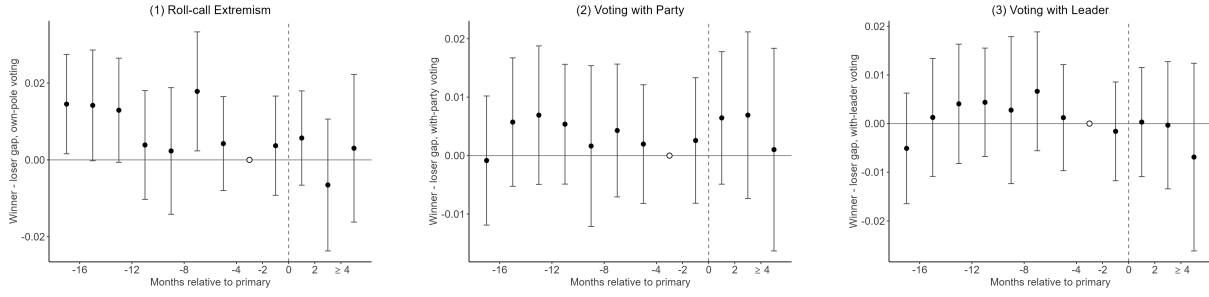
* $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$

Note: Coefficients are expressed as the effect of losing the primary. In the difference-in-differences panel, the “Primary Loss” row reports the $\text{PrimaryLoss} \times \text{AfterPrimary}$ interaction (equation (4); legislator and bill-by-party-by-Congress fixed effects) for the three vote-level columns and the Primary Loss coefficient (aggregated legislator-period specification with legislator and party-Congress fixed effects) for the absolute-change column, which has no vote-level analog; with legislator fixed effects, that coefficient is identified within legislators whose primary outcome varies across their terms, not by a cross-sectional comparison of winners with losers. Winner Mean and Loser Mean give the average within-term change for terms won versus lost at the legislator–Congress level, computed on the ± 0.2 estimation sample in the regression-discontinuity panel and on all contested primaries in the difference-in-differences panel (0 indicates no change). For the extremism columns the dependent variable is the own-party-pole indicator extreme_{iv} . Difference-in-differences standard errors are clustered by legislator; regression-discontinuity standard errors are conventional OLS standard errors.

(RDD columns of Table 4), the vote-level own-party-pole, with-party, or with-leader indicators with roughly 2.9–3.3 million bill-vote observations each (DiD columns of Table 4), or a simple pre/post mean difference computed over equal-length windows without any regression adjustment (Table A.11). Nor is the null an artifact of imprecision: the estimates are precise enough that a with-party or with-leader shift of two votes in a hundred would have been detected with conventional power, so the data affirmatively bound any repositioning below that mark (Table F.1). This is the standard applied to null results throughout the paper: a null carries evidential weight only where the design could have detected an effect of the size at issue. The position nulls meet that standard and are read as evidence of absence. Nulls from designs without that power—the raw-count level estimates at the cutoff, whose standard errors are an order of magnitude larger than the changes at issue (Table A.3)—are read as absence of evidence, and no conclusion in the paper rests on them. A final check for the RDD asks whether the null change could conceal a winner–loser gap that is already present before the primary and simply persists afterward, since a constant gap differences away to zero. I estimate separate RDDs on the pre-primary and the post-primary voting levels (subsection A.2): at the cutoff, the pre- and post-primary levels are small and at most marginally significant for both with-party and with-leader voting. There is no meaningful gap before the primary and none after, so the null reflects genuine stability rather than a constant winner–loser gap that the change would difference away. The same roll-call data, by contrast, yields a statistically precise and substantively large effect on the effort margin of participation, indicating the null on policy positions reflects actual stability in legislators’ within-term voting rather than insensitivity of the measures.

The stability also has no hidden time structure. Figure 6 applies the same event-time lens as Figure 4 to the three vote-based position measures: the winner–loser gaps are flat throughout, before and after the primary alike, where participation breaks discretely at the primary date. The joint tests confirm what the panels show—post-primary gaps are jointly null for all three measures. The far leads of the extremism measure are marginally non-flat

Figure 6: Event-Time Stability of Policy Positions around the Primary



Note: Each panel traces the winner–loser gap in a vote-based position measure by event time relative to each legislator’s own primary date, with 95% confidence intervals and the open circle marking the omitted reference bin; the specification is identical to the participation event study of Figure 4 (close primaries, two-month bins, post-primary window ending at each legislator’s general election), estimated on cast votes for legislators with at least ten cast votes on each side of the primary. Estimates are plotted at the centers of their two-month windows, the dashed line at the primary date. The final bin pools weeks 16 and beyond: later event times exist only for legislators with early primaries, and the weeks-32-and-beyond tail holds only eight legislators in this sample, two of them losers, too few to estimate separately. Joint tests: pre-primary leads are flat for with-party ($p = 0.90$) and with-leader voting ($p = 0.75$) and marginally non-flat for extremism ($p = 0.04$), driven by bins more than six months before the primary; post-primary gaps are jointly null at $p = 0.47$, 0.59 , and 0.88 for extremism, with-party, and with-leader voting. The term-level tests of Table 4 carry the precision of the stability claim; bin-level confidence intervals are wider by construction.

($p = 0.04$), echoing the small pre-primary own-pole gap already documented in the level decomposition (subsection A.2); the bins adjacent to the primary are null. The precision of the stability claim rests on the term-level estimates and equivalence bounds above; the event studies add that the term-level null is not averaging away a short-lived repositioning around the primary.

6 Heterogeneous Effects of Electoral Incentives

6.1 How Effort Effects Vary with General-Election Competitiveness

If legislators who lose their primaries withdraw effort, a natural reading is that the forgone post-primary effort was strategic preparation for the upcoming general election, where such

activity can enhance a candidate’s vote share. In some districts, however, the general election is expected to be minimally competitive, especially when a legislator or their party holds a dominant position. In such districts, legislators may worry less about reelection, and the reelection incentive that a primary loss removes may be correspondingly weaker (Burden, 2004). This suggests that the effort effect could depend on anticipated competition in the general election.

To test this, I divide the sample by whether the upcoming general election is expected to be competitive. Because incumbents in uncompetitive districts typically win by large margins, I use the incumbent’s vote share in the most recent general election: districts where the incumbent received more than 75% form the uncompetitive group, and those below form the competitive group.

Table 5 shows that the withdrawal is general rather than confined to competitive environments. Roll-call participation falls in both district types under both designs. Constituent communication—the most campaign-facing of the four activities—falls in safe districts as well: losers there send 0.22 fewer e-newsletters per week on the weekly-panel specification of Table 3, a decline that is stable at every competitiveness cutoff between 0.60 and 0.75 and survives a wild cluster bootstrap over the ten losing terms that identify it ($p = 0.005$). Behind the coefficient stands a simpler fact: the eight of those losers who used the medium at all sent 227 newsletters before their primaries and 4 after. Cosponsorship shows nearly identical proportional-margin estimates in the two district types (-0.114 and -0.115), though the smaller safe-district sample renders the latter marginal ($p = 0.06$), and its uncompetitive additive cell is too imprecise to be informative. Sponsorship is detected only in competitive districts; its uncompetitive cells cannot distinguish an equal effect from none at all, with a minimum detectable effect on the proportional margin more than twice the full-sample estimate.

Formal tests agree with what the table shows: across sixteen interaction and cutoff tests—every outcome, both margins, both designs—none detects a competitiveness gradient

Table 5: Effort Effects by General-Election Competitiveness: Two Designs, Two Margins

Competitive Districts ($V_{t-1} < 0.75$)				
	Roll-call Participation (share of votes cast)	Bill Sponsorship (count per session-week)	Bill Cosponsorship (count per session-week)	E-Newsletters (count per week)
<i>Additive margin: change in rate</i>				
Primary Loss (RD)	-0.088*** (0.023)	-0.051 (0.076)	-1.632*** (0.534)	-0.914*** (0.233)
Observations	660	487	424	179
Primary Loss × After (DiD)	-0.110*** (0.015)	-0.063*** (0.021)	-1.072*** (0.162)	-0.803*** (0.258)
Observations	2,567,939	171,634	149,164	69,497
<i>Proportional margin: share of activity / change relative to own rate</i>				
Primary Loss (RD, proportion)	-0.040 (0.032)	-0.149* (0.077)	-0.114*** (0.035)	-0.219*** (0.082)
Observations	682	417	421	155
Primary Loss × After (Poisson)	-0.131*** (0.020)	-0.608*** (0.181)	-0.415*** (0.053)	-1.170*** (0.244)
Observations	2,567,939	161,502	150,720	59,748
Uncompetitive Districts ($V_{t-1} > 0.75$)				
	Participation	Sponsorship	Cosponsorship	E-Newsletters
<i>Additive margin: change in rate</i>				
Primary Loss (RD)	-0.199*** (0.039)	0.042 (0.205)	-0.791 (1.519)	0.230 (0.476)
Observations	266	178	154	66
Primary Loss × After (DiD)	-0.085** (0.035)	0.007 (0.071)	-0.754 (0.499)	-0.224*** (0.070)
Observations	952,477	64,791	57,132	26,566
<i>Proportional margin: share of activity / change relative to own rate</i>				
Primary Loss (RD, proportion)	-0.134** (0.060)	-0.206 (0.161)	-0.115* (0.060)	0.327* (0.190)
Observations	278	147	154	55
Primary Loss × After (Poisson)	-0.101** (0.044)	0.136 (0.385)	-0.291* (0.160)	-2.240*** (0.580)
Observations	949,426	55,913	57,400	21,861

* $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$

Note: Coefficients are the effect of losing the primary; standard errors are conventional in the discontinuity rows, clustered by legislator in the panel rows. Panels split districts at 75% of the incumbent's most recent general-election vote share. Margins follow Table 3: additive rows are changes in weekly rates under that table's measurement convention (session-week rates and session-week panels for sponsorship and cosponsorship), with difference-in-differences rows estimated on Table 3's weekly-panel specification; proportional rows are post-primary shares of activity at the discontinuity and fixed-effects Poisson estimates in the panel; the participation proportional cells translate the additive response through the base rate, as in Table 3. Count-panel subgroup samples omit terms without a recorded previous vote share and one 116th-Congress party switcher whose mid-term party coding is undefined; full-sample estimates are unchanged at printed precision. Sixteen formal gradient tests—continuous and cutoff interactions across all outcomes, margins, and designs—are null at the five-percent level. The uncompetitive sponsorship and cosponsorship panel nulls are uninformative rather than contrary: their minimum detectable effects (1.08 proportional, 1.40 additive) exceed the corresponding full-sample estimates. The uncompetitive e-newsletter panel estimate rests on ten losing terms, eight with any newsletters (227 before the primary, 4 after); it is stable at every cutoff between 0.60 and 0.75 and survives a wild cluster bootstrap over those terms ($p = 0.005$; randomization inference $p = 0.08$ at this cut, ≤ 0.03 at every broader one). The uncompetitive e-newsletter proportional cells rest on the same eight legislators and disagree in sign across implementations, describing those legislators rather than supporting an inference.

at the five-percent level.⁵ The point estimates leave room for the communication response to be larger where campaigns loom larger—the e-newsletter decline is roughly three and a half times the size in competitive districts (0.80 versus 0.22 newsletters per week), and sponsorship registers only there—but the data do not order these magnitudes statistically, and for participation the point pattern runs, if anything, the other way. What the design does establish is presence: wherever it has the power to look, in the safest districts and the most contested alike, losing the primary is followed by withdrawal.

Notably, the regression-discontinuity participation decline is larger in uncompetitive districts (-0.199 , about twenty percentage points of participation) than in competitive ones (-0.088). This pattern is specific to the regression discontinuity, however: in the more highly powered difference-in-differences design the two are statistically indistinguishable (-0.110 versus -0.085), and the uncompetitive regression-discontinuity estimate rests on a small close-primary sample of 266 legislators. I therefore read it cautiously. If it reflects a real difference, a plausible account is that a substantial share of legislators’ motivation to attend roll-calls is non-electoral, such as personal ideological commitments, consistent with the earlier finding that losers still take part in most roll-calls, albeit less frequently than winners; on this account, primary winners in competitive districts attend somewhat fewer roll-calls because they must devote more time to campaigning and visiting home districts ahead of the general election. But because the two designs disagree, and a formal interaction test detects no gradient ($p = 0.31$), the safer conclusion is that the participation effect is broadly comparable across district types.

The pattern speaks to what the incentive is. If effort tracked the marginal electoral return of the upcoming general election, safe-district losers—whose defeat changes little about a foregone conclusion—should barely respond; instead they withdraw as sharply as anyone. What loss removes in both district types alike is candidacy itself: the extensive

⁵One of the sixteen reaches the ten-percent level (the e-newsletter volume interaction, $p = 0.059$, pointing toward a larger response in competitive districts)—within what chance alone would produce across sixteen tests. The tests also have limited power against moderate gradients: they establish that the withdrawal is present in both district types, not that its magnitude is everywhere equal.

margin of having a next election at all, not the intensive margin of how close it will be. Nor does the withdrawal track the length of the career ahead: age and seniority splits are uniform—losers below the median age of 55, for whom careers beyond the House loom largest, withdraw no less than their older colleagues (Appendix A.17)—which tells against accounts in which effort protects a remaining career rather than a candidacy. One alternative the heterogeneity cannot reach remains: because a primary defeat ends the House career along with the campaign, the design cannot separate the electoral incentive from the prospect of another term—for House members the two coincide. What separates the withdrawal from a generic end-of-career wind-down is instead its timing, established earlier: effort holds flat through the final year and breaks discretely at each legislator’s own primary date.

6.2 Effects on Policy Positions Remain Null for Important Bills

The null on policy positions might stem from a narrower channel: legislators may respond to electoral incentives only on a small subset of salient or important bills. To test this, I restrict the analysis to bills deemed important.

I take three approaches to identifying “important” bills. First, I examine bills “whipped” by party leadership to test whether legislators respond to party pressure. Party leaders occasionally conduct whip counts — informal surveys on significant agenda items — typically reserved for high-priority bills. If electoral incentives operate through party pressure, the effects should be more pronounced on whipped bills. This analysis leverages whip count data from the Democratic Party (1955–1986) and the Republican Party (1975–1980) provided by Evans (2012).

Second, I focus on bills previous scholars deemed especially significant or salient. I pool two standard measures of legislative importance, each linked to specific bill numbers: priority agenda items from Curry and Lee (2020) and landmark enactments compiled from CQ Press.⁶

⁶The priority-agenda series covers only Congresses 103–115 and supplies too few within-window votes per legislator to support the regression-discontinuity change-score once the minimum-vote requirement below is imposed; the landmark series spans Congresses 84–113 and supplies the needed coverage. The two measures

I apply the same measure of position change to these bills. If legislators are more responsive to high-profile legislation, the effects should be more pronounced on them than across all bills.

Third, I focus on contested bills — those close to passing or failing. Such bills are especially informative because each legislator is more likely to cast a pivotal vote. If electoral incentives heighten legislators’ attention to legislative outcomes, the effects should be more pronounced on these narrowly decided bills. To operationalize this, I calculate the difference between yea and nay votes for each bill and repeat the analysis for bills in the lowest 30% of that difference.⁷

are largely redundant, so the results are nearly identical using the landmark series alone, while the priority-agenda series alone is underpowered.

⁷30% is arbitrary; results are unchanged at the 20% or 50% thresholds. This measure of “contestedness” is posterior, capturing realized vote outcomes after electoral incentives may have influenced legislators’ votes, but realized contestedness likely approximates expected contestedness.

Table 6: Effects of Primary Loss on Voting on Important Bills

	Regression Discontinuity			Difference-in-Differences		
	Whipped Bills	Significant/ Landmark Bills	Contested Bills	Whipped Bills	Significant/ Landmark Bills	Contested Bills
Change in Roll-call Extremism	0.051 (0.075)	0.013 (0.022)	0.000 (0.016)	0.072 (0.048)	0.015** (0.008)	0.010 (0.006)
Change in Voting with Party Majority	0.046 (0.081)	0.012 (0.021)	−0.010 (0.016)	0.029 (0.047)	0.014* (0.008)	0.010 (0.006)
Change in Voting with Party Leaders	0.137 (0.086)	0.000 (0.023)	0.004 (0.018)	0.078 (0.049)	0.018* (0.009)	0.015** (0.007)
Change in Roll-call Participation	−0.118** (0.052)	−0.107*** (0.027)	−0.112*** (0.020)	−0.161*** (0.038)	−0.086*** (0.018)	−0.099*** (0.013)
Observations	124–136	656–684	899–909	28k–36k	496k–623k	942k– 1,101k
Specification	Bandwidth = 0.2 Specification = Linear			Legislator + Bill-Party-Congress FE		

* $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$

Note: Each cell reports the effect of losing the primary on the change in the respective voting behavior; standard errors are in parentheses and clustered by legislator in the difference-in-differences design. The difference-in-differences and regression-discontinuity specifications are the same as in Table 3, and the roll-call extremism outcome is the own-party-pole indicator extreme_{iv} defined in subsection 3.1; the with-party and with-leader outcomes are the corresponding vote-level indicators, so coefficients are changes in the share of votes. The three bill subsamples are whipped bills (Evans 2012; Congresses 84–99), significant/landmark bills (priority agenda items from Curry & Lee 2020 pooled with landmark enactments compiled from CQ Press; Congresses 84–115), and contested bills (bottom 30% of $|\text{yea} - \text{nay}|$; Congresses 84–117). To estimate a legislator’s within-subsample position or participation change, the regression-discontinuity columns require at least five subsample votes on each side of the primary (cast votes for the position outcomes, available roll calls for participation; Appendix A.6); the difference-in-differences columns pool all votes and impose no such requirement. Regression-discontinuity sample sizes therefore differ across rows (positions versus participation) and are reported as ranges; vote-level sizes likewise differ within each column due to outcome-specific missingness. Because the difference-in-differences samples for the significant/landmark and contested columns exceed half a million votes, coefficients there can be statistically distinguishable from zero while remaining substantively negligible in magnitude ($|\hat{\beta}| < 0.02$).

Each cell in Table 6 reports the effect of losing a primary on subsequent voting for whipped, significant/landmark, and contested bills. Across both designs, coefficients for roll-call extremism and voting alignment (with party majority or party leaders) are small

and substantively indistinguishable from zero. A few of the difference-in-differences position estimates reach conventional significance in the significant/landmark and contested columns, but only because those pooled vote-level samples exceed half a million observations; the coefficients themselves are negligible, all below 0.02 in absolute value. This pattern reinforces the earlier evidence on the rigidity of legislators’ policy positions: primary winners and losers do not systematically differ in how they adjust positions on major votes after primary elections. The bottom row provides a power check. On all three subsamples—each restricted to legislators with at least five available roll calls per side, so the change-scores are well estimated—roll-call participation falls sharply and significantly for primary losers, so the flat policy positions reflect genuine stability rather than a lack of statistical power. Nor does the null stem from losers disengaging—their participation falls only about nine percentage points after the primary while winners’ is essentially unchanged (Table 3, note), leaving them active on a substantial share of votes.

7 Conclusion

Electoral accountability lies at the heart of democratic governance, compelling representatives to remain responsive to constituents. Alexander Hamilton underscored this during the debates over term limits, arguing that removing the prospect of reelection would diminish legislators’ motivation: “One ill effect of the exclusion [of reelection] would be a diminution of the inducements to good behavior” (Hamilton, 2003).

Despite the recognized importance of electoral incentives, identifying precisely which “good behavior” they induce remains empirically challenging. Representatives strategically choose whether to seek reelection or retire, making it difficult to isolate the influence of electoral incentives from selection bias. This study addresses this challenge by leveraging the post-primary period, when incumbents face a critical juncture: continuing their reelection campaigns or exiting after a primary defeat. Legislators who lose their primaries withdraw

legislative effort: roll-call participation falls sharply, in a decline timed to the primary itself, and bill sponsorship, cosponsorship, and constituent communication fall in step, each by roughly a third or more of a legislator’s own rate. Yet legislators’ policy positions in roll-call voting do not significantly change, whether they won the primary or not.

Taken together, the results show that electoral incentives discipline legislative effort but not legislative position-taking. The effort findings reinforce classic theories of electoral accountability (Barro, 1973; Ferejohn, 1986; Mayhew, 2004): legislators sustain floor participation, bill sponsorship and cosponsorship, and constituent communication so long as they remain candidates for another term—and they do so in safe districts and competitive ones alike, whatever the length of the career ahead. Contrary to the worry that voters’ inattention blunts accountability (Achen and Bartels, 2017), what disciplines representatives is not the intensity of the contest but the prospect of facing voters at all.

Their roll-call positions, by contrast, barely change. Although position-taking is classically treated as central to the electoral connection (Mayhew, 2004) and electoral competition (Downs, 1965), positions hold steady after a primary defeat, even when electoral pressure should produce the sharpest response: whipped bills, significant agenda items, and narrowly contested ones. The estimates are precise enough to detect a party-line shift as small as two votes in a hundred, so the stability is affirmatively bounded, not merely undetected. I interpret this divergence as reflecting a difference in kind between the two margins. Effort is an individually controlled, observable signal that can pay off in the next campaign; roll-call positions are anchored by more durable commitments—members’ own ideological convictions and the established policy records they were elected on (Poole and Rosenthal, 2014; Hall, 2019; Lee et al., 2004; Alesina, 1988; Osborne and Slivinski, 1996)—that the lifting of reelection pressure does not dislodge.

The same stability informs how scholars read roll-call data. Parties control resources legislators need to win (Cox and McCubbins, 2007; Krehbiel, 1993), yet party-line voting is unchanged when a member’s reelection incentive vanishes, so within-term electoral pressure

is not the primary channel through which parties pull members into line. For the same reason, because ideological scores such as DW-NOMINATE (Poole and Rosenthal, 1985; McCarty, 2016) are stable across the primary cutoff even on the strongest-prediction bills, they are unlikely to be mere artifacts of immediate electoral pressure operating through party discipline or donors (Snyder and Groseclose, 2000; Bronars and Lott, 1997).

The findings also bear on the roots of polarization. Because sitting incumbents do not shift their positions in their term’s closing months, regardless of primary outcome, widening partisan divides are more plausibly driven by who enters and leaves office—replacement, sorting, and nomination—than by repositioning among those already in office (Moskowitz et al., 2024).

This null is subject to one qualification: it describes the months after a primary, not elections in general. The design isolates the within-term effect of removing future reelection incentives; it does not preclude electoral incentives from shaping policy at a slower pace over a longer time span. Albouy (2011) finds that U.S. Senators, who serve six-year terms, moderate their voting more in the years closer to reelection, and Caughey and Warshaw (2018) document multi-year responsiveness of state policy to public opinion—evidence that electoral pressure may accumulate over a longer horizon. A null effect over a relatively short period in this design does not necessarily contradict accounts in which electoral pressure builds slowly and positions, once taken, are costly to abandon, so that its abrupt removal over a few months leaves recorded votes unchanged. Nor can the design detect positioning that adjusted *before* the primary—a member casting tough votes in anticipation of not running again—or change deterred by the very threat of a challenge; the null accordingly bounds the post-primary, within-term channel rather than every path by which electoral competition could discipline positions.

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Sign convention. Main-text tables and the appendix tables labeled “Primary Loss” report effects of *losing* the primary. Diagnostic and robustness tables in this appendix whose rows are labeled by outcome retain the winner-relative convention in which they were estimated, unless their notes state otherwise; those winner-relative coefficients are the negatives of the corresponding primary-loss effects.

Window conventions. Headline estimates in the main text measure all outcomes from the start of the Congress through each legislator’s general election date (Section 2 of the main text). Appendix diagnostics that predate this convention and are marked neither as re-based nor as count-based use the full-Congress vote window, including the post-general lame-duck session; for the participation headline the two conventions differ by about one percentage point (main text, Section 2), and the labeled full-term extension is reported in Appendix G.

A Appendix

A.1 Balance at the Threshold

This section collects the at-threshold balance tests underlying the design’s central comparison, all at the ± 0.2 bandwidth. Table A.1 reports local-linear RDDs of pre-determined legislator and district characteristics on the primary margin: age, party, ideology in level and extremity, redistricting exposure, district presidential lean, the previous general-election vote share, the previous primary winning margin (with an uncontested primary coded as a margin of one), freshman status, seniority, and the length of the equidistant measurement window itself. Nothing jumps at the cutoff; every p -value exceeds 0.3. In particular, the measurement window over which the count outcomes are constructed is itself balanced, so the pre-primary activity imbalance documented in Table A.3 is not an artifact of losers being measured over systematically longer or shorter windows.

Table A.2 then asks whether any pre-primary activity difference at the cutoff is a stable trait of the legislators involved or a feature of the contest season, by estimating the pre-primary level RDDs over progressively longer measurement windows: the equidistant pre-window in raw counts and per-week rates, the full pre-primary period from the start of the Congress, and the previous Congress over both its start-to-primary window and its full two years, in weekly rates, raw two-year counts, and logs; roll-call participation joins the table as the share of votes cast over the same windows. No contemporaneous cell detects a significant difference. In the equidistant pre-window the cosponsorship and e-newsletter point estimates are positive—close losers more active—but imprecise (+9.4 cosponsored items, $p = 0.28$; +9.1 e-newsletters, $p = 0.18$; the rate cells agree), the full pre-period cells are null for every activity, and the same-sample rows show that on exactly the subsample with previous-Congress records the contemporaneous cells remain insignificant. One Congress earlier, every cell that reaches or approaches significance runs in the *opposite* direction—eventual losers the less active group (sponsorship, -0.17 per session-week before that Congress’s own primary, $p = 0.04$, and -0.14 over its full two years, $p = 0.07$; e-newsletters, -0.40 per week, marginal, with its raw and log cells negative but insignificant; participation, about two percentage points of votes cast, marginal on both windows)—the direction that works against, not for, a manufactured post-primary decline; no cosponsorship cell approaches significance on any window or unit in either Congress.

The within-legislator change designs absorb any stable level component of behavior by construction, and the previous-Congress change placebos (Table A.21) test for systematic differences outside the loss term; movement within the loss term itself is what the event-study leads and the trajectory decomposition (Appendix C) examine, and the pre-level adjustments (subsection A.3), taken over both the full pre-period and the outcome’s own pre window, close the residual of either channel.

Table A.1: Balance at the Threshold: Pre-Determined Characteristics

	Estimate	SE	p	N
Age	1.175	(1.681)	0.485	979
Democrat (0/1)	−0.000	(0.074)	0.996	979
DW-NOMINATE (dim. 1)	0.023	(0.061)	0.705	979
Ideological extremity	−0.025	(0.029)	0.387	979
Redistricted (0/1)	0.052	(0.071)	0.465	979
District presidential Dem. share	−0.014	(0.024)	0.567	979
Previous general vote share	−0.004	(0.024)	0.868	979
Previous primary winning margin	−0.070	(0.070)	0.314	788
Freshman (0/1)	−0.034	(0.057)	0.551	943
Seniority (Congresses served)	0.146	(0.602)	0.808	943
Window length (days)	−4.250	(9.089)	0.640	979

* $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$

Note: Each row reports a local-linear RDD of the pre-determined characteristic on the primary winning margin at bandwidth ± 0.2 (separate slopes), expressed as the effect of losing the primary; conventional standard errors. Freshman and seniority are computed from first appearance in the 84th–117th member roster; Congress 84 is dropped (left-censoring). The previous primary winning margin is measured among legislators serving in the previous Congress, with an uncontested primary coded as a winning margin of one. Window length is each legislator’s primary-to-general gap in days, the length of the equidistant measurement window.

Table A.2: Balance at the Threshold: Pre-Primary Activity Levels by Window and Unit

	Sponsorship	Cosponsorship	E-Newsletters	Roll-call Participation
Raw count, equidistant pre-window	0.23 (0.80) [665]	9.42 (8.80) [578]	9.08 (6.80) [245]	—
Per-week rate, equidistant pre-window	0.075 (0.074) [665]	0.350 (0.659) [578]	0.348 (0.334) [245]	−0.011 (0.025) [926]
Weekly rate, full pre-period	−0.013 (0.067) [665]	0.222 (0.681) [578]	0.300 (0.253) [245]	−0.018 (0.013) [926]
Weekly rate, previous Congress (start–primary)	−0.171** (0.085) [515]	0.174 (0.824) [449]	−0.400* (0.237) [168]	−0.022* (0.013) [750]
Weekly rate, previous Congress (two-year)	−0.138* (0.075) [515]	0.088 (0.751) [449]	−0.359 (0.235) [168]	−0.023* (0.013) [750]
Previous Congress, raw count (two-year)	−7.24 (5.08) [512]	32.64 (43.17) [449]	−37.47 (24.47) [168]	—
Previous Congress, log(1 + count) (two-year)	−0.172 (0.189) [512]	0.119 (0.154) [449]	−0.764 (0.567) [168]	—
Same-sample control: raw count (t , equidistant)	0.02 (1.00) [512]	13.18 (10.74) [449]	10.98 (10.19) [151]	—
Same-sample control: log(1 + count) (t)	0.188 (0.197) [512]	0.204 (0.196) [449]	−0.302 (0.491) [151]	—

* $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$

Note: Each cell is the same local-linear RDD applied to the pre-primary activity level under the stated window and unit. Raw and per-week cells use the equidistant window ending at the primary; full-pre rates average from the start of the Congress to the primary; previous-Congress cells re-measure the same legislators one Congress earlier, over that Congress’s own start-to-primary window (start–primary rows) or its full two years (two-year rows). Sponsorship and cosponsorship rate cells are per session-week, following the measurement convention of Table 3; raw-count and log cells are unaffected by that convention, and the log cells are differences in $\log(1 + \text{count})$, approximately proportional differences in the two-year counts. Roll-call Participation cells report the share of votes cast over the row’s window, among legislators with at least ten available votes on each side of the primary; the count-specific rows are not defined for it. The same-sample rows repeat the contemporaneous test on exactly the subsample with previous-Congress records. DCinbox e-newsletter coverage begins in 2010, so the previous-Congress e-newsletter cells draw on 2010–2020 (Congresses 111–116). E-newsletter cells count legislators absent from DCinbox as sending zero, the convention of the e-newsletter cells throughout the article; conditioning instead on use would select on the outcome—close losers are about 22 percentage points less likely than close winners to have sent any e-newsletter in the pre-primary window ($p = 0.056$), and among users alone the equidistant gap is 14.9 e-newsletters ($p = 0.048$). Standard errors in parentheses; sample sizes in brackets.

A.2 Placebo Decomposition of the Change Coefficient

As an additional check on the RDD identification, I conduct a placebo decomposition: separately estimating local-linear RDDs on the pre-primary and post-primary outcome *levels* at the same ± 0.2 bandwidth (Table A.3). The pre-primary RDD is a placebo: because the primary outcome is realized only after the primary date, narrow winners and losers should not differ in pre-primary behavior at the cutoff. The placebo passes for every outcome. For roll-call participation the decomposition is sharpest: the pre-primary level coefficient is -0.02 ($p = 0.18$), while the post-primary level shows a large, significant drop (-0.13^{***}), so the headline change coefficient (-0.119^{***}) reflects a clean post-primary divergence. The count outcomes are decomposed in the share’s own ingredients, the raw counts over the equidistant windows. Cosponsorship shows no pre-primary gap (9.4 items, $p = 0.28$) and a negative but individually insignificant post-primary deficit (-9.6 items, $p = 0.24$); e-newsletters carry a positive but insignificant pre-primary and a negative but insignificant post-primary coefficient (9.1, $p = 0.18$; -7.1 , $p = 0.13$); sponsorship’s counts are null on both sides of the primary, as its base rate implies. The level RDDs are less precise than the change estimates—a cross-legislator comparison of raw counts carries the full spread of baseline volumes—so their pre-primary nulls discipline the design’s starting line rather than substitute for the change estimates; the previous-Congress rates (subsection A.21) test for systematic differences outside the loss term, and the trajectory decomposition (Appendix C) for movement within it.

The decomposition sits alongside the other identification checks, which point the same way: the McCrary density test detects no manipulation of the running variable (Figure A.1); a placebo-window test confirms that the participation effect is a sharp outlier relative to estimates from randomly placed no-election windows ($p < 0.002$; subsection A.20); and the non-challenged three-group natural placebo confirms that legislators whose re-election incentive is *not* removed behave like primary winners, not losers, across twelve effort and ideology outcomes (Table A.14). As a final check on residual level dependence, each headline change coefficient is re-estimated with the pre-primary level entered as a covariate: participation strengthens slightly ($-0.129 \rightarrow -0.135^{***}$) and the position nulls hold (Table A.4),

Table A.3: Decomposing the Change Coefficient: Pre-Primary and Post-Primary Levels

	Pre-Primary Level	Post-Primary Level	Change (post share)	N
Roll-call Participation	−0.018 (0.013)	−0.135*** (0.023)	−0.119*** (0.020)	926
Sponsorship	0.23 (0.80)	−0.05 (0.66)	−0.165** (0.068)	665
Cosponsorship	9.42 (8.80)	−9.57 (8.13)	−0.118*** (0.030)	578
E-Newsletters	9.08 (6.80)	−7.12 (4.65)	−0.170** (0.072)	245

* $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$

Note: Each cell reports the local-linear OLS RDD coefficient on Primary Loss at bandwidth ± 0.2 , with conventional standard errors in parentheses. The Pre-Primary Level and Post-Primary Level columns are each change measure’s own ingredients: for the three count outcomes, the raw activity counts over the equidistant windows that enter the post-primary share; for Participation, the raw participation rate entering the rate change. For the three count outcomes the Change column uses the post-primary share, $\text{post}/(\text{pre} + \text{post})$, the proportion measure of Table D.2; because the share is undefined for legislators with no activity in either window, its samples are 564, 575, and 210, while the level columns retain all legislators (N column). For participation it is the within-term change in the participation rate, the post-primary minus the pre-primary level.

while the three count proportions move by at most a few thousandths ($-0.165 \rightarrow -0.158$, $-0.118 \rightarrow -0.117$, $-0.170 \rightarrow -0.168$; Table D.2). Sponsorship’s additive weekly-rate margin shows no response in any design; the two margins are compared directly in Appendix D.

Applying the same decomposition to the three vote-level position outcomes returns null at both levels: for signed extremism, pre -0.014 ($p = 0.54$) and post -0.029 ($p = 0.24$); for with-party voting, pre -0.016 ($p = 0.33$) and post -0.024 ($p = 0.17$); for with-leader voting, pre -0.020 ($p = 0.22$) and post -0.032 ($p = 0.08$), all estimated at bandwidth ± 0.2 on the same legislator–Congress samples ($N = 918, 918$, and 911 respectively). The absolute-extremity change is an aggregated legislator–Congress measure with no vote-level pre/post-primary analog, so it is not decomposed here. Because both pre- and post-primary levels are small and at most marginally significant at the cutoff, the null on the headline change coefficient (RDD columns of Table 4) cannot be masking a constant winner–loser gap

that is present both before and after the primary and that differences away to zero in the change.

A.3 Robustness of Change Coefficients to Pre-Primary Level

Because the count outcomes’ headline change measures are within-term contrasts, a change coefficient could in principle absorb part of any difference in pre-primary levels. I verify directly that none does: I re-estimate the change-outcome RDD adding the pre-primary level as an additional covariate, restricting both the baseline and the augmented specification to the same observations:

$$\Delta y_{ic} = \beta_0 + \beta_1 \text{PrimaryLoss}_{ic} + \gamma_1 rv_{ic} + \gamma_2 \text{PrimaryLoss}_{ic} \times rv_{ic} + \delta y_{ic}^{\text{pre}} + \varepsilon_{ic}. \quad (1)$$

Here y_{ic}^{pre} is the pre-primary level entering the outcome’s own change measure—the equidistant pre-window count for the count proportions, the pre-primary rate for the vote-based outcomes. If the change coefficient β_1 is largely unchanged when y_{ic}^{pre} is partialled out, the headline estimate is robust to pre-primary levels. If β_1 shrinks substantially, the headline change is partly mechanical from the absolute-level difference. Table A.4 reports the vote-based outcomes; the count outcomes’ adjusted estimates appear with their margin battery in Table D.2 and are summarized below.

The participation coefficient *strengthens* from 0.129 to 0.135 (both $p < 0.001$) when the pre-primary participation rate is partialled out, and the position outcomes (signed extremism, with-party, and with-leader) remain null in both specifications. For the three count outcomes the same adjustment, applied to the proportion measure on the equidistant windows, moves the estimates by at most a few thousandths—sponsorship -0.165 to -0.158 , cosponsorship -0.118 to -0.117 , e-newsletters -0.170 to -0.168 , all retaining significance (Table D.2). Controlling instead for the level over the entire pre-primary period, as a raw count or a weekly rate, moves them no more (-0.164 , -0.115 , and -0.170 under the count control). No headline change coefficient rests on pre-primary levels; cosponsorship’s timing qualification is assessed in Appendix C.

Table A.4: Change-Coefficient Robustness to Pre-Primary Level (Bandwidth ± 0.2)

Outcome	Coefficient on <i>Primary Loss</i>		Pre-Level	N
	Baseline	+ Pre Control	Coefficient	
Roll-call Participation	−0.129*** (0.019)	−0.135*** (0.018)	−0.298*** (0.045)	926
Roll-call Extremism	−0.003 (0.009)	−0.004 (0.009)	−0.070*** (0.013)	918
With Party	−0.003 (0.008)	−0.004 (0.008)	−0.074*** (0.016)	918
With Leader	0.000 (0.008)	−0.002 (0.008)	−0.057*** (0.016)	911

* $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$

Note: Each row reports the coefficient on *Primary Loss* from a local-linear OLS RDD at bandwidth ± 0.2 , with the change outcome as dependent variable. The Baseline column omits the pre-primary level term ($\delta = 0$ in equation (1)); the + Pre Control column includes it. Both columns are estimated on the same per-outcome restricted sample, so any difference reflects the pre-primary level partialling rather than sample composition. The participation row's change outcome is the within-term change in the participation rate; the position rows use the within-term change scores β_{it} of equation (2) in the main text, on each outcome's own vote-share scale. Conventional standard errors in parentheses; no clustering. Replication materials are provided with the paper.

A.4 Post-congress Career Trajectory of Primary Losers

Table 1 in the main article presents the post-Congress career trajectories of primary election losers. It is notable that most legislators do not pursue elected office again after losing a primary election although these are ex-post realized decisions and do not directly reveal candidates' true incentives during their lame-duck periods. To probe this as far as the data allow, I conduct a separate analysis excluding all primary losers who later sought an elected office. The results remain consistent, as shown in Table A.5.

Table A.5: Results without Re-running Primary Losers

Effort Behavior				
	Regression Discontinuity			
	Participation	Sponsored Bills	Cosponsored Bills	E-Newsletters
Primary Loss	−0.132*** (0.020)	−0.033 (0.082)	−1.349** (0.595)	−0.713*** (0.219)
Observations	897	642	555	235
Bandwidth	0.2	0.2	0.2	0.2
Specification	Linear	Linear	Linear	Linear
	Difference in Differences			
	Participation	Sponsored Bills	Cosponsored Bills	E-Newsletters
Primary Loss	−0.106*** (0.021)	−0.114** (0.047)	−1.978*** (0.424)	−0.988** (0.431)
Observations	3,685	2,845	2,483	995
Specification	Aggregated	Aggregated	Aggregated	Aggregated
Fixed Effects	Legislator, Party-Cong.	Legislator, Party-Cong.	Legislator, Party-Cong.	Legislator, Party-Cong.
Policy Position				
	Regression Discontinuity			
	Extremism	With Party	With Leader	Abs. Extremism
Primary Loss	−0.003 (0.010)	−0.004 (0.009)	−0.002 (0.009)	0.010 (0.007)
Observations	889	889	882	889
Bandwidth	0.2	0.2	0.2	0.2
Specification	Linear	Linear	Linear	Linear
	Difference in Differences			
	Extremism	With Party	With Leader	Abs. Extremism
Primary Loss	−0.010 (0.009)	−0.009 (0.009)	−0.003 (0.009)	0.014* (0.007)
Observations	3,665	3,665	3,605	3,665
Specification	Aggregated	Aggregated	Aggregated	Aggregated
Fixed Effects	Legislator, Party-Cong.	Legislator, Party-Cong.	Legislator, Party-Cong.	Legislator, Party-Cong.

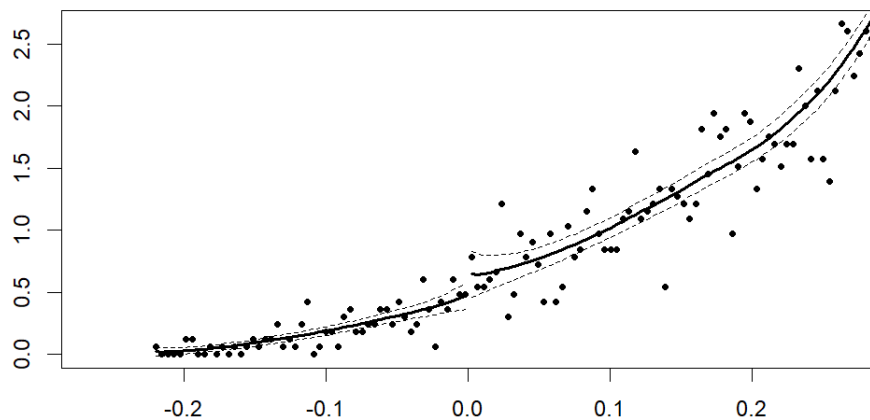
* $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$

Note: These results are based on the sample without any House representatives who lost in primary elections but ran for an election later. Effort outcomes and units follow Table 3 (participation as the share of votes cast; session-week rates for sponsorship and cosponsorship, e-newsletters per calendar week); the position outcomes are the within-term change scores β_{it} of equation (2), on each outcome's own vote-share scale. The results are mostly the same as the main findings. The sponsorship difference-in-differences cell reaches nominal significance in this subsample; the cutoff discontinuity and the level-based panel estimates for sponsorship remain null, consistent with the transient dip around the primary described in Appendix C.

A.5 Manipulation Test

The McCrary density test is used in regression discontinuity designs to check for potential manipulation of the running variable around the threshold (McCrary, 2008). The idea is that if individuals can manipulate the running variable, there might be a discontinuity in its density function at the cutoff, indicating possible sorting or manipulation. The test produces Figure A.1; the estimated log difference in the height of the running-variable density at the cutoff is 0.30 (SE = 0.23), statistically indistinguishable from zero ($p = 0.18$). This suggests that manipulation of the running variable is unlikely.

Figure A.1: Plot from McCrary Test



A.6 Excluded Legislators

The roll-call sample begins with 14,582 Congress-legislators whose primary election can be matched to a roll-call voting record. I drop 269 who participated in fewer than half of their seat’s roll-call votes in their respective Congresses. Because the research designs require voting behavior on both sides of the primary, I further require at least ten roll calls per side—ten available roll calls for the participation analyses and ten cast votes for the position analyses, mirroring how each outcome is measured. Almost all member-terms failing the availability requirement fall short on the post-primary side (549 of 558; 414 faced no post-primary roll calls at all), for two distinct reasons. Over half are concentrated in the 1956, 1958, and 1960 cycles, when Congress adjourned by late summer, before those states’ largely September primaries, so that no post-primary roll calls existed to observe; once sessions began running into the fall (from 1962 onward) this case nearly disappears. The remainder are members who left office before the primary (through resignation, death, or appointment to other office) or who, though still serving, faced only a handful of roll calls between a late primary and the autumn campaign recess. Only nine member-terms fall short on the pre-primary side alone, all of them members who entered the Congress after their primary date, typically via special election. Because those dropped for sparse post-primary voting are, by construction, among the least active after the primary, their exclusion is more likely to understate than to inflate the post-primary participation decline the analysis identifies. These restrictions leave 13,755 Congress-legislators for the participation analyses and 13,700 for the position analyses, the panels underlying both the aggregated legislator-period and vote-level specifications.

The same principle applies, at the subsample level, to the analyses of important bills (Section 6.2). A legislator’s change in position or participation *on a particular set of bills* can only be estimated when enough such votes fall on each side of the primary; when only one or two do, the change-score is mechanically driven toward ± 1 and a handful of these cells can dominate the estimate. I therefore require at least five subsample votes per side for the regression-discontinuity change-scores in those analyses—five rather than the full panel’s ten because the bill subsamples are far sparser (whipped and landmark votes are a small

fraction of all roll calls), so the lower threshold removes the unestimable one- and two-vote cells while retaining adequate power. Positions are filtered on cast votes and participation on available roll calls, mirroring the full-panel rule. The pooled vote-level difference-in-differences estimates do not form per-legislator cells and are left unfiltered.

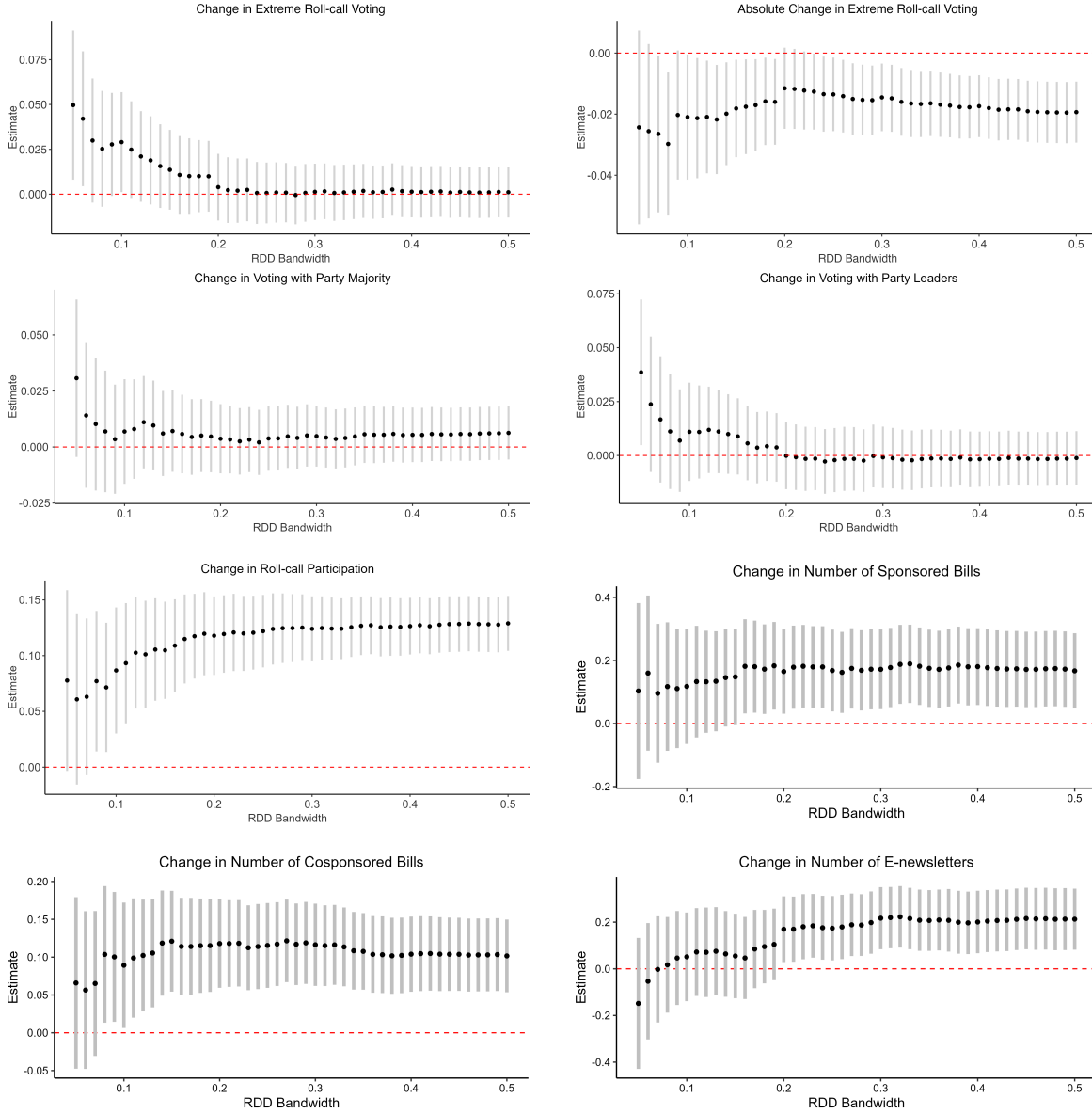
A.7 Regression Discontinuity Results with Different Bandwidths

The choice of bandwidth in a regression discontinuity design can be arbitrary, so it is important to demonstrate that conclusions are not driven by an particular choice of the specification. Figure A.2 shows the RDD estimate over different sizes of the bandwidths. The results are rather unstable for very narrow bandwidths due to the small sample sizes.

Due to the shorter time spans of the data for e-newsletters, sponsorship, and cosponsorship, the sample sizes are limited when using narrow bandwidths. As a result, many coefficients are statistically insignificant and fluctuate with small adjustments in bandwidth sizes within these narrow ranges. Nonetheless, the signs of the coefficients generally remain largely stable and align with the main conclusion of this article. Where the sample sizes are sufficiently large, the coefficients are largely consistent with the results from the difference-in-differences design.

Data-driven bandwidth selection has become a standard part of regression discontinuity practice. Table A.6 therefore reports estimates obtained with the mean-squared-error-optimal bandwidth of Calonico, Cattaneo and Titiunik (2014), using a local-linear specification with a uniform kernel. The optimal bandwidths are narrow—roughly $h = 0.04$ to 0.08 —which leaves very few observations near the cutoff: the effective samples for the effort outcomes range from 186 for participation down to about 50 for e-newsletters, whose data cover the shortest span. At these bandwidths most coefficients are no longer statistically significant, and the participation estimate (0.05) is smaller than its ± 0.2 counterpart (0.129); the e-newsletter estimate rests on only 52 observations and is too imprecise to interpret, even reversing sign. The policy-position estimates remain statistically insignificant at conventional levels throughout, with the sole exception of the with-leader cell, which is marginal

Figure A.2: RDD Estimates from Different Bandwidths



Note: Each graph visualizes the RDD estimates of changes in legislator behavior over different bandwidth. The x -axis in each graph is for the bandwidth, and the y -axis for the estimate size. The vertical gray lines for each bandwidth represents 95% confidence interval. The count-outcome panels (sponsorship, cosponsorship, e-newsletters) sweep the post-primary proportion measure of Table D.2; the corresponding weekly-rate estimates at the optimal bandwidth appear in Table A.6.

at the 10% level on only 157 observations. These results reinforce the bandwidth-sensitivity pattern in Figure A.2: the close-primary discontinuity is identified from a genuinely small number of competitive races, and the design's statistical power rests on the wider window

Table A.6: RD Estimates at the MSE-Optimal Bandwidth

	Coefficient	Robust SE	<i>p</i> -value	Optimal <i>h</i>	<i>N</i>
<i>Effort-related behavior</i>					
Change in Roll-call Participation	0.051	0.048	0.474	0.048	186
Change in Weekly Bill Sponsorship	0.045	0.102	0.798	0.057	146
Change in Weekly Bill Cosponsorship	0.296	0.904	0.793	0.045	101
Change in Weekly E-Newsletters	−0.294	0.827	0.518	0.058	52
<i>Policy positions</i>					
Change in Roll-call Extremism	0.031	0.027	0.153	0.068	250
Absolute Change in Roll-call Extremism	−0.022	0.025	0.319	0.054	204
Change in Voting with Party Majority	0.008	0.026	0.778	0.081	302
Change in Voting with Party Leader	0.052*	0.034	0.059	0.041	157

Note: Estimates use the mean-squared-error-optimal bandwidth of Calonico et al. (2014), implemented with `rdrobust` (local-linear, uniform kernel). “Coefficient” is the conventional RD point estimate; “Robust SE” and “*p*-value” are the bias-corrected robust standard error and *p*-value. *N* is the number of observations within the optimal bandwidth. Effort outcomes and units follow Table 3; the position outcomes are the within-term change scores β_{it} of equation (2), on each outcome’s own vote-share scale. Coefficients retain the winner-relative sign convention (see the sign-convention note at the head of this appendix). * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

used in the main analysis. I therefore report the optimal-bandwidth estimates transparently as a limitation of the close-primary RD rather than as corroborating evidence.

A.8 Regression Discontinuity Results from Redistricting Years

The regression discontinuity design only derives local average treatment effect and is not externally valid. The specific concern is that those legislators who are subject to close primary elections might be systematically and significantly different from the other legislators who face non-close primaries or no primary challenges at all. To address this concern indirectly, I implement the same research design focusing on only redistricting years, assuming that in redistricting years, legislators are more likely to face substantial primary competition in an as-if random fashion. It suffers from fewer sample and weaker statistical power as it includes fewer election years, but the results in Table A.7 show that the sizes and direction of the coefficients remain mostly consistent with the main findings.

Table A.7: Regression Discontinuity Results from Redistricting Years

	Change in Roll-call Participation	Change in Sponsorship	Change in with Cosponsorship	Change in E-Newsletter
Primary Loss	−0.123*** (0.029)	0.120 (0.104)	−1.202 (0.786)	−0.435** (0.183)
Observations	280	195	195	84
Bandwidth	0.2	0.2	0.2	0.2
Specification	Linear	Linear	Linear	Linear

	Change in Roll-call Extremism	Change in Voting with Party Majority	Change in Voting with Party Leaders	Absolute Change in Roll-call Extremism
Primary Loss	−0.024 (0.015)	−0.011 (0.012)	−0.002 (0.013)	0.008 (0.011)
Observations	279	279	277	279
Bandwidth	0.2	0.2	0.2	0.2
Specification	Linear	Linear	Linear	Linear

* $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$

Note: Standard errors are in the parentheses and are conventional OLS standard errors. Each column is a separate local-linear regression discontinuity estimate at bandwidth ± 0.2 restricted to redistricting-cycle years (primary years 1962, 1972, 1982, 1992, 2002, 2012, 2022). Effort outcomes and units follow Table 3 (participation as the share of votes cast; session-week rates for sponsorship and cosponsorship, e-newsletters per calendar week); the position outcomes are the within-term change scores β_{it} of equation (2), on each outcome’s own vote-share scale.

A.9 Policy Position Change in Competitive and Uncompetitive Districts

In the main article, I assess whether the effects on effort-related behavior vary with the expected competitiveness of the general election, finding that the withdrawal appears in competitive and uncompetitive districts alike wherever the designs have the power to detect it. To examine whether similar heterogeneous effects exist for legislators’ policy positions, I compare changes in policy positions after the primary between competitive and uncompetitive districts.

Table A.8 shows estimates of legislators’ changes in policy positions—the within-term

change scores β_{it} of equation (2), on each outcome’s own vote-share scale—in both competitive and uncompetitive districts following the primary elections. Across both research designs, the directional measures—signed extremism, voting with the party majority, and voting with party leaders—show no significant changes in either group. The absolute change in roll-call extremism is the one exception: losers reposition more than winners in uncompetitive districts (0.038, $p < 0.01$ in the regression discontinuity; 0.028, $p < 0.10$ in the difference-in-differences), the same loser-repositioning pattern documented for the full contested-primary sample (cf. the 0.014 difference-in-differences estimate in Table 4), here concentrated where general-election pressure is weakest. Directional position-taking, by contrast, remains unaffected by the expected competitiveness of the upcoming general election, reinforcing the argument that adjusting policy stances is unlikely to be a predominant strategy for legislators seeking reelection.

Table A.8: Changes in Policy Positions in Competitive and Uncompetitive Districts

Competitive Districts ($V_{t-1} < 0.75$, $N = 2,706$)				
	Regression Discontinuity			
	Change in Roll-call Extremism	Absolute Change in Roll-call Extremism	Change in Voting With Party Majority	Change in Voting With Party Leaders
Primary Loss	-0.005 (0.010)	0.000 (0.007)	-0.006 (0.009)	-0.000 (0.009)
Observations	653	653	653	649
Bandwidth	0.2	0.2	0.2	0.2
Specification	Linear	Linear	Linear	Linear
	Difference in Differences			
	Change in Roll-call Extremism	Absolute Change in Roll-call Extremism	Change in Voting With Party Majority	Change in Voting With Party Leaders
Primary Loss	-0.004 (0.009)	-0.000 (0.006)	-0.011 (0.007)	0.001 (0.007)
Observations	2,705	2,705	2,705	2,662
Specification	Aggregated	Aggregated	Aggregated	Aggregated
Fixed Effects	Legislator, Party-Cong.	Legislator, Party-Cong.	Legislator, Party-Cong.	Legislator, Party-Cong.
Uncompetitive Districts ($V_{t-1} > 0.75$, $N = 989$)				
	Regression Discontinuity			
	Change in Roll-call Extremism	Absolute Change in Roll-call Extremism	Change in Voting With Party Majority	Change in Voting With Party Leaders
Primary Loss	-0.000 (0.019)	0.038*** (0.014)	0.001 (0.017)	-0.008 (0.017)
Observations	265	265	265	262
Bandwidth	0.2	0.2	0.2	0.2
Specification	Linear	Linear	Linear	Linear
	Difference in Differences			
	Change in Roll-call Extremism	Absolute Change in Roll-call Extremism	Change in Voting With Party Majority	Change in Voting With Party Leaders
Primary Loss	-0.016 (0.018)	0.028* (0.014)	0.009 (0.023)	0.002 (0.018)
Observations	989	989	989	972
Specification	Aggregated	Aggregated	Aggregated	Aggregated
Fixed Effects	Legislator, Party-Cong.	Legislator, Party-Cong.	Legislator, Party-Cong.	Legislator, Party-Cong.

* $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$

Note: The top table shows the results from competitive districts where the incumbent legislator's previous vote share was less than 75%. The bottom table shows the results from less competitive districts where the previous vote share was greater than 75%.

A.10 Results Over Different Periods

An additional analysis was conducted to examine whether the effects vary across different time periods. Given the limited sample size, it was not feasible to segment the data into very fine time intervals, and I divided the sample into two periods: before and after 1994, which marks the beginning of a significant rise in ideological polarization and stricter party discipline.

The results in Table A.9—again the within-term change scores β_{it} of equation (2), on each outcome’s own vote-share scale—show that the main policy-position conclusions are stable across both periods: changes in roll-call extremism and party-line voting are small and statistically insignificant in nearly all cells, with only a few marginal exceptions that do not overturn the overall null on policy positions.

Table A.9: Policy Position Results in the Periods Before and After 1994

After 1994 ($N = 1,566$)				
	Regression Discontinuity			
	Change in Roll-call Extremism	Absolute Change in Roll-call Extremism	Change in Voting With Party Majority	Change in Voting With Party Leaders
Primary Loss	0.009 (0.008)	-0.000 (0.006)	-0.000 (0.009)	-0.001 (0.008)
Observations	360	360	360	358
Bandwidth	0.2	0.2	0.2	0.2
Specification	Linear	Linear	Linear	Linear
	Difference in Differences			
	Change in Roll-call Extremism	Absolute Change in Roll-call Extremism	Change in Voting With Party Majority	Change in Voting With Party Leaders
Primary Loss	0.004 (0.007)	0.006* (0.003)	-0.014** (0.006)	-0.010* (0.006)
Observations	1,566	1,566	1,566	1,537
Specification	Aggregated	Aggregated	Aggregated	Aggregated
Fixed Effects	Legislator, Party-Cong.	Legislator, Party-Cong.	Legislator, Party-Cong.	Legislator, Party-Cong.
Before 1994 ($N = 2,128$)				
	Regression Discontinuity			
	Change in Roll-call Extremism	Absolute Change in Roll-call Extremism	Change in Voting With Party Majority	Change in Voting With Party Leaders
Primary Loss	-0.010 (0.013)	0.013 (0.009)	-0.005 (0.012)	-0.001 (0.012)
Observations	558	558	558	553
Bandwidth	0.2	0.2	0.2	0.2
Specification	Linear	Linear	Linear	Linear
	Difference in Differences			
	Change in Roll-call Extremism	Absolute Change in Roll-call Extremism	Change in Voting With Party Majority	Change in Voting With Party Leaders
Primary Loss	-0.013 (0.011)	0.020** (0.010)	-0.007 (0.013)	0.004 (0.012)
Observations	2,128	2,128	2,128	2,097
Specification	Aggregated	Aggregated	Aggregated	Aggregated
Fixed Effects	Legislator, Party-Cong.	Legislator, Party-Cong.	Legislator, Party-Cong.	Legislator, Party-Cong.

* $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$

Note: Comparison of changes in policy positions before and after 1994. The top panel covers election years after 1994 (from 1996 onward); the bottom panel covers election years through 1994.

A.11 Legislators' Anticipation of Presidential Appointment

A potential concern for primary losers is non-electoral career prospects, such as presidential appointments to federal or local office (Jenkins and Nokken, 2008). If losers anticipate such appointments in exchange for adhering to the party line, their policy positions could remain stable for reasons unrelated to electoral incentives—making the null on policy positions an artifact of substitute motivations, not evidence against the electoral-incentives channel.

To test this, I restrict the analysis to legislators in the president's party during midterm-election years. Because the president serves at least two more years and rarely appoints across party lines, these lawmakers are substantially more likely than others to anticipate a post-Congress appointment. If the appointment-anticipation channel drives the null on policy positions, these legislators should show the smallest post-primary changes. The results (Table A.10) are mixed: the regression discontinuity design shows three statistically significant shifts (losers are 3.4 percentage points less likely to cast an extreme roll-call vote, show a 3.7-point larger absolute extremity shift, and are 3.5 points less likely to vote with party leaders), but the difference-in-differences estimates are small and statistically insignificant. The pattern does not strongly support the appointment-anticipation channel as the driver of the null on policy positions.

Table A.10: Policy Positions Among Legislators in the President’s Party in Midterm Years

	Regression Discontinuity			
	Change in Roll-call Extremism	Absolute Change in Roll-call Extremism	Change in Voting With Party Majority	Change in Voting With Party Leaders
Primary Loss	−0.034* (0.019)	0.037*** (0.014)	−0.031 (0.019)	−0.035** (0.018)
Observations	226	226	226	226
Bandwidth	0.2	0.2	0.2	0.2
Specification	Linear	Linear	Linear	Linear
	Difference in Differences			
	Change in Roll-call Extremism	Absolute Change in Roll-call Extremism	Change in Voting With Party Majority	Change in Voting With Party Leaders
Primary Loss	0.015 (0.043)	0.025 (0.029)	0.007 (0.037)	−0.009 (0.033)
Observations	889	889	889	878
Specification	Aggregated Legislator	Aggregated Legislator, Party-Cong.	Aggregated Legislator, Party-Cong.	Aggregated Legislator, Party-Cong.
Fixed Effects				

* $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$

Note: These results are from legislators who face primary elections during midterm election years and belong to the president’s party. These legislators are more likely than others to anticipate a post-Congress appointment by the president, as he will remain in office for at least two more years and rarely appoints legislators from the opposing party. The difference-in-differences estimate for Change in Roll-call Extremism includes legislator fixed effects only (no Party-Congress fixed effects).

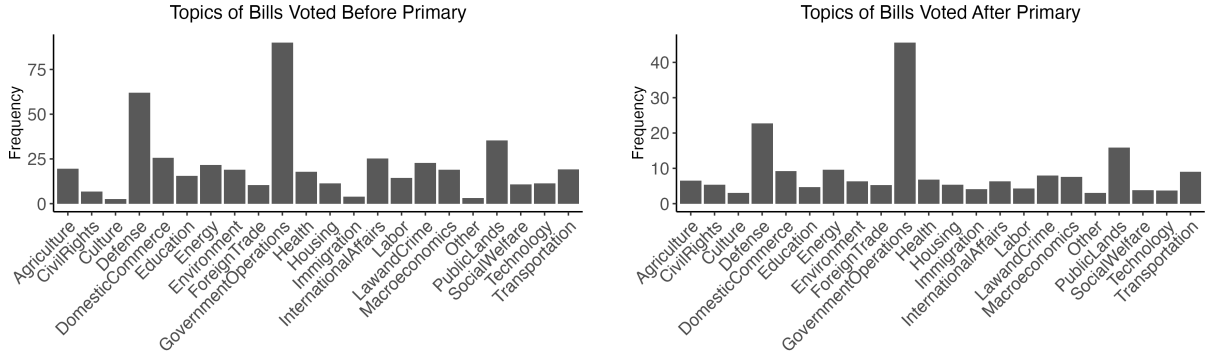
A.12 Descriptive Bill Differences Before and After Primary Elections

In the main article, I account for potential differences in the types of bills voted before and after primary elections by using regression models with bill fixed effects, thereby isolating changes in legislators’ voting behavior from potential variations in bill content.

Nevertheless, to ensure that the composition of bills is not dramatically different before and after primary elections, I conduct several descriptive comparisons, examining the distribution of bill topics and legislators’ general voting patterns.

Figure A.3 presents the average number of topics of bills voted on per congressional

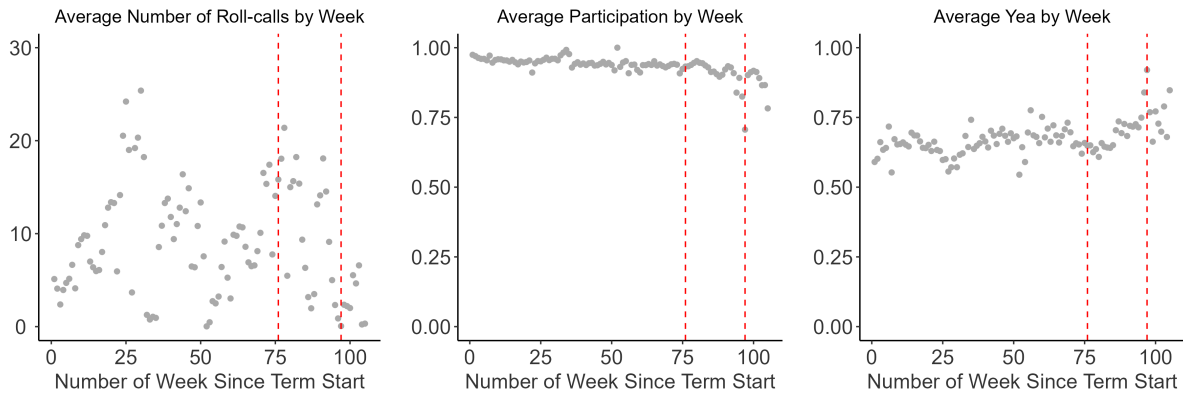
Figure A.3: Distribution of Topics of Bills Before (left) and After Primary (right)



term before (left) and after (right) primary elections. The topics are classified according to the Congressional Bills Project (Adler and Wilkerson, 2020). The distributions appear largely consistent across both periods, with government operations being the most frequently addressed topic, followed by defense, public lands, and other policy areas in a similar order. These patterns suggest that the legislative agenda remains relatively stable before and after primary elections.

Figure A.4 illustrates three aggregate characteristics of bills that reach a roll-call vote across a congressional term: the number of bills voted on per week (left), legislators' partici-

Figure A.4: Average Weekly Roll-call Voting Characteristics



Note: The x-axis of each figure represents the number of weeks since the start of a Congress. The first figure plots the average number of roll-call votes per week, the second displays the average participation rate, and the third shows the average yea vote rate. The two red vertical dotted lines indicate the time window between the average primary election date and the general election.

pation rates in those votes (center), and the frequency with which bills pass (right). The two red vertical lines indicate the average dates of the primary election and the general election, serving as reference points to assess whether the nature of bills considered changes before and after primary elections. While the plots reveal natural week-to-week fluctuations and a sharp shift near the general election, there is no clear evidence of a systematic or substantial change in bill characteristics around the primary election period.

A.13 Policy Position Results by Mean Differences Between Pre- and Post-Primary Periods

In the main article, I assess legislators' changes in roll-call voting using a regression approach, estimating them through OLS coefficients. As a robustness check, I additionally measure changes in legislators' voting behavior using a simpler method – calculating the mean difference in their voting patterns between the pre- and post-primary periods. This alternative approach ensures that the results are not driven by the specific estimation method used in the main analysis.

Specifically, for each legislator i in t , I measure a legislator's change in voting pattern after primary election by the difference of average voting patterns in the pre- and post-primary periods. For example, if legislator i in t , participated 90% and 70% of roll-calls in the pre-primary period and post-primary period, respectively, then the change is 20% point decrease and I denote it as $Y_{it} = -0.2$.

In the second analysis, to ensure that the time windows are directly comparable, I restrict the sample to roll-calls occurring within the same specific time window in each year. For instance, if a primary election is held on August 1 and the general election on November 1, I compare the participation rate from August 1 to November 1 in the election year with the rate during that same period in the preceding year. As shown in Table A.11, the results remain largely consistent in both specifications.

Table A.11: Result on Policy Position by Mean Difference Before and After Primary Elections

Difference in Mean Before and After Primary					
	Regression Discontinuity				
	Change in Roll-call Participation	Change in Roll-call Extremism	Absolute Change in Roll-call Extremism	Change in Voting With Party Majority	Change in Voting With Party Leaders
Primary Loss	−0.117*** (0.019)	−0.015 (0.012)	0.021** (0.008)	−0.009 (0.009)	−0.012 (0.009)
Observations	926	918	918	919	912
Bandwidth	0.2	0.2	0.2	0.2	0.2
Specification	Linear	Linear	Linear	Linear	Linear
	Difference in Differences				
	Change in Roll-call Participation	Change in Roll-call Extremism	Absolute Change in Roll-call Extremism	Change in Voting With Party Majority	Change in Voting With Party Leaders
Primary Loss	−0.091*** (0.018)	−0.010 (0.009)	0.021*** (0.007)	−0.013 (0.009)	−0.008 (0.008)
Observations	3,714	3,694	3,694	3,697	3,637
Specification	Aggregated	Aggregated	Aggregated	Aggregated	Aggregated
Fixed Effects	Legislator, Party-Cong.	Legislator, Party-Cong.	Legislator, Party-Cong.	Legislator, Party-Cong.	Legislator, Party-Cong.
Difference in Mean Before and After Primary in Same Time Windows					
	Regression Discontinuity				
	Change in Roll-call Participation	Change in Roll-call Extremism	Absolute Change in Roll-call Extremism	Change in Voting With Party Majority	Change in Voting With Party Leaders
Primary Loss	−0.106*** (0.021)	−0.008 (0.014)	0.015 (0.010)	0.004 (0.011)	0.007 (0.011)
Observations	920	913	913	914	907
Bandwidth	0.2	0.2	0.2	0.2	0.2
Specification	Linear	Linear	Linear	Linear	Linear
	Difference in Differences				
	Change in Roll-call Participation	Change in Roll-call Extremism	Absolute Change in Roll-call Extremism	Change in Voting With Party Majority	Change in Voting With Party Leaders
Primary Loss	−0.103*** (0.018)	−0.009 (0.010)	0.021*** (0.007)	−0.006 (0.011)	0.000 (0.011)
Observations	3,699	3,679	3,679	3,682	3,622
Specification	Aggregated	Aggregated	Aggregated	Aggregated	Aggregated
Fixed Effects	Legislator, Party-Cong.	Legislator, Party-Cong.	Legislator, Party-Cong.	Legislator, Party-Cong.	Legislator, Party-Cong.

* $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$

Note: The table presents estimates of the difference in legislators' roll-call voting behavior before and after primary elections. The top panel compares all votes cast after the primary to those cast before. The bottom panel refines the comparison by examining votes cast post-primary against those from the same time window in the preceding year.

A.14 Aggregated Difference-in-Differences for Vote-Based Outcomes (Cross-Check)

The main-text difference-in-differences results for roll-call-based outcomes—participation, signed roll-call extremism, voting with the party majority, and voting with party leaders—are estimated at the legislator–bill vote level using equation (4) in the main text. As a cross-check, Table A.12 reports the corresponding estimates from an aggregated legislator–period specification identical in form to equation (5), with the dependent variable constructed in two steps: (i) for each legislator–Congress pair, a within-term regression of the vote-level outcome on an `AfterPrimary` indicator produces a legislator-specific coefficient; (ii) that coefficient is then used as the dependent variable in a DiD regression with `Primary Loss` as the regressor, legislator fixed effects, and party–Congress fixed effects. It is reported here for readers who prefer the legislator-level summary.

Table A.12: Aggregated (β_{it} -Based) Difference-in-Differences for Vote-Based Outcomes

	Change in Roll-call Participation	Change in Roll-call Extremism	Change in Voting with Party Majority	Change in Voting with Party Leaders
Primary Loss	−0.092*** (0.017)	−0.007 (0.008)	−0.010 (0.008)	−0.002 (0.008)
Observations	3,714	3,694	3,694	3,634
Winner Mean	0.027	−0.000	−0.000	0.001
Loser Mean	−0.086	0.007	−0.007	−0.003
Specification	Aggregated	Aggregated	Aggregated	Aggregated
Fixed Effects	Legislator, Party-Cong.	Legislator, Party-Cong.	Legislator, Party-Cong.	Legislator, Party-Cong.

* $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$

Note: Each column reports the Primary Loss coefficient from an aggregated legislator-period specification ($Y_{ipt} = \beta_0 + \beta_1 \text{PrimaryLoss}_{it} + \delta_i + \gamma_{pt} + \varepsilon_{ipt}$, with legislator and party-Congress fixed effects), applied to vote-based outcomes whose main-text estimates are reported at the vote level. Standard errors (in parentheses) are clustered by legislator. Unit of observation is the legislator-period; N refers to the number of legislator-Congress pairs for which the β_{it} outcome could be constructed. The aggregated participation estimate of -0.092 and the vote-level interaction of -0.105 in Table 3 are directly comparable: both identify the same within-legislator shift in post-primary participation between winners and losers. The same correspondence holds for the signed-extremism, voting-with-party, and voting-with-leader columns, where both specifications produce null estimates.

A.15 Non-Challenged Incumbents: Scope and Natural Placebo

Incumbents who face contested primaries may differ systematically from those who do not, which bears both on the scope of the main results and, by extension, on the lame-duck mechanism itself. I address this along two dimensions. First, I compare the two groups on the district-political characteristics most relevant to selection into a contested primary (general-election margin, district presidential vote) and on standard legislator covariates (Table A.13). Second, I extend the difference-in-differences specifications to a three-group categorical variable with primary losers as the reference and two non-loser dummies for narrow winners and non-challenged incumbents (Table A.14). The behavioral comparison provides a particularly sharp placebo: the lame-duck mechanism predicts that non-challenged incumbents—who never had a re-election threat to remove—should behave like narrow primary winners and unlike narrow primary losers. If they instead resembled losers post-primary, the mechanism would be in question.

The non-challenged group consists of incumbents who sought renomination without a contested primary, defined as those recorded as having a single-candidate primary or an unopposed primary (the latter recorded under a data convention that omits the runner-up entry, equivalent in substance), excluding Louisiana and Connecticut to match the main analysis ($N = 8,234$ legislator–Congresses, all with non-missing primary dates). For roll-call-based outcomes I apply the vote-level specification of equation (4) with $\text{Winner} \times \text{AfterPrimary}$ and $\text{NonChallenged} \times \text{AfterPrimary}$ interactions. For non-vote effort outcomes (sponsorship, cosponsorship, e-newsletters), pre- and post-primary weekly rates for the non-challenged group are constructed from the same record series used in the main analysis, applying the identical window construction (Congress start to primary; primary to general); sanity-check correlations against the existing main-analysis values exceed 0.9996 for all three outcomes.¹ The aggregated specifications follow equation (5) without the legislator fixed effects (consistent with the footnote there), which would otherwise absorb the `NonChallenged` dummy:

¹For β_{it} -based outcomes (extremity, conservative voting, voting with party, voting with leader, abstention), within-Congress regression coefficients are already produced for both the contested and non-challenged samples by the existing pipeline, since the underlying fixed-effect regression operates on the full *MemberVotesCleaned* panel.

Table A.13: Scope Comparison on Observable Characteristics: Contested-Primary Incumbents vs. Non-Challenged Incumbents

	Group A: contested ($N = 3,860$)		Group B: non-challenged ($N = 8,234$)		
Variable	Mean	SD	Mean	SD	Difference
<i>Legislator characteristics</i>					
Age (years)	54.69	11.02	52.23	10.13	+2.47***
Tenure (prior Congresses)	4.71	4.36	4.01	3.87	+0.71***
Democrat (0/1)	0.622	0.485	0.515	0.500	+0.107***
DW-NOMINATE (1st dim.)	−0.062	0.400	0.002	0.378	−0.064***
DW-NOMINATE 1st dim. (extremity)	0.370	0.165	0.343	0.159	+0.027***
Redistricted (0/1)	0.246	0.431	0.209	0.406	+0.038***
<i>District political environment</i>					
Prior general-election vote share	0.698	0.148	0.688	0.149	+0.011***
District Dem. presidential vote share	0.520	0.154	0.494	0.134	+0.026***

* $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$ (two-sided Welch t -test)

Note: Group A is the paper’s treatment sample (narrow primary winners and losers). Group B is the non-challenged sample defined as incumbents seeking renomination with a recorded primary date and either a single-candidate or an unopposed primary record (both indicate an uncontested primary; the latter arises from an unopposed single-candidate record that does not produce a runner-up entry). Louisiana and Connecticut are excluded in both groups to match the main analysis. The two district-environment variables (general-election margin, district presidential vote) are the central scope concern; their absolute differences across groups are within 1–2.6 percentage points.

every non-challenged incumbent is always coded as such, yielding no within-legislator variation on that indicator.

Table A.13 reports the comparison on observable characteristics. Statistically, contested-primary incumbents are modestly older, more senior, more likely to be Democrats, marginally more ideologically extreme, and more likely to have been redistricted, with all gaps reaching conventional significance under the very large combined sample. Substantively, the magnitudes are small: the largest gap is the Democrat share at +10.7 percentage points; the rest correspond to roughly 0.1–0.2 standard deviations. On the two district-political variables of central concern, the gaps are minimal. Prior general-election vote share differs by +1.1 percentage points (0.698 vs. 0.688) and district presidential Democratic vote share by +2.6 percentage points (0.520 vs. 0.494). The implicit concern that contested-primary incumbents might come from systematically more competitive electoral environments is not supported by the data; if anything, contested-primary incumbents represent slightly safer general-election seats.

Table A.14 reports the behavioral comparison across twelve outcomes. The pattern is consistent and theoretically expected. On the effort-related outcomes that respond to the primary—roll-call participation, abstention, and the weekly rates of cosponsoring and newsletter activity—both narrow primary winners and non-challenged incumbents differ sharply from primary losers, with similar magnitudes across the two non-loser groups; sponsorship, the sparsest activity, shows correspondingly small gaps here. On the directional ideology outcomes—voting with the party majority, voting with party leaders, change in conservative voting, and the signed extremity change—all three groups are statistically and substantively indistinguishable from zero. The lone exception is the absolute extremity change, where winners and non-challenged incumbents differ from losers by a small -0.013 and -0.012 : this gap is statistically detectable only under the large legislator–Congress sample, and winners and non-challenged remain indistinguishable from each other ($p = 0.136$). A formal test of equality between the Winner and NonChallenged interactions, $H_0 : \beta_W = \beta_{NC}$, fails to reject ($p > 0.10$) for ten of the twelve outcomes. The two rejections—bill sponsorship ($p < 0.001$, a gap of 0.031 bills per session-week) and the signed extremity change—are small-magnitude differences in the same theoretical direction: both non-loser groups sit on the same side of the losers, behaving like primary winners rather than losers. Taken together with the scope comparison, the natural-placebo evidence supports the lame-duck interpretation: the post-primary effort decline is concentrated specifically among legislators whose re-election incentive disappears, not generated by some specific characteristic from contested primaries.

Table A.14: Three-Group Difference-in-Differences across Twelve Outcomes: Primary Losers, Winners, and Non-Challenged Incumbents

Outcome	Winner vs. Loser	Non-Challenged vs. Loser	Winner – Non-Challenged	<i>N</i>
<i>Panel A: Vote-level DiD (eq. 4). Coefficients on Status × AfterPrimary.</i>				
Roll-call participation	0.106*** (0.014)	0.104*** (0.014)	0.002 (0.215)	10,937,174
Voting with party majority	−0.002 (0.004)	−0.002 (0.004)	0.000 (0.616)	10,295,321
Voting with party leader	−0.004 (0.004)	−0.004 (0.004)	0.000 (0.999)	8,970,645
<i>Panel B: Aggregated DiD on non-vote outcomes (eq. 5). DV is the change in weekly activity.</i>				
Bill sponsorship	0.055* (0.033)	0.086*** (0.032)	−0.031*** (0.001)	8,757
Bill cosponsorship	1.549*** (0.288)	1.625*** (0.289)	−0.076 (0.189)	7,677
E-newsletter	0.653*** (0.186)	0.677*** (0.187)	−0.024 (0.147)	2,166
<i>Panel C: Aggregated β_{it} DiD (eq. 5). DV is the within-Congress regression coefficient.</i>				
Change in roll-call abstention	−0.116*** (0.014)	−0.114*** (0.014)	−0.003 (0.157)	11,663
Change in conservative voting	−0.007 (0.006)	−0.005 (0.006)	−0.002 (0.127)	11,609
Change in extremity (signed)	−0.007 (0.006)	−0.009 (0.006)	0.002** (0.023)	11,609
Change in extremity	−0.013*** (0.005)	−0.012*** (0.005)	−0.001 (0.136)	11,609
Change in voting with party	0.007 (0.006)	0.006 (0.006)	0.001 (0.505)	11,609
Change in voting with party leader	0.005 (0.006)	0.003 (0.006)	0.002 (0.166)	11,444

* $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$

Note: Each row reports estimates from a single regression with primary losers as the reference category. Coefficients are clustered at the legislator level; standard errors in parentheses. The third column reports the Winner – Non-Challenged difference (column 1 – column 2); the p -value in parentheses is from a cluster-robust Wald test of $\beta_W = \beta_{NC}$, and stars on the difference denote that test’s significance. Sample restricted to incumbents seeking renomination with a recorded primary date, excluding Louisiana and Connecticut (matching the main analysis). The non-challenged sample is defined as in Table A.13 ($N = 8,234$). Per the footnote to equation (5), the aggregated specifications drop legislator fixed effects, which would otherwise absorb the NonChallenged dummy. Each count outcome is restricted to the years its source data cover (sponsorship 1973–2022, cosponsorship 1979–2022, e-newsletters 2010–2022).

A.16 Heterogeneity by Career Stage

A natural question is whether the within-Congress behavioral response to losing reelection incentives differs by legislator career stage—whether the effect of losing a close primary early in one’s career differs from the effect later in tenure. I construct seniority directly from the Voteview congressional member file by counting prior House terms for each legislator-Congress, splitting at *junior* (1st–2nd term, 0–1 prior terms) versus *senior* (3rd term or later, 2+ prior). I re-estimate both the main RDD specification and the simpler vote-level DiD on each subgroup, with a pooled fully-interacted model providing a formal differential test.

Table A.15 reports the results. In the RDD (Panel A, $|rv| < 0.2$), the participation effect is essentially identical across career stages: junior $\hat{\beta} = -0.134$ ($p < 0.001$, $N = 244$) versus senior $\hat{\beta} = -0.126$ ($p < 0.001$, $N = 682$), with a fully null differential ($p = 0.86$). None of the other seven outcomes display any meaningful differential by seniority, including the signed and absolute roll-call extremism measures. The vote-level DiD (Panel B), which trades some internal validity for substantially larger samples ($N \approx 3$ million vote-bill observations), agrees: the junior and senior participation responses are essentially the same ($\hat{\beta}_{\text{junior}} = -0.102$ vs. $\hat{\beta}_{\text{senior}} = -0.119$; differential $p = 0.44$), with no significant differential on any of the three ideology outcomes. Together the two specifications yield a clear answer: the within-Congress effect on participation is robust across career stages, and neither design detects a meaningful seniority differential. The close-primary sample is a binding power constraint for any sharper test of heterogeneity, as the wide differential confidence interval of the preferred RDD shows; but the much higher-powered vote-level DiD likewise finds no seniority differential, so the data rule out large qualitative differences across career stage.

Table A.15: Career-Stage Heterogeneity by Seniority

Outcome	Junior (1st–2nd term)	Senior (3rd term or later)	Diff. ($\beta_{\text{Loss} \times \text{Sr}}$)	N
<i>Panel A: RDD ($rv < 0.2$, local-linear OLS, change / weekly-rate outcome).</i>				
Roll-call participation	−0.134*** (0.027)	−0.126*** (0.023)	0.008 (0.043)	926
Roll-call extremism	0.020 (0.014)	−0.011 (0.011)	−0.031 (0.021)	918
Absolute extremity	0.001 (0.010)	0.009 (0.008)	0.008 (0.015)	918
Voting with party	0.005 (0.013)	−0.005 (0.010)	−0.011 (0.019)	918
Voting with leader	0.010 (0.012)	−0.004 (0.010)	−0.014 (0.019)	911
Bill sponsorship	0.002 (0.095)	−0.016 (0.099)	−0.018 (0.160)	665
Bill cosponsorship	−1.227 (0.889)	−1.566** (0.666)	−0.339 (1.122)	578
E-newsletter	−0.270 (0.309)	−0.789*** (0.262)	−0.519 (0.460)	245
<i>Panel B: Vote-level DiD (eq. (4), legislator + bill-by-party-by-Congress FE, cluster icpsr).</i>				
Roll-call participation	−0.102*** (0.021)	−0.119*** (0.016)	−0.021 (0.027)	3,622,179
Roll-call extremism	0.004 (0.005)	0.005 (0.005)	−0.002 (0.007)	3,296,336
Voting with party	−0.002 (0.006)	0.002 (0.005)	0.004 (0.008)	3,290,805
Voting with leader	0.000 (0.005)	0.006 (0.005)	0.007 (0.008)	2,892,750

* $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$

Note: Seniority constructed from the Voteview congressional member file as the count of prior House terms (same legislator) before the focal Congress. Standard errors in parentheses; Panel B is cluster-robust at the legislator level. The Diff. column reports the $\text{PrimaryLoss} \times \text{Senior}$ coefficient from a fully-interacted model on the pooled sample. Subgroup N 's sum to less than pooled N because the differential model uses additional observations with non-missing outcomes that drop out of one or the other subgroup-only regression. Replication materials are provided with the paper.

A.17 Heterogeneity by Age at the Primary

Accounts in which post-primary effort protects a remaining career make an age-graded prediction: a losing legislator with decades of professional life ahead—future runs for office, appointments, reputation beyond the House—has more to protect and should withdraw less, while a legislator near retirement has little left to protect and should withdraw more. If instead the withdrawal is keyed to candidacy itself, age should not matter. I construct each legislator’s age at the primary from the Voteview member file and split the contested sample at the median age of 55 (interquartile range 47–64; 61 losing terms below the median, 99 above), re-estimating the specifications of the main heterogeneity table on each subgroup; continuous age-interaction versions of all sixteen gradient tests provide the formal battery.

Table A.16 reports the results, and they mirror the competitiveness split: the withdrawal is uniform. Participation falls in both age groups under both designs and on both margins. Sponsorship’s thin cells scatter without a coherent age structure: the additive panel leans toward the younger group, the proportional panel estimates are nearly identical in the two groups, the proportional cutoff estimate is larger among the older, and every sponsorship subgroup cell sits near its minimum detectable effect. Cosponsorship and e-newsletter declines appear in both groups on the weekly-panel specifications—the cosponsorship cells are nearly identical (-0.76 versus -0.75 least squares, -0.38 versus -0.38 Poisson). None of the sixteen age-gradient tests is significant at the five-percent level, and none reaches even the ten-percent level. The prediction of the career-protection account—sharper withdrawal among older losers—appears nowhere. Together with the seniority splits of subsection A.16, which are likewise uniform, the data indicate that the withdrawal does not track the length of the career ahead.

Table A.16: Effort Effects by Age at the Primary: Two Designs, Two Margins

Younger than 55 at the Primary				
	Roll-call Participation (share of votes cast)	Bill Sponsorship (count per week)	Bill Cosponsorship (count per week)	E-Newsletters (count per week)
<i>Additive margin: change in rate</i>				
Primary Loss (RD)	−0.115*** (0.032)	−0.016 (0.066)	−0.563 (0.442)	−1.089*** (0.352)
Observations	386	293	254	91
Primary Loss × After (DiD)	−0.129*** (0.024)	−0.055* (0.029)	−0.765*** (0.196)	−1.106** (0.533)
Observations	1,595,777	125,750	106,809	32,218
<i>Proportional margin: share of activity / change relative to own rate</i>				
Primary Loss (RD, proportion)	−0.065 (0.048)	−0.073 (0.105)	−0.083* (0.046)	−0.178* (0.102)
Observations	405	241	252	84
Primary Loss × After (Poisson)	−0.154*** (0.032)	−0.447* (0.237)	−0.378*** (0.083)	−1.412*** (0.242)
Observations	1,595,752	96,283	91,628	28,199
55 and Older at the Primary				
	Participation	Sponsorship	Cosponsorship	E-Newsletters
<i>Additive margin: change in rate</i>				
Primary Loss (RD)	−0.124*** (0.025)	−0.020 (0.058)	−1.066** (0.435)	−0.385 (0.234)
Observations	540	372	324	154
Primary Loss × After (DiD)	−0.094*** (0.017)	−0.011 (0.025)	−0.753*** (0.204)	−0.469*** (0.143)
Observations	1,924,639	148,508	132,817	63,845
<i>Proportional margin: share of activity / change relative to own rate</i>				
Primary Loss (RD, proportion)	−0.074** (0.034)	−0.239*** (0.091)	−0.142*** (0.039)	−0.156 (0.102)
Observations	555	323	323	126
Primary Loss × After (Poisson)	−0.112*** (0.022)	−0.447* (0.245)	−0.381*** (0.081)	−1.071*** (0.312)
Observations	1,924,633	119,637	116,062	53,358

* $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$

Note: Coefficients are the effect of losing the primary; standard errors are conventional in the discontinuity rows, clustered by legislator in the panel rows. Age at the primary is the primary year minus the Voteview birth year; panels split the contested sample at the median of 55. Margins, specifications, and sample conventions follow the competitiveness table in the main text (additive difference-in-differences rows are the weekly-panel specification; participation proportional cells translate the additive response through the base rate; Poisson coefficients are on the log scale, with $\exp(b) - 1$ the proportional change in a legislator's own rate). Sixteen formal gradient tests—continuous age interactions in the panels and at the cutoff, across all outcomes, margins, and designs—are null at the five-percent level, and none reaches the ten-percent level. The e-newsletter cells rest on 13 losing terms below the median and 27 above; every sponsorship subgroup cell sits near its minimum detectable effect.

A.18 Heterogeneity by Primary-to-General Window Length

A further question is whether the lame-duck effect persists beyond the short period between the primary and the general election. Persistence into a subsequent Congress is not directly estimable in my within-Congress identification: by construction, primary losers are not in $t+1$ as House members. A related and design-compatible question—whether the post-primary effect might be a brief blip rather than a sustained pattern—is whether the magnitude of the lame-duck response varies with the *length* of the post-primary window. State law sets primary dates anywhere from March to September, so close-primary losers face anywhere from about 8 weeks to about 8 months of lame-duck status before the general election.

I construct the number of days from each legislator’s primary date to the relevant November general-election day, restrict to plausible values (30–300 days; this drops two outlier rows), and split the close-primary RDD sample at the median of 147 days (≈ 5 months). I then re-estimate both the main RDD specification and the simpler vote-level DiD on each subgroup, with a pooled fully-interacted model providing a formal differential test. Table A.17 reports the results. The participation effect is statistically indistinguishable across the two window-length subgroups in both specifications: in the RDD, -0.119^{***} (late) versus -0.131^{***} (early), differential $p = 0.75$; in the vote-level DiD, -0.129^{***} (late) versus -0.106^{***} (early), differential $p = 0.29$. The two specifications give numerically opposing point estimates but neither differential is significant. The remaining outcomes are largely null or of negligible magnitude, including signed roll-call extremism in both panels. Absolute extremity’s early-window RDD cell is a small significant positive (0.019, $p < 0.05$), while the late-window cell is essentially zero and the timing differential is null (0.021, $p > 0.10$), indicating marginally greater absolute ideological movement among long-window losers but no significant timing heterogeneity. Measured as weekly rates, the count outcomes show no window-length sensitivity for cosponsorship or e-newsletters (both significant in the subgroups where their data are dense, differentials null); sponsorship—null in both subgroups—shows a marginal timing differential (0.291, $p = 0.05$), the early-window point estimate sitting slightly above the late-window one, with neither subgroup estimate distinguishable from zero. In the high-powered vote-level DiD, the with-leader early-window and differential coef-

ficients are nominally significant (0.012 and 0.022, $p < 0.05$) but substantively trivial—one to two percentage points—reflecting the very large vote-level sample rather than a meaningful positional shift. The substantive bottom line is that the lame-duck effect is largely *robust* to window length: a short window of weeks already delivers most of the response detectable in my data, and a longer window of several months does not produce a substantially larger one.

I caveat that primary date is set by state law and correlates with state political environment (Southern runoff states tend to hold later primaries, etc.), so the early-versus-late split is not orthogonal to underlying state characteristics. I re-estimate both panels including a $\text{South} \times \text{PrimaryLoss}$ (and, for the DiD, $\times \text{AfterPrimary}$) interaction to absorb the dominant region confound; the differentials are essentially unchanged. Other state-level confounders may remain. The differential should be read as descriptive of timing heterogeneity rather than the causal effect of additional lame-duck time.

Table A.17: Heterogeneity by Primary-to-General Window Length

Outcome	Late primary (≤ 147 d to general)	Early primary (> 147 d)	Diff. ($\beta_{\text{Loss} \times \text{Early}}$)	N
<i>Panel A: RDD ($rv < 0.2$, local-linear OLS, change / weekly-rate outcome).</i>				
Roll-call participation	−0.119*** (0.027)	−0.131*** (0.025)	−0.012 (0.037)	926
Roll-call extremism	−0.015 (0.013)	0.009 (0.012)	0.025 (0.018)	918
Absolute extremity	−0.002 (0.009)	0.019** (0.009)	0.021 (0.013)	918
Voting with party	−0.003 (0.012)	−0.004 (0.011)	−0.001 (0.016)	918
Voting with leader	−0.004 (0.012)	0.003 (0.011)	0.007 (0.016)	911
Bill sponsorship	−0.169 (0.121)	0.122 (0.088)	0.291* (0.150)	665
Bill cosponsorship	−1.353* (0.810)	−1.647** (0.699)	−0.294 (1.073)	578
E-newsletter	−0.859*** (0.323)	−0.429*** (0.151)	0.430 (0.401)	245
<i>Panel B: Vote-level DiD (eq. (4), legislator + bill-by-party-by-Congress FE, cluster icpsr).</i>				
Roll-call participation	−0.129*** (0.020)	−0.106*** (0.018)	0.029 (0.027)	3,622,179
Roll-call extremism	0.002 (0.005)	0.008 (0.006)	0.005 (0.008)	3,296,336
Voting with party	−0.007 (0.007)	0.006 (0.005)	0.012 (0.009)	3,290,805
Voting with leader	−0.009 (0.007)	0.012** (0.006)	0.022** (0.010)	2,892,750

* $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$

Note: Window length is the number of days from each legislator’s primary date to the November general election (first Tuesday after the first Monday). Sample restricted to legislators with window length between 30 and 300 days (drops two implausible-date rows). Median split at 147 days (≈ 5 months). Standard errors in parentheses; Panel B is clustered at the legislator level. The Diff. column reports the $\text{PrimaryLoss} \times \text{Early}$ coefficient from a fully-interacted model. Robustness controlling for $\text{South} \times \text{PrimaryLoss}$ (Panel A) and $\text{South} \times \text{PrimaryLoss} \times \text{AfterPrimary}$ (Panel B) leaves all differentials essentially unchanged. **Caveat:** primary date is endogenous to state political environment; the differential is descriptive, not causal. Replication materials are provided with the paper.

A.19 Session Time versus Calendar Time

The main text measures the chamber-dependent count outcomes—bill sponsorship and cosponsorship—over session time rather than calendar time, equalizing windows in days the House is actually in session. This section documents the session calendar behind that convention and reports the calendar-time and trimmed-window comparisons. The premise is real: averaged across the 3,860 contested-primary legislator-Congresses, the pre-primary windows contain a median of 64 House-active days while post-primary windows contain a median of 57 (median asymmetry +8 days; 69% of rows have shorter post windows; 24% exceed an absolute asymmetry of 20 days). If the estimates from the symmetric pre/post-primary design reflected fewer session days available for voting rather than reduced electoral incentive, equalizing session days should reduce the effect.

For each date from 1955 to 2024 ($n = 10,319$ dates), I construct a binary variable indicating whether the House was in active session on that date, combining Voteview House roll-call dates (a roll-call by definition occurs on a session day) with Library of Congress Congressional Record API queries filtered to records carrying a House Section. The resulting series matches the U.S. House Historian’s official Legislative Days within $\pm 5\%$ for all 68 Congress-Sessions in my analysis range (median absolute error 1.0 day, $r = 0.997$). For each (legislator, congress), I project the original symmetric calendar window onto this binary and trim each leg to $\min(\text{sd_pre}, \text{sd_post})$ session days. I then re-aggregate all six outcomes within the trimmed window and re-run the within-Congress RDD on the session-day-restricted measures. For the β_{it} -based ideology scores, I apply the same within-term score-construction procedure to the full Congress 84–117 roll-call panel restricted to the same per-legislator window, yielding session-day analogues of the change in conservative voting, in voting with the party majority, in voting with party leaders, and in participation. As a sanity check, with no window restriction the procedure reproduces the scores used elsewhere in the paper to machine precision over Congresses 84–92, the range for which a direct comparison is available.

Table A.18 reports the central comparisons. The participation effect is preserved and

slightly amplified under session-day equalization across both legislator-level estimators. In the within-Congress simple-mean RDD on $|rv| < 0.2$, the change-in-participation coefficient is -0.119^{***} at the calendar window and -0.125^{***} at the trimmed window; and in the β_{it} RDD on the participation outcome, the coefficient is -0.127^{***} at the calendar window and -0.132^{***} at the session-day window, again slightly amplified under session-day equalization. I do not subject the vote-level difference-in-differences to this windowing: it is estimated on individual roll-call votes rather than on a window-aggregated legislator-level outcome, so the session-day question (how many in-session days fall in each leg of the window) does not enter its construction, and the headline participation effect it records (Table 3) is a per-vote rate that is mechanically invariant to the number of session days. The ideology nulls central to the paper’s interpretation hold in the preferred specifications: across all window definitions the β_{it} change in signed roll-call extremism (own-pole), in voting with the party majority, and in voting with party leaders each satisfy $|\hat{\beta}| < 0.05$ and $p > 0.10$ (signed extremism -0.008 at the calendar window and -0.007 at the session-day window, both $p > 0.45$), consistent with the precise full-sample null (Table 4, with-party -0.003 , with-leader 0.000). The Conservative-shift row reports the raw conservative-direction change in β_{it} , not signed by the legislator’s own party pole as the paper’s extremism measure is; its small coefficient is essentially identical at the two windows (0.022 versus 0.024), so it too shows no session-day sensitivity, which is the quantity this robustness check evaluates.

For the count outcomes, the calendar-time and trimmed-window cutoff estimates agree with the session-week cells the main text prints: cosponsorship holds under every construction (-0.965^{***} calendar, -1.208^{***} trimmed, -1.649^{***} per session-week), e-newsletters, measured in calendar time by design, hold under the trim (-0.688^{***} calendar, -0.696^{**} trimmed), and sponsorship’s cutoff cells are null under all three constructions (-0.074 calendar, -0.030 trimmed, -0.112 per session-week)—the cutoff power boundary discussed in the main text, whose panel design detects the session-time decline. The absolute change in the β_{it} ideology score (an auxiliary measure of overall ideological movement distinct from the headline within-party and leader-following nulls) is a precise null at both windows (0.008 at the calendar window, 0.003 at the session-day window; $p > 0.30$). I report both cells

Table A.18: Pre/Post-Primary Outcomes under Calendar versus Session-Day-Equalized Windows

Outcome	Calendar window	Session-day window	Δ (sd – cal)	N
<i>Panel A: Simple-mean RDD ($rv < 0.2$, local-linear OLS, change / weekly-rate outcome).</i>				
Roll-call participation	–0.119*** (0.025)	–0.125*** (0.027)	–0.006	940
Voting with party	–0.021* (0.012)	–0.008 (0.014)	+0.013	936
Voting with leader	–0.019 (0.012)	–0.014 (0.013)	+0.005	929
Bill sponsorship	–0.074 (0.046)	–0.030 (0.047)	+0.044	665
Bill cosponsorship	–0.965*** (0.293)	–1.208*** (0.408)	–0.243	578
E-newsletter	–0.688*** (0.225)	–0.696** (0.270)	–0.008	245
<i>Panel B: β_{it} RDD ($rv < 0.2$, OLS; Cong 84–117).</i>				
Conservative shift	0.022** (0.011)	0.024** (0.012)	+0.002	918
Signed extremism (β_{it} , own-pole)	–0.008 (0.011)	–0.007 (0.012)	+0.001	918
Absolute change ($ \beta_{it} $)	0.008 (0.008)	0.003 (0.009)	–0.005	918
With Party (β_{it})	–0.017 (0.011)	–0.011 (0.012)	+0.006	918
With Leader (β_{it})	–0.015 (0.011)	–0.017 (0.012)	–0.002	911
Participate (β_{it})	–0.127*** (0.028)	–0.132*** (0.029)	–0.005	940

Notes: Coefficient on PrimaryLoss in both panels under each window definition. The calendar window runs from the pre-primary date equidistant to the general election through the general-election date; the session-day window is trimmed to the smaller of the pre- and post-primary counts of House-active days, centered on the primary date. Heteroskedastic OLS standard errors. Significance: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Replication materials are provided with the paper.

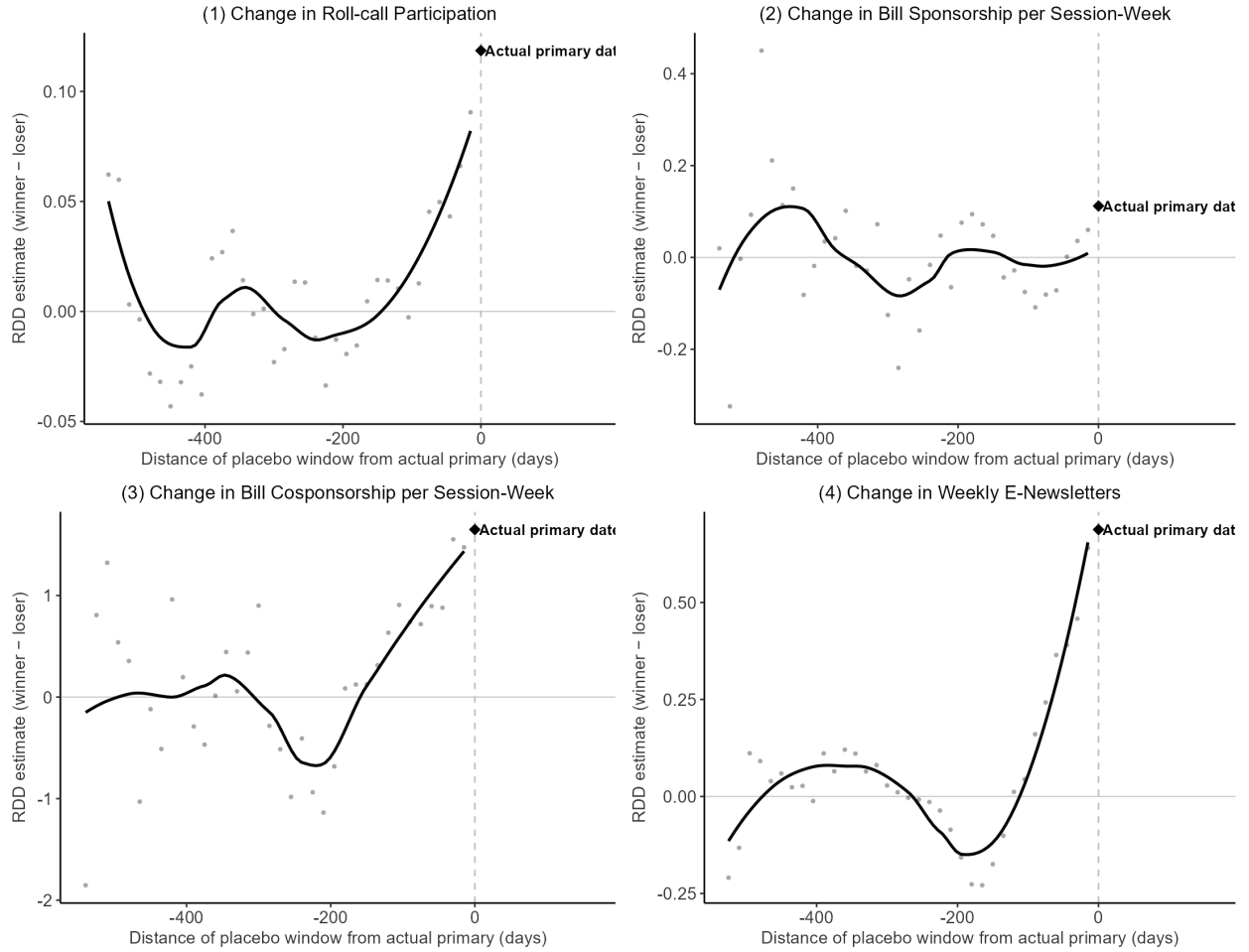
transparently; the headline participation decline and the headline ideology nulls hold across the windowing.

A.20 Placebo Window Test

Because the pre- and post-primary windows are relatively short, the estimated effects could in principle reflect idiosyncratic variability in a particular short window rather than the primary date itself. To assess whether the estimated effects are specific to the actual primary date rather than an artifact of an arbitrary short window, I conduct a placebo-window test. For each placebo draw I displace every legislator’s primary date by a random amount toward the no-election portion of the term—holding each legislator’s own window length fixed—reconstruct the effort measures within the displaced window, and re-estimate the close-primary RD. If the effect is tied to the actual primary, windows placed well before it, when no election has yet occurred, should yield null estimates, and the actual-date estimate should be an outlier in the resulting distribution.

Figure A.5 plots, for each effort outcome, the RD estimate against the average distance between the placebo and actual primary dates. The pattern matches the identifying logic: estimates are largest when the window still straddles the actual primary and decay toward zero as it is moved into the no-election period, with the actual-date estimate (the diamond at distance zero) sitting at the top of each panel. Table A.19 quantifies the test against the null formed by placebo windows placed well before the primary (average displacement exceeding 150 days). The actual participation estimate is never matched by such a placebo window ($p < 0.002$), e-newsletters likewise ($p < 0.002$), and cosponsorship is an outlier as well ($p = 0.006$). Sponsorship is the exception ($p = 0.210$), consistent with its null weekly-rate effect throughout the article. Because every window here is short and symmetric around its center date, the actual-date estimates use the symmetric-window construction rather than the full pre-window of the headline design; the participation anchor (0.119) is nonetheless close to the headline estimate (0.129***).

Figure A.5: Placebo Window Test: RD Estimates Across Displaced Windows



Note: Each panel plots the close-primary RD estimate (winner – loser) for one effort outcome against the average distance, in days, between the placebo and actual primary dates. Gray points are individual placebo draws (random window displacements toward the no-election period); the solid line is a local-linear smooth. The diamond marks the actual-date estimate (distance zero). Sponsorship and cosponsorship are measured per session-week within each displaced window, per the measurement convention of Table 3.

Table A.19: Placebo Window Test: Actual-Date Estimate vs. the No-Election Null

Outcome	Actual estimate	Null mean	RI p -value
Change in Roll-call Participation	0.119	−0.003	< 0.002
Change in Weekly Bill Sponsorship	0.112	0.012	0.210
Change in Weekly Bill Cosponsorship	1.649	−0.502	0.006
Change in Weekly E-Newsletters	0.688	0.006	< 0.002

Note: The null distribution comprises placebo windows with an average displacement exceeding 150 days from the actual primary (windows placed in the no-election portion of the term). The RI p -value is the share of those placebo draws whose estimate is at least as large in absolute value as the actual-date estimate; $p < 0.002$ indicates that no placebo draw (of 500) matched the actual estimate. All outcomes use the symmetric-window construction: participation as the within-window participation-rate change, count outcomes as the within-window weekly-rate change, with sponsorship and cosponsorship rates per session-week per the measurement convention of Table 3.

A.21 Previous Congress Placebo

As a further placebo, I repeat the analysis with outcomes from one Congress earlier. For each legislator at Congress t , I recompute the outcomes for the *previous* Congress $t - 1$ —a term in which that legislator had not yet faced the close primary I exploit—and re-estimate each design on them: the within-Congress RDs of the $t - 1$ outcome on the Congress- t treatment, and the weekly count panels rebuilt one Congress earlier with each legislator assigned the following Congress’s primary outcome. If the discontinuities and panel estimates I report were artifacts of fixed differences between narrow winners and narrow losers rather than the post-primary removal of the reelection incentive, they would appear at $t - 1$ as well. They do not. Table A.20 collects the placebo for every design-margin cell of Table 3. Table A.21 then decomposes the discontinuity placebos—the seven effort and position outcomes—into pre-primary and post-primary levels and changes, in two samples: every legislator with a $t - 1$ record (Panel A), and the subset who also contested a primary in $t - 1$ (Panel B). The $t - 1$ outcomes are built from the same sources and design as the main analysis.

Thirteen of the fourteen change cells are null, and the one exception—sponsorship’s Panel A change—is marginal at the ten-percent level ($+0.143$, $p = 0.076$) and runs *opposite* in sign to the post-primary estimates, a direction that cannot produce the post-primary decline. The placebo therefore confirms that the post-primary behavioral change is specific to the Congress in which the close primary occurs. Two features of the level columns are worth noting and are not placebo violations. First, the pre-primary participation level runs marginally against eventual losers (-0.022 , $p \approx 0.10$), echoing the small baseline participation advantage that narrow winners hold over narrow losers even before the primary; the within-term *change* it implies is nonetheless zero. Second, measured as weekly rates the count-based $t - 1$ levels are small: the cosponsorship level gap is statistically zero on every unit, sponsorship’s pre-primary cell is significant in the direction of eventual losers being the *less* active group one Congress earlier—the direction that works against, not toward, a manufactured post-primary decline (Table A.2)—and e-newsletters show at most a marginal gap in the same direction.

Table A.20: Previous-Congress Placebos across Designs and Margins

	Roll-call Participation	Bill Sponsorship	Bill Cosponsorship	E-Newsletters
<i>Panel A: Regression Discontinuity</i>				
Additive: change in rate	0.013 (0.019)	0.143* (0.080)	0.224 (0.607)	0.106 (0.168)
<i>N</i>	750	515	449	168
Proportional: proportion	0.026 (0.030)	−0.062 (0.081)	−0.013 (0.032)	0.021 (0.099)
<i>N</i>	779	455	446	141
<i>Panel B: Difference in Differences</i>				
Additive: least squares	—	0.037 (0.031)	0.085 (0.364)	−0.100 (0.088)
<i>N</i>		88,820	78,186	36,290
Proportional: Poisson	—	0.153 (0.158)	−0.008 (0.097)	−0.160 (0.240)
<i>N</i>		79,667	79,128	30,775

* $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$

Note: Each cell re-estimates the corresponding specification of Table 3 one Congress earlier, shown under the loss framing (winner–loser sign negated); standard errors in parentheses. Discontinuity cells regress Congress $t - 1$ outcomes on the Congress- t treatment within $|rv| < 0.2$: the additive cells are the Change column of Table A.21, and the proportional cells are the previous-Congress proportion column of Table D.2. Difference-in-differences cells rebuild the weekly count panel one Congress earlier and assign each legislator the following Congress’s primary outcome; Poisson coefficients are on the log scale. Roll-call participation’s proportional placebo is the previous-Congress votes-cast proportion (its two margins are mechanically linked; see Table 3); its difference-in-differences placebos are not constructed—a vote-level previous-Congress panel would require a separate pseudo-date convention—and its remaining discontinuity placebos, including the decomposed levels, appear in Table A.21. All cells are statistically null except sponsorship’s additive discontinuity cell, marginal at the ten-percent level and opposite in sign to the post-primary estimates: one Congress earlier, future losers show a slight relative *increase* in the rate change, a direction that cannot produce the post-primary decline. Additive cells for sponsorship and cosponsorship follow the session-time measurement of Table 3.

Table A.21: Previous-Congress Placebo: Decomposed RD on Congress $t - 1$ Outcomes

Outcome	Pre-Primary Level	Post-Primary Level	Change (Headline)	N (Post)
<i>Panel A: Hybrid (all close-primary legislators with a $t - 1$ record)</i>				
Participation	-0.022* (0.013)	-0.009 (0.022)	0.013 (0.019)	750
Signed Extremism	-0.040 (0.025)	-0.039 (0.026)	0.001 (0.013)	747
With Party	0.004 (0.018)	0.003 (0.019)	-0.002 (0.009)	747
With Leader	-0.002 (0.019)	-0.010 (0.019)	-0.010 (0.009)	738
Sponsorship	-0.171** (0.085)	-0.028 (0.069)	0.143* (0.080)	515
Co-sponsorship	0.174 (0.824)	0.398 (0.675)	0.224 (0.607)	449
E-Newsletter	-0.400* (0.237)	-0.294 (0.271)	0.106 (0.168)	168
<i>Panel B: Contested-in-both (the $t - 1$ Congress also featured a contested primary)</i>				
Participation	-0.033* (0.020)	-0.019 (0.031)	0.014 (0.027)	365
Signed Extremism	-0.044 (0.033)	-0.045 (0.034)	-0.001 (0.016)	363
With Party	0.000 (0.024)	-0.001 (0.025)	-0.002 (0.013)	363
With Leader	0.001 (0.024)	-0.014 (0.026)	-0.015 (0.013)	361
Sponsorship	-0.219* (0.122)	-0.085 (0.093)	0.135 (0.121)	252
Co-sponsorship	0.354 (1.206)	0.401 (1.040)	0.047 (0.885)	216
E-Newsletter	-0.243 (0.208)	-0.221 (0.285)	0.023 (0.184)	89

Note: For each close-primary legislator at Congress t , outcomes are measured at Congress $t - 1$ and regressed on the Congress- t treatment within $|rv| < 0.2$ (local-linear OLS, conventional standard errors in parentheses). Coefficients are expressed as the effect of losing the primary. Vote-based outcomes (participation, signed extremism, with-party, with-leader) use afterprimary-split means—participation over ≥ 10 available roll calls per side, positions over ≥ 10 cast votes—and the Change column is the post-minus-pre difference. Signed extremism is the own-party-pole indicator (higher = more own-pole voting); absolute extremity has no pre/post analog and is omitted, as in the contemporaneous decomposition (subsection A.2). Count-based outcomes (sponsorship, cosponsorship, e-newsletters) are weekly rates, as in the main text: the Pre column is the $t - 1$ rate from the start of the Congress to the primary, the Post column the rate from the primary to the general election, and the Change column their difference; sponsorship and cosponsorship rates are per session-week, following the measurement convention of Table 3. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

A.22 Bootstrap Standard Errors for Headline Estimates

Several of my dependent variables are themselves estimated quantities — in particular, the within-term regression coefficients β_{it} that summarize a legislator’s ideology in a Congress are first-stage estimates that carry measurement error — so the analytical standard errors may understate the true uncertainty. As a check, I bootstrap the standard errors. I implement two bootstrap procedures at the same fixed bandwidth (± 0.2) used in the main RDD analysis, with $B = 1,000$ replications each:

- **Nonparametric bootstrap:** resample legislator–Congresses with replacement from the full close-primary sample and re-fit the local-linear RDD on the resampled data.
- **Wild bootstrap:** keep the original observations fixed and perturb fitted residuals by Rademacher weights (± 1 with probability 0.5 each), then re-fit the RDD on the perturbed dependent variable.

Table A.22 reports the resulting standard errors side by side with the analytical OLS SE and the corresponding two-sided p -values. For the four ideology outcomes the bootstrap SEs are roughly 55–65% larger than the analytical SEs, consistent with the concern that analytical OLS understates uncertainty when the dependent variable is itself estimated. None of the substantive conclusions changes: the participation effect remains highly significant ($p < 0.001$ under all three SEs); both ideology nulls (with-party, with-leader) remain null with bootstrap $p > 0.8$; and the auxiliary extremity nulls (signed and absolute) likewise remain null. For the three weekly-rate count outcomes, the bootstrap SEs sit close to the analytical ones for sponsorship (null under all three SEs) and cosponsorship (significant at the 1% level under all three); e-newsletters carry the largest bootstrap inflation but remain significant at the 5% level under both bootstraps ($p = 0.027$ nonparametric, $p = 0.021$ wild).

Table A.22: Bootstrap Standard Errors at Bandwidth ± 0.2 ($B = 1,000$)

Outcome	Coefficient	Standard Error			Two-sided p -value		
		Analytical	NP Boot	Wild Boot	Analytical	NP Boot	Wild Boot
Roll-call Participation	−0.129***	0.019	0.025	0.024	<0.001	<0.001	<0.001
Weekly Bill Sponsorship	−0.024	0.075	0.074	0.074	0.749	0.746	0.747
Weekly Bill Cosponsorship	−1.521***	0.537	0.471	0.476	0.005	0.001	0.001
Weekly E-Newsletters	−0.639***	0.198	0.290	0.276	0.001	0.027	0.021
Change in Extremity (signed)	−0.003	0.009	0.014	0.014	0.742	0.832	0.833
Change in Extremity (abs.)	0.008	0.006	0.010	0.011	0.195	0.428	0.431
Change in Voting With Party	−0.003	0.008	0.013	0.013	0.715	0.818	0.815
Change in Voting With Leader	−0.0004	0.008	0.013	0.013	0.952	0.971	0.971

* $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$ (analytical SE)

Note: Each row reports the local-linear OLS RDD coefficient on Primary Loss at fixed bandwidth ± 0.2 , with three SE estimates: (i) conventional analytical OLS, (ii) nonparametric bootstrap at $B = 1,000$ (resample legislator–Congresses with replacement and re-fit), (iii) wild bootstrap at $B = 1,000$ (Rademacher perturbation of residuals; observations held fixed). Two-sided p -values are computed from the z -statistic $\hat{\beta}/\text{SE}$ under the standard normal. Significance stars on the coefficient column are based on the analytical SE for consistency with Table 3 and Table 4. Replication materials are provided with the paper.

B Timing of the E-Newsletter Effect

Table B.1 reports a corrected two-group event study of weekly e-newsletter counts, with event-time main effects and party-by-Congress-by-week fixed effects, displayed at the eight-week bins of the printed event-study figures. The estimates are the winner–loser gap in weekly e-newsletter counts by event-time bin relative to the primary; no Primary Win/Loss reframing applies because the quantity is already a winner–loser difference. No individual lead bin differs from the reference; over the display’s long pre-window the leads are jointly non-flat ($p = 0.046$), and the four-week bins specified in advance locate the only individually resolvable pre-primary movement in the final campaign month before the primary (-0.20 , $p = 0.03$), consistent with embattled incumbents intensifying constituent communication in the closing weeks of the campaign. A discrete post-primary break follows: the gap rises to between 0.66 and 0.82 additional e-newsletters per week across the three post-primary bins. This timing underlies the campaign-month caveat noted in the main text.

Table B.1: Event-Study Estimates for Weekly E-Newsletter Counts

Bin (rel. to primary)	Winner – loser gap (weekly e-newsletters)
≤ -9	–0.01 (0.10)
–8	0.11 (0.15)
–7	0.05 (0.10)
–6	0.14 (0.14)
–5	0.05 (0.11)
–4	0.16 (0.12)
–3	0.10 (0.15)
–2	<i>reference</i>
–1	–0.08 (0.13)
0	0.66*** (0.20)
+1	0.82** (0.33)
$\geq +2$	0.72** (0.34)

* $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$

Note: Each row reports the winner–loser gap in weekly e-newsletter counts at an eight-week event-time bin relative to the primary, from a two-group event study with event-time main effects and party-by-Congress-by-week fixed effects. Bin –2 is the omitted reference; negative bins precede the primary, bin 0 begins the day after it, and the edge bins pool all earlier and later event time. Standard errors in parentheses. No individual lead differs from the reference, though the display’s leads are jointly non-flat ($p = 0.046$). At the four-week resolution specified in advance, the only individually significant pre-primary cell is the final campaign month (–0.20 per week, $p = 0.03$), a deviation the eight-week bins average away (the displayed final pre-primary bin is –0.08, $p = 0.50$); that four-week localization is the evidence behind the main text’s campaign-month reading.

Table C.1: Cosponsorship: Pre-Specified Criteria under the Loss Framing

Quantity	Estimate (SE)	p	N
<i>Criterion (i): weekly-rate discontinuity</i>			
Discontinuity (Primary Loss)	−0.825*** (0.307)	0.007	578
Previous-Congress placebo	−0.125 (0.320)	0.70	449
<i>Criterion (ii): weekly-panel estimate</i>			
Full sample	−0.778*** (0.147)	< 0.001	239,626
Close races	−0.841*** (0.183)	< 0.001	55,169
Full sample, Poisson	−0.387*** (0.061)	< 0.001	209,231
Close races, Poisson	−0.443*** (0.072)	< 0.001	47,022
<i>Criterion (iii): event-study leads (jointly flat?)</i>			
Joint-leads test — FAILS	—	0.005	

* $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$

Note: Estimates are shown under the loss framing (winner–loser sign negated). Criterion (i) is the weekly-rate regression-discontinuity estimate with its previous-Congress placebo; criterion (ii) is the weekly legislator-period panel estimate on the full sample and on close races; Poisson rows re-estimate the weekly panel by fixed-effects Poisson regression with the same fixed effects and clustering, and their coefficients are on the log scale, with $\exp(b) - 1$ the proportional change in a legislator’s own rate; criterion (iii) is the joint test that the event-study leads are flat. All three criteria were specified in advance and all are required to advance the outcome as evidence.

C Cosponsorship: Records and Timing Assessment

The cosponsorship records cover Congresses 96–117 (1979–2022): records for 1979–2004 come from the Fowler-era cosponsorship files, and those for 2005–2022 are parsed from official Congress.gov bulk data, covering all four House bill types (bills and simple, joint, and concurrent resolutions). The two sources overlap in the 108th Congress, where per-legislator counts correlate at 0.98 (0.997 among legislators present in both) with a stable level ratio of 1.07.

I specified three criteria in advance, all required for the outcome to be advanced as evidence: (i) a significant discontinuity in the weekly rate with a flat previous-Congress placebo; (ii) a significant weekly-panel estimate; and (iii) event-study leads that are jointly flat with a discrete break at the primary. Table C.1 reports the results under the loss framing.

The event study is decisive against criterion (iii). Table C.2 shows that several lead bins

Table C.2: Event-Study Estimates for Weekly Cosponsorship Counts

Bin (rel. to primary)	Winner – loser gap (weekly cosponsorships)
≤ -9	0.70** (0.28)
-8	0.33 (0.27)
-7	0.51** (0.22)
-6	0.47** (0.19)
-5	0.76*** (0.14)
-4	0.10 (0.17)
-3	0.04 (0.15)
-2	<i>reference</i>
-1	0.38*** (0.14)
0	1.50*** (0.21)
+1	1.21*** (0.23)
$\geq +2$	1.44*** (0.31)

* $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$

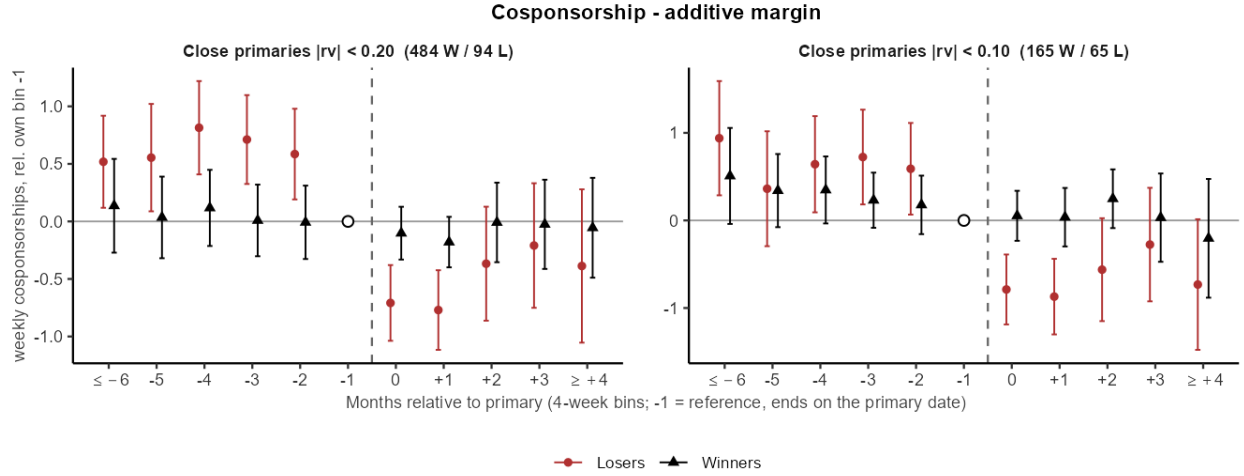
Note: Each row reports the winner–loser gap in weekly cosponsorship counts at an eight-week event-time bin relative to the primary, from the two-group event study on Congresses 96–117 (in-session weekly panel, per the session-time convention of the main text). Bin -2 is the omitted reference; negative bins precede the primary, bin 0 begins the day after it, and the edge bins pool all earlier and later event time. Standard errors in parentheses. The joint test that the leads are flat is rejected at these eight-week bins ($p < 0.001$) and at four-week bins with the final pre-primary month as reference ($p = 0.005$).

differ significantly from the reference, so the leads are not jointly flat: the criterion, specified and evaluated on four-week bins, puts the joint test at $p = 0.005$, and the failure is invariant to the display—it holds at the eight-week bins shown ($p < 0.001$) and on either week grid—so cosponsorship does not clear the criteria specified in advance. The differential gaps alone, however, do not reveal which group moves; the decomposition below traces the violation to the losers’ own path.

Figure C.1 separates the two groups’ own paths—the loser path from the event-time main effects, the winner path from the main effects plus the interactions, each relative to that group’s final pre-primary month. Winners’ cosponsoring shows no campaign run-up and no primary-anchored change at either bandwidth. Eventual losers rise above their final pre-primary pace over the contest season, peak three to four months out, and converge back to that pace as the primary nears; at the primary they drop discretely, to nearly one cosponsorship per week below it. The trough attenuates over the following months but does not close. Measured against each group’s own rate, the pooled gap from four months after the primary through each legislator’s general election leaves losers roughly a third below their own pace (-0.43 , SE 0.15 , $p = 0.004$, at the ± 0.20 bandwidth; -0.42 , SE 0.16 , $p = 0.008$, at ± 0.10); the absolute gap points the same way at both bandwidths (-0.39 and -0.73 cosponsorships per week), the narrower marginal ($p = 0.054$). The below-baseline destination is the informative feature: mean reversion returns unusually active legislators to their own normal pace—fixed from outside the contest season by the prior-year comparison in the main text—and losers end the term well below theirs. At the ± 0.10 bandwidth the two groups’ pre-primary paths are approximately parallel and the divergence concentrates at the primary itself, though with 65 losing terms this is a failure to reject rather than a confirmation.

Restricting the series to House bills (HR) alone gives estimates within sampling variation of those for all four bill types. Bill sponsorship’s level-margin cutoff cells remain null, and its event-study leads are flat. One feature stands out sharply: eventual losers introduce fewer bills after the primary while winners’ sponsorship is flat throughout. Measured over calendar weeks the absolute interruption appears confined to roughly eight weeks around the primary; measured over session weeks—the convention of the main text—the absolute gap persists through the post-primary period (joint post-primary $p = 0.056$; the pooled final bin alone, $p = 0.01$), consistent with the session-time panel estimate. Sponsorship’s proportional-margin estimates, significant in both designs, are reported with the margin comparison in Appendix D; the main text claims the decline on the proportional margin, with the session-time panel detecting it on the additive margin as well, its cutoff nulls the

Figure C.1: Winner and Loser Cosponsorship Trajectories around the Primary



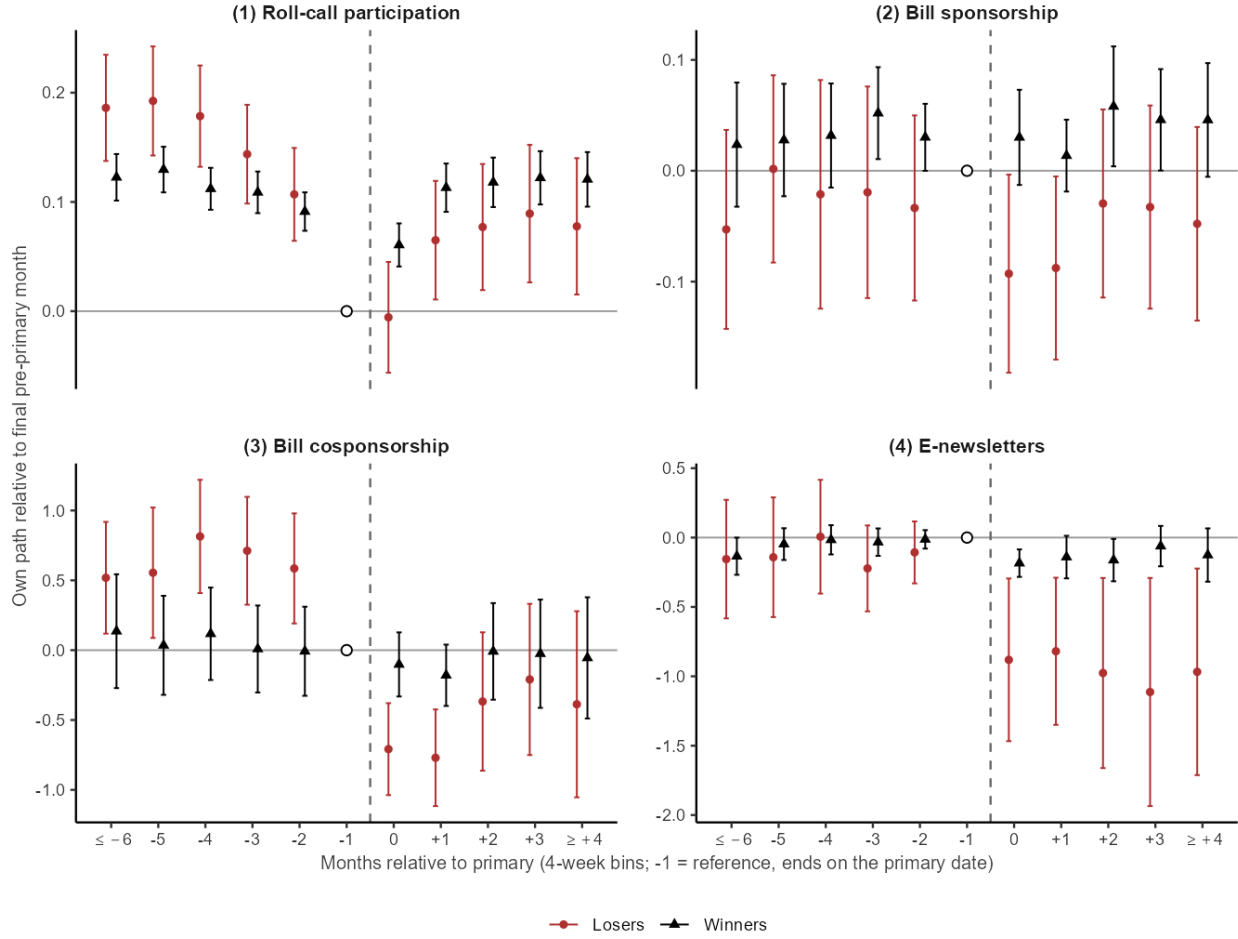
Note: Weekly cosponsorship counts by event-time bin relative to the primary, separately for eventual losers (event-time main effects) and winners (main effects plus interactions), each relative to that group’s own final pre-primary month (bin -1 , open circle). Estimates from the two-group event study on Congresses 96–117 with legislator-Congress and party-by-Congress-by-week fixed effects; on the week grid of the main text the reference month ends on the primary date and the first post-primary bin begins the day after it; the ≤ -6 and $\geq +4$ edges pool, and the post window ends at each legislator’s general election. Error bars are 95% confidence intervals, clustered by legislator.

arithmetic consequence of the outcome’s low base rate.

Figure C.2 extends the same own-path decomposition to all four effort activities on the additive margin (close primaries, $|rv| < 0.20$). The reference month is not a common baseline across activities: for roll-call participation it is the bottom of the campaign-season dip, from which both groups rebound after the primary—winners by roughly half again as much, which is the discontinuity the main text documents. For the self-initiated activities the pattern mirrors the cosponsorship decomposition above: eventual losers sit significantly below their own final pre-primary pace in every post-primary bin for e-newsletters, and in the first two post-primary months for sponsorship and cosponsorship, while winners’ movements around the primary are far smaller—at most about a fifth of the losers’ in the e-newsletter series.

The sponsorship estimates carry one further decomposition, specified in advance: whether the proportional response reflects ceremonial output alone. Bills are classified as commemorative or substantive by a rule-based title classifier (ceremonial patterns and private relief)

Figure C.2: Winner and Loser Own Paths around the Primary: All Four Effort Activities



Note: Each panel plots the two groups' own paths in the activity—the loser path from the event-time main effects, the winner path from the main effects plus the interactions, each relative to that group's own final pre-primary month (bin -1, open circle)—on the additive margin among close primaries ($|rv| < 0.20$).

Bins are four weeks; the ≤ -6 and $\geq +4$ edges pool, the reference month ends on the primary date, and the post window ends at each legislator's general election. Sponsorship, cosponsorship, and e-newsletters are weekly counts on the calendar weekly panel with legislator-Congress and party-by-Congress-by-week fixed effects; participation is the vote-level analogue anchored on the primary date, for legislators with at least ten cast votes on each side. Error bars are 95% confidence intervals, clustered by legislator.

Both-margin versions of all four panels are included in the replication package.

built from the record series' own sources and validated against the Center for Effective Lawmaking's legislator-level commemorative counts (Volden and Wiseman, 2014): the correlation is 0.95, and commemorative bills are 7.1 percent of the records. The expectation before estimation was that cheap ceremonial position-taking would carry the post-primary

withdrawal, mirroring cosponsorship. The data answer otherwise (Table C.3). The proportional decline sits in substantive bills: the fixed-effects Poisson estimate on substantive bills alone is -0.444 ($p = 0.016$; a decline of about 36 percent of a legislator’s own substantive-bill rate), the location of the aggregate signal, and the substantive-only proportion at the cutoff agrees in direction (-0.131 , $p = 0.064$), with null previous-Congress placebos. On the additive margin the commemorative cell is a precise zero and the substantive cell, like the aggregate calendar-week panel, reaches only the ten-percent level. Commemorative bills are too sparse to carry information on any margin: most close-race legislator-terms introduce no commemorative bill in either equidistant window (498 of 665 close observations drop from the proportion), and the commemorative Poisson estimate (-0.444 , $p = 0.098$), while directionally consistent, carries a minimum detectable effect larger than the estimate itself. The transient eight-week dip described above likewise concentrates in substantive bills, with commemorative sponsorship flat throughout the term: a campaign squeeze on the bills that demand the member’s own time, not ceremonial withdrawal.

Table C.3: Sponsorship by Bill Class: Commemorative versus Substantive

	Commemorative	Substantive
RDD, additive: change in rate	−0.007 (0.006)	−0.013 (0.042)
<i>N</i>	665	665
RDD, proportional: post-primary share	−0.174 (0.156)	−0.131* (0.071)
<i>N</i>	167	554
Previous-Congress proportion placebo	−0.200 (0.216)	−0.035 (0.082)
<i>N</i>	125	446
DiD, additive: least squares	−0.002 (0.002)	−0.030* (0.018)
<i>N</i>	274,258	274,258
DiD, proportional: Poisson	−0.444* (0.268)	−0.444** (0.183)
<i>N</i>	81,599	225,162

* $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$

Note: Coefficients are the effect of losing the primary, estimated separately on each bill class under the specifications of the paper’s sponsorship column: the RDD additive cell is the weekly-rate change over the full pre- and post-primary windows, the proportional cell the post-primary share on equidistant windows (legislator-terms with no bills of the class in either window drop), and the panel cells the weekly count panel of Equation 5 by least squares and fixed-effects Poisson (Poisson coefficients on the log scale, with $\exp(b) - 1$ the proportional change in a legislator’s own rate). Classes from a rule-based title classifier (ceremonial patterns and private relief = commemorative/symbolic) validated against the Center for Effective Lawmaking’s legislator-level counts (correlation 0.95; commemorative bills are 7.1 percent of records). Previous-Congress placebos for the additive cells are likewise null. Standard errors in parentheses (RDD conventional; panel clustered by legislator).

D Window and Margin Robustness for the Count Outcomes

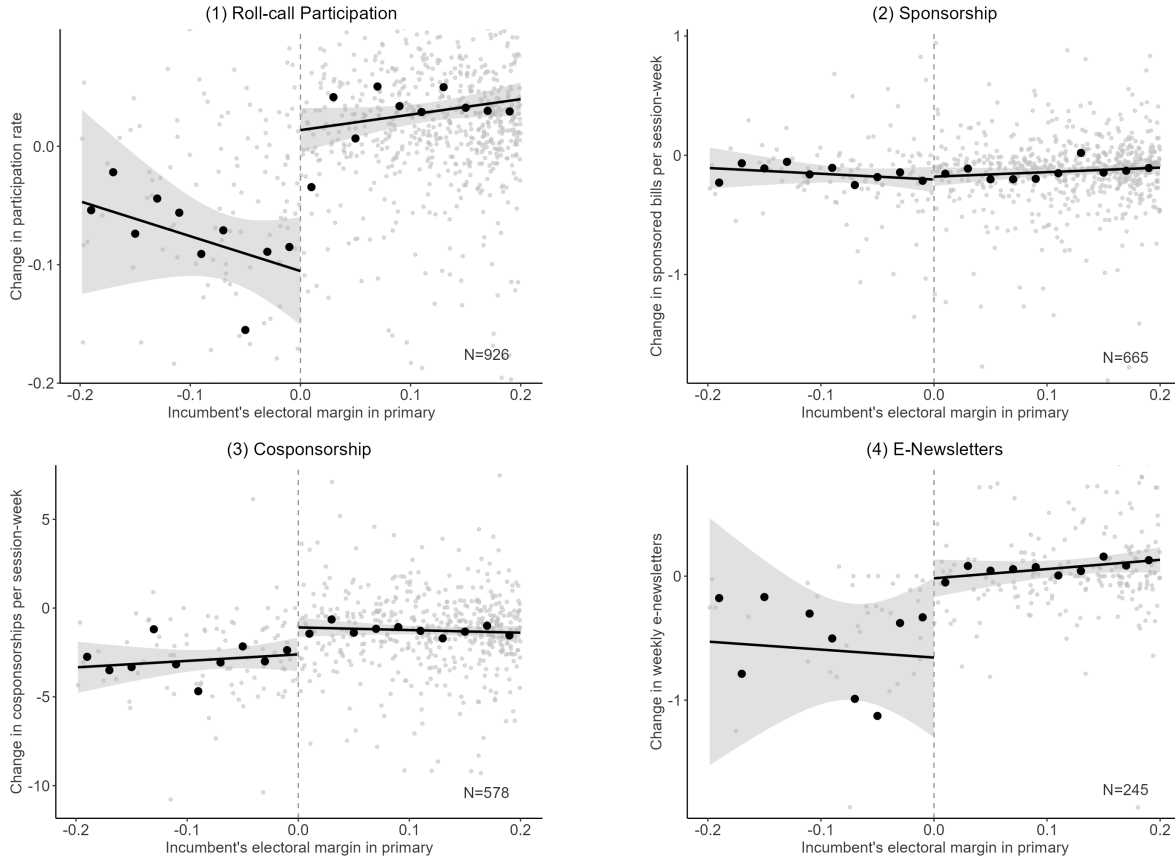
This section varies the two construction choices behind the count discontinuities—the measurement window and the outcome’s normalization—one at a time, holding everything else fixed.

Figure D.1 is the additive-margin counterpart to the main text’s Figure 3: the same four activities and the same discontinuity, plotted as changes in absolute rates rather than as post-primary shares. Participation, e-newsletters, and cosponsorship break at the threshold on this margin as they do on the other. Sponsorship does not, and the flat panel is what the low base rate implies: at a median of about 0.16 bills per week, the proportional decline the main text reports amounts to a few hundredths of a bill per week, which lies inside the confidence band this display can resolve.

Figure D.2 repeats the main text’s count event studies on the additive margin, under the session-time measurement for sponsorship and cosponsorship. The shapes agree with the proportional panels: cosponsorship and e-newsletters break at the primary and stay down, and sponsorship’s session-week gap opens at the primary with jointly flat leads—the panel-power counterpart of its null cutoff cells.

The count outcomes are measured as weekly rates over each legislator’s full pre-primary and post-primary windows, which differ in length across legislators. A natural alternative construction equalizes the two windows within each legislator, measuring both rates over windows of equal length on either side of the primary. Table D.1 re-estimates the count discontinuities under this equidistant construction, with previous-Congress placebos. The conclusions are invariant to the window choice: the cosponsorship and e-newsletter discontinuities hold or strengthen under equidistance with clean placebos, and sponsorship remains statistically indistinguishable from zero under both constructions, though its equidistant point estimate is larger (-0.112 , $p = 0.18$, versus -0.024).

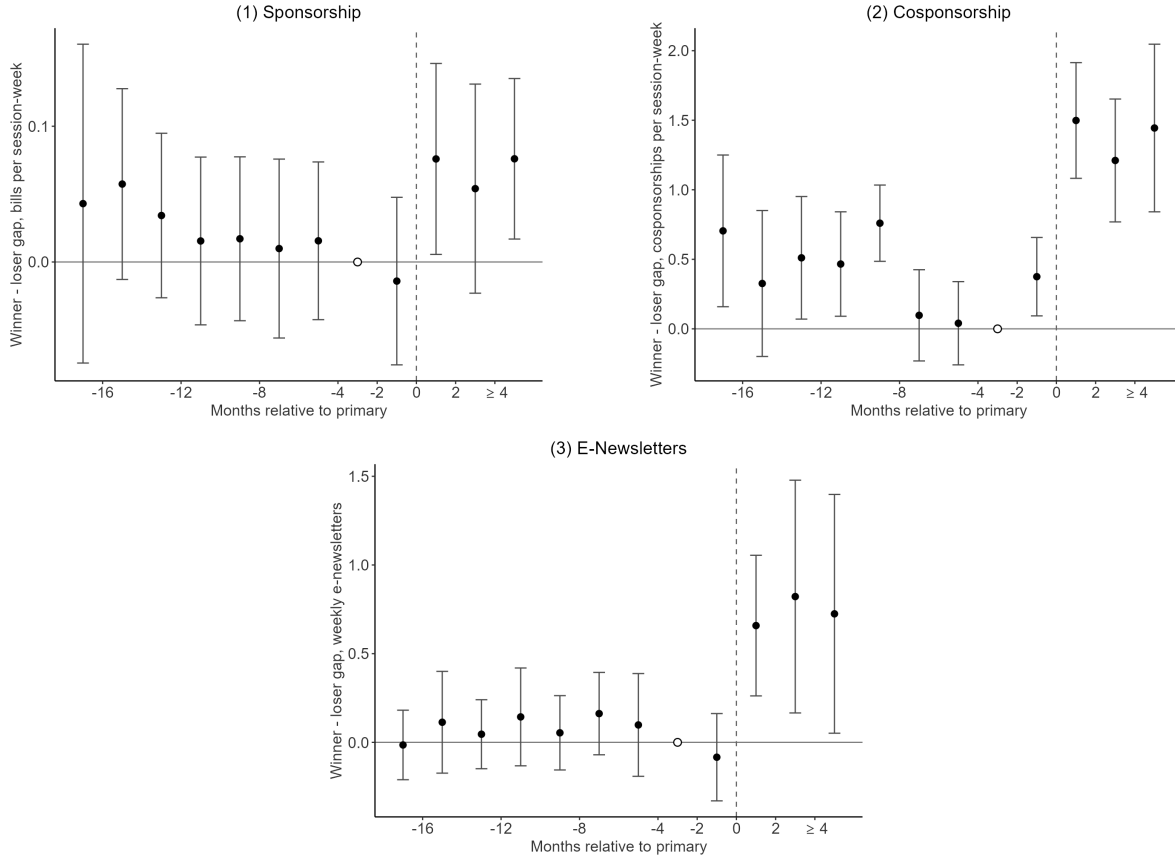
Figure D.1: Regression Discontinuity Results on Legislators' Effort-Related Activities: Additive Margin



Note: Each panel plots the within-term change in the activity against the incumbent's primary winning margin; legislators to the left of the threshold lose the primary. Participation is the change in the participation rate; the count outcomes are changes in weekly rates, measured from the start of the Congress to the primary and from the primary to the legislator's general election, with sponsorship and cosponsorship rates per session-week per the measurement convention of Table 3. Large black points are averages within 0.02-point bins, small gray points the raw data, and the lines are OLS fits estimated separately on each side; each panel's display range is set around the binned means and fitted lines so the estimated discontinuity is legible, and raw points outside that range are not shown (fits use all observations). Sample sizes match the additive-margin cells of Table 3.

Table D.2 holds these equidistant windows and record series fixed and instead varies the outcome's normalization; it is the controlled counterpart of the margin columns in Table 3, which reports each margin under its own headline construction. The rate difference measures the absolute change in weekly activity; the proportion of activity occurring after the primary, $\text{post}/(\text{pre} + \text{post})$, measures the change relative to each legislator's own volume. Because a proportion is exposed to pre-primary level imbalance where one exists (subsection A.1),

Figure D.2: Count Event Studies on the Additive Margin



Note: Additive-margin counterparts of the proportional count panels in Figure 4, in the same event-time frame (two-month bins, reference bin months -4 to -2 , edges pooled). Sponsorship and cosponsorship are weekly counts on the in-session weekly panel (session-time convention of Table 3); e-newsletters are calendar weekly counts. Joint tests, leads/post: sponsorship 0.87/0.056, cosponsorship < 0.001 / < 0.001 , e-newsletters 0.05/0.002; standard errors clustered by legislator.

the table tests that exposure directly, with a previous-Congress proportion placebo and a pre-level-adjusted estimate for each outcome.

All three proportion estimates are significant at the cutoff, all three previous-Congress placebos are null, and all three estimates are essentially unchanged when the pre-primary level is partialled out. Cosponsorship's two margins agree at the cutoff (-1.649 and -0.118 , both $p < 0.01$); its event-study leads fail the flatness criterion specified in advance, a timing qualification assessed in Appendix C. Sponsorship's additive margin remains null at the cutoff while its proportional margin is significant: for so sparse an activity the cross-sectional cutoff lacks the power that the session-time panel has, which detects the decline on both

Table D.1: Window-Choice Robustness: Full-Window versus Equidistant Count Discontinuities

Outcome	Full-window RD	Equidistant RD	Equidistant previous-Congress placebo
Weekly Bill Sponsorship	−0.024 (0.075)	−0.112 (0.082)	0.076 (0.089)
<i>N</i>	665	665	515
Weekly Bill Cosponsorship	−1.521*** (0.537)	−1.649*** (0.514)	0.228 (0.544)
<i>N</i>	578	578	449
Weekly E-Newsletters	−0.639*** (0.198)	−0.688*** (0.225)	−0.021 (0.178)
<i>N</i>	245	245	168

* $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$

Note: Coefficients are expressed as the effect of losing the primary on the weekly-rate change measure; sponsorship and cosponsorship rates are per session-week, following the measurement convention of Table 3. The full-window column reproduces the headline construction (pre-primary window from the start of the Congress, post-primary window to the general election); the equidistant columns measure both rates over equal-length windows on either side of the primary (t^* construction) and report the previous-Congress placebo under the same construction. Sponsorship uses the 1973–2022 records (Congresses 93–117), cosponsorship 1979–2022 (Congresses 96–117), e-newsletters 2010–2022 (Congresses 111–117); $|rv| < 0.2$, local-linear OLS.

margins (main text). The proportional margin is where sponsorship’s response concentrates, and it is on that margin that the main text states sponsorship’s claim.

One residual reading survives these checks in principle: mean reversion. Even a pre-primary elevation too imprecise to detect at the cutoff could, if real, let a within-term contrast register losers’ return to normal levels as a post-primary decline. Table D.3 confronts this reading with a baseline that lies outside the contest season altogether: each legislator’s post-primary window is compared with the same calendar window one year earlier, in the same Congress’s off year. The baseline is seasonally matched and of identical length by construction, and it cannot contain a contest-season elevation. The two readings then make opposite predictions. Under mean reversion, losers return to their own normal level, so the winner–loser gap in year-over-year change is zero; under a genuine response, losers fall below their own prior-year pace.

Table D.2: Margin Robustness at the Cutoff: Rate Difference versus Post-Primary Proportion

Outcome	Rate difference (equidistant)	Proportion (equidistant)	Proportion, previous-Congress placebo	Proportion, adjusted for pre-primary level
Bill Sponsorship	−0.112 (0.082)	−0.165** (0.068)	−0.062 (0.081)	−0.158** (0.065)
<i>N</i>	665	564	455	564
Bill Cosponsorship	−1.649*** (0.514)	−0.118*** (0.030)	−0.013 (0.032)	−0.117*** (0.030)
<i>N</i>	578	575	446	575
E-Newsletters	−0.688*** (0.225)	−0.170** (0.072)	0.021 (0.099)	−0.168** (0.073)
<i>N</i>	245	210	141	210

* $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$

Note: Both margin columns use the same equidistant windows and record series as Table D.1; only the outcome’s normalization differs. The rate difference measures the absolute change in weekly activity; the proportion, post/(pre + post), measures the change relative to each legislator’s own volume. Coefficients are discontinuities at the cutoff, so zero marks no effect; read as a level, the proportion’s benchmark of unchanged activity is 0.5 (Table 3). The two margins have power in different places: a proportional decline in a sparse activity such as sponsorship (median base about 0.16 bills per week) implies an absolute change of only a few hundredths of a bill per week, inside the rate column’s confidence interval, while the proportion is exposed to pre-primary level imbalance where one exists. The previous-Congress placebo and pre-level-adjusted columns test that exposure directly: all three proportion estimates carry insignificant previous-Congress placebos and are essentially unchanged by the pre-level adjustment. Displaced-window randomization inference for the proportion measure returns $p = 0.002$ (sponsorship), $p < 0.002$ (cosponsorship), and $p = 0.04$ (e-newsletters). Cosponsorship’s timing caveat and the panel’s primacy for the count conclusions are unchanged.

The comparison finds a response. Losing the primary lowers the year-over-year proportion by 0.089 for cosponsorship ($p = 0.004$) and by 0.297 for e-newsletters ($p = 0.002$); the rate-difference margin agrees (−0.63 and −0.55 per week), and the previous-Congress placebos show no corresponding gap (the e-newsletter placebo is marginal with the opposite sign, which if anything works against the estimate). In the close window the means are direct: winners’ e-newsletter activity holds its prior-year pace (mean proportion 0.54, where 0.5 marks the prior-year level) while losers fall to 0.20, about a quarter of their prior-year season; cosponsorship means are 0.40 against 0.30, putting losers at well under half of their prior-year volume and winners at about two-thirds; both sit below one-half because bill activity declines over the course of a Congress for winners and losers alike—a common component the cutoff

contrast nets out. Sponsorship is null on both year-over-year margins, but the design cannot arbitrate here: its minimum detectable effect on the proportion, roughly 0.18, exceeds the proportional decline the equidistant estimates detect (0.165), so the null is uninformative rather than contrary, consistent with the power-bounded nulls reported in the main text.

Table D.3: Year-over-Year Robustness at the Cutoff: Post-Primary Activity against the Same Calendar Window One Year Earlier

Outcome	Year-over-year rate difference	Year-over-year proportion	Proportion, previous-Congress placebo
Bill Sponsorship	−0.021 (0.047)	−0.102 (0.065)	−0.006 (0.079)
<i>N</i>	665	577	458
Bill Cosponsorship	−0.634* (0.350)	−0.089*** (0.031)	0.005 (0.030)
<i>N</i>	578	574	446
E-Newsletters	−0.554** (0.218)	−0.297*** (0.094)	0.237* (0.124)
<i>N</i>	245	211	143

* $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$

Note: Coefficients are expressed as the effect of losing the primary. The post-primary window (primary to general election) is compared with the same calendar window shifted back one year, into the same Congress’s off year: the year-over-year rate difference is the post-primary weekly rate minus the prior-year rate, and the year-over-year proportion, $\text{post}/(\text{prior-year} + \text{post})$, is bounded, with 0.5 marking the prior-year pace; legislator-Congress cells with no activity in either window drop from the proportion. Windows are of identical length by construction, and the prior-year baseline cannot contain a contest-season elevation. This comparison is deliberately kept in calendar time: its identification is seasonal alignment of the two windows, which session-time conversion would not sharpen. The placebo column re-runs the proportion one Congress earlier. E-newsletter estimates are robust to dropping the 111th Congress, whose prior-year window partially precedes DCinbox coverage (−0.541** rate difference, −0.207** proportion). Record series as in Table D.1; $|rv| < 0.2$, local-linear OLS.

E Sponsorship Records: Sources and Curation

The sponsorship series merges the Congressional Bills Project’s records for 1973–2014 with sponsor records parsed from official Congress.gov bulk data for 2015–2022. Three curation steps precede the merge. First, records are restricted to members serving in the House in the relevant Congress. Second, two classes of identifier defects are repaired: legislators to whom Voteview assigns distinct ICPSR numbers across service spells are consolidated under a single identifier, recovering 1,104 sponsorship records, and one identifier with a corrupted digit (39337, for ICPSR 29337) is corrected—an edit accepted only because the corrected identifier uniquely matches the member’s chamber, state, and service years, affecting roughly 300 records. Third, the Congressional Bills Project’s files for the 114th and 115th Congresses are replaced with parses of the official Congress.gov records: the raw 114th file omits most of the freshman class (16.6 percent of House member-terms record zero sponsored bills, against 0.9 percent in the official records), and the raw 115th file includes Senate members and joint resolutions. Estimates for outcomes other than sponsorship are unaffected by these corrections at printed precision.

Table F.1: Precision of the Ideology Null: Minimum Detectable Effects and Equivalence Bounds

Outcome	Estimate	SE	MDE (80% power)	Equivalence bound
Voting with party	0.002	0.004	0.012	0.009
Voting with leader	0.005	0.004	0.013	0.013

Note: Estimates and standard errors are from the vote-level difference-in-differences specification. The minimum detectable effect (MDE) is the effect size detectable with 80% power at the 5% level; the equivalence bound is the tightest symmetric bound not rejected. Both are computed from the vote-level difference-in-differences standard errors, and both are expressed in the outcome’s own unit, the share of a legislator’s roll-call votes cast with the party or with the leader.

F Precision of the Ideology Null

Table F.1 quantifies the precision of the ideology null. A with-party or with-leader shift of even two votes in a hundred would have been detected: the minimum detectable effects (0.012–0.013) and equivalence bounds (0.009–0.013) all lie below that mark. This is the basis for the main text’s claim that the ideology null is a powered null rather than an absence of evidence.

G The Pre-Primary Onset and the Margin Window

The participation event study (Figure 4) shows a modest widening of the winner–loser gap in the final weeks before the primary. If this onset reflects anticipatory withdrawal as the outcome becomes foreseeable, it should scale with the electoral margin: in races decided within a few points, neither side can foresee the outcome, so little differential onset should appear near the cutoff. Table G.1 re-estimates the event study across margin windows, holding the specification fixed.

The pattern matches the prediction. The onset declines monotonically as the window tightens—from 0.086 in the full contested sample to 0.047 within twenty points, and to statistically indistinguishable from zero within ten points—while the discontinuity at the primary persists in every window and the pre-campaign leads are jointly flat throughout. Two further results sharpen the interpretation. First, incumbents essentially never lose primaries by wide margins: only one of 152 losing legislator-terms in the vote panel exceeded a twenty-point margin, so the onset cannot be attributed to a separate population of foregone-conclusion losers. Second, retaining all winners while restricting losers to close races leaves the onset largely intact (0.077), whereas restricting both sides to close races removes it. The pre-primary component therefore reflects the campaign-intensity asymmetry between closely contested candidates—who spend the final weeks campaigning at home, whether they go on to win or lose—and safe winners, rather than behavior specific to eventual losers. At the cutoff, the comparison the regression discontinuity design actually uses, the pre-primary gap is essentially zero and only the discontinuity at the primary remains.

Table G.1: Participation Event Study Across Margin Windows

Sample	Onset β_{-1} (final pre-primary bin)	Jump β_0 (primary bin)	Joint leads (≤ -3), p	Legislator-terms (W / L)
All contested margins	0.086*** (0.015)	0.121*** (0.016)	0.104	3,714 (3,562 / 152)
$ V_{it} < 0.20$	0.047*** (0.015)	0.108*** (0.017)	0.450	926 (775 / 151)
$ V_{it} < 0.10$	0.032 (0.020)	0.123*** (0.024)	0.579	399 (284 / 115)
$ V_{it} < 0.05$	0.024 (0.030)	0.085** (0.035)	0.484	197 (129 / 68)
All winners + losers $ V_{it} < 0.10$	0.077*** (0.017)	0.137*** (0.020)	0.188	3,677 (3,562 / 115)

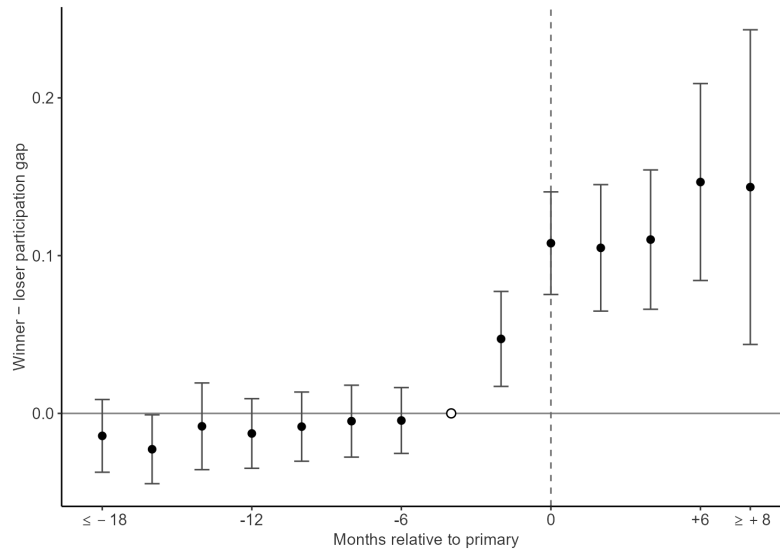
* $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$

Note: Each row re-estimates the corrected participation event study of Figure 4 (two-month bins over the full Congress, reference bin -2 , event-time main effects plus interactions, legislator-Congress and party-by-Congress-by-month fixed effects, standard errors clustered by legislator) on the indicated sample. Cells report the winner–loser gap (the negative of the primary-loss effect). Onset is the final two-month pre-primary bin; Jump is the first post-primary bin. The joint-leads test covers bins -9 through -3 , excluding the onset bin.

Extension: The Gap through the Post-General Lane-Duck Weeks

The headline event study ends each legislator’s post-primary window at the general election, because after the general the winner group itself mixes in general-election losers whose reelection incentives are also extinguished. As an explicitly labeled extension beyond that estimand, the figure below re-runs the close-race event study with the post-primary bins continuing through the end of the Congress, including the post-general lane-duck session. The onset–jump pattern is unchanged (onset 0.047, jump 0.108), and the winner–loser gap persists—indeed grows slightly—through the lane-duck weeks ($+0.143^{***}$ in the final bins), indicating that the participation response does not fade once the general election has passed.

Figure G.1: Full-Term Extension of the Participation Event Study



Note: Identical specification to the right panel of the main text’s participation figure, except that post-primary event-time bins continue through the end of the Congress rather than ending at the general election. Estimates are winner–loser gaps (the negative of the primary-loss effect), as in that figure.

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